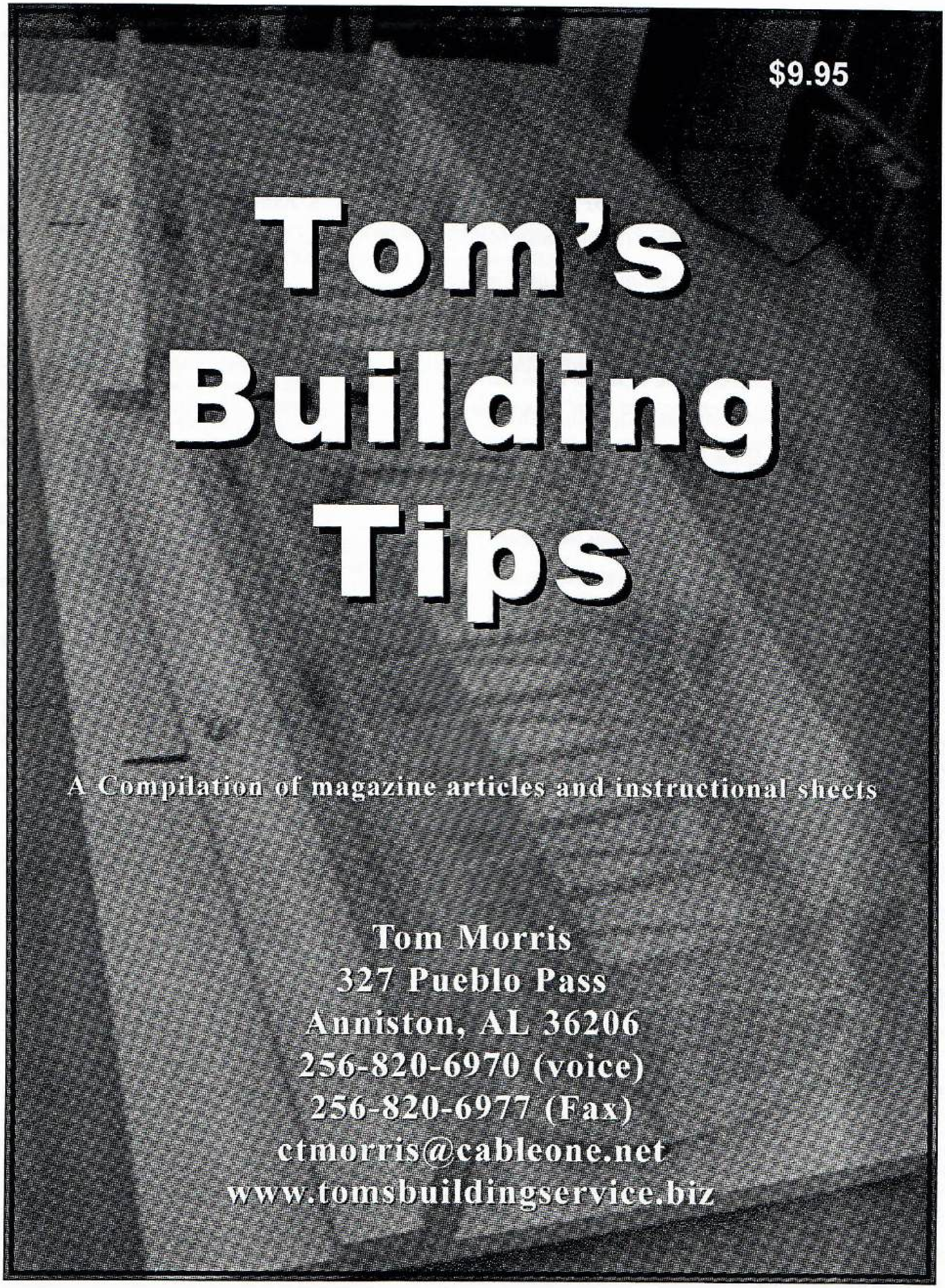


$9.95

Tom's Building Tips

A Compilation of magazine articles and instructional sheets

Tom Morris
327 Pueblo Pass
Anniston, AL 36206
256-820-6970 (voice)
256-820-6977 (Fax)
ctmorris@cableone.net
www.tomsbuildingservice.biz

Table of Contents

The "Lincoln Log" Wing

By Tom Morris

Remember how much fun it was to build log cabins with Lincoln Logs? Remember how nicely the pieces locked together without any glue? Well, now you can build a Lincoln log wing and have just as much fun! And you'll also get a lighter, stronger, straighter and cheaper wing. Once you've built a wing using this method, you'll never build a wing any other way!

Introduction

In the early 80's when I returned to C/L Stunt, the foam wing was the "in" thing. So, most of the wings I built had foam cores. After an eight year hiatus from building and flying I started back by building John Simpson's ***Cavalier***, a classic era ship which demands either an I-beam or C/D tube wing construction. I built two I-beamers last year. They were a little heavy, a little warped, and a lot of trouble to build. So, this year I decided to build two more Cavaliers with traditional C/D tube wings.

Now, I don't know about you, but there are a lot of things about traditional C/D tube wings I don't like. I've tried all the traditional jigging methods and found them time consuming and awkward to use. Building the traditional C/D tube wing can be a little frustrating trying to line up the ribs and having to pin everything together. I never could get all the ribs lined up vertically and horizontally. The leading edge tended to be too sharp and hard to sand with the overlapping sheeting and all that glue. I always had difficulty getting tight fits where the ribs meet the leading and trailing edges. And, many times there were gaps between the sheeting and the ribs. Then, when I took it off the jig it had poor torsional rigidity. It still twisted easily, which meant when I covered it, it would likely get some warps. Also, the traditional way of mounting the bellcrank and landing gear was weak.

So, I set about in a logical way to solve every one of those problems associated with building the traditional C/D tube wing. The criteria I set was that the building method had to be simple, easy, quick, inexpensive, light, straight, ridgid, not require the use of pins or Cya glue, and use materials readily available to everyone. Additionally, the jigging method had to be such that it could be used on any wing, not just one specific wing and would allow the wing to be easily turned over to work on the bottom side.

I took each problem and tried alternative solutions and then used the best solution. I'm sure a variation of most of these ideas have been used by someone. So, I'm not claiming to have invented anything new. But, this method puts it all together. And, it works! The two 540 square inch wings I built were a lot of fun to build, are dead straight, extremely strong, and super light. One weighed 8.9 ounces complete; the other 9.1. And, that's not using my best lumber.

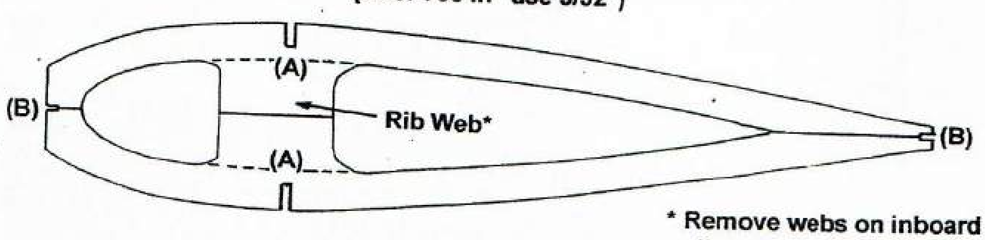
How large a wing could you build using 1/16" ribs and spars? I'd say you could go to 650 square inches safely; 700 if you keep the rib spacing to two inches and use Polyspan or Thermalspan and generously overlap in the middle. Over 700, substitute spruce for the spars and 3/32" for the ribs and sheeting.

Principles

Strength comes from using leverage and tying stress points to the spars and to the sheeting. Tying everything to the wing skin gives tremendous strength. Torsional rigidity comes from offsetting the cross bracing on each side of the spars. Ribs are for shape only. Don't rely on ribs to support landing gear and bellcrank.

Typical Rib in "Lincoln Log" Wing

All ribs are 1/16" Balsa
(Over 700 in^2 use 3/32")



* Remove webs on inboard ribs after cross bracing is installed

(A) Spar Slots are 1/16" (W) X 1/4" (D)

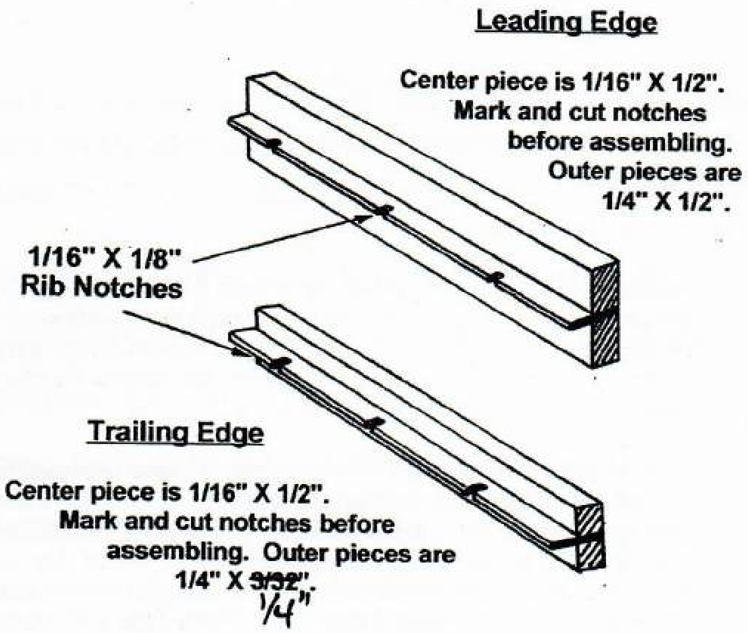
(B) End Slots are 1/16" (W) X 1/8" (D)

Tools

You'll need a flat work surface, a good balsa stripper and something straight and heavy to hold your jigged wing in place. I'm going to get some 2" by 2" angle iron for the next wing; a piece long enough for the entire trailing edge and a piece for each leading edge. You could use straight pieces of 2 X 4 lumber or build wooden "angle irons" out of plywood or boards. Just as long as it's straight and has a little weight.

Since there are a lot of 1/16" wide by 1/8" deep notches to be cut, you could make up some kind of tool to cut these. But, I find that if I mark the notch location with a felt tip pin, I can "eyeball" a 1/16 wide by 1/8" notch real accurately

Three Piece Leading and Trailing Edges



Notches are 1/16" wide by 1/8" deep.

or at least accurately enough and cut them with a #11 blade. A piece of hacksaw blade also makes a nice 1/16 wide cut. A razor plane is convenient for shaping the leading and trailing edges. A flat Perma-Grit file/sander is really convenient for shaping flaps and elevators as well as leading and trailing edges. You'll also find that square key stock from 3/8" to 1/8" in 1/16" increments is very handy when shaping flaps and elevators (See Photo).

Glue

I absolutely hate Cya glue! I don't like what it does to my eyes and lungs. I don't like what it does to balsa (turns it to rock). I don't like having to use kicker. I don't like the mess it makes at the tip of tl bottle. So, I use Pica sandable aliphatic resin glue. Tl tip of their 4 ounce bottle is just right. Four ounces wi build three planes for $2! It's stronger than needed, tacl enough to hold balsa vertically while it dries (about 1 minutes), and fills small gaps just fine. I use a cut o tooth pick to apply the glue in the corners around the ril and spars. It puts on just the right amount of glue.

Building Sequence

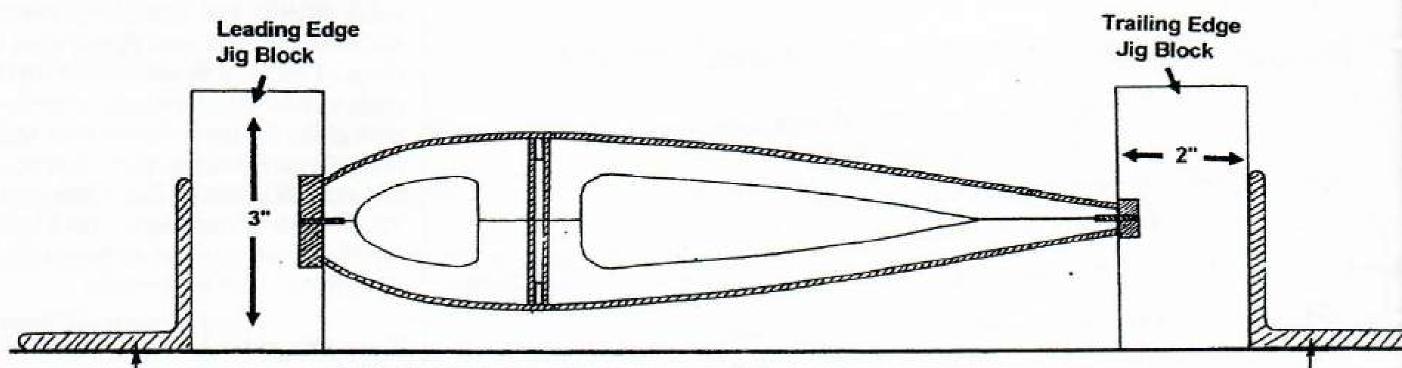
Ribs

Cut out a set of accurate ribs. I use See Temp to mal accurate templates. I'll go into more detail in the ne issue on how I cut dead accurate ribs. Mark spar locatio using the top view of the plan; don't use the individu ribs patterns. I find it's much more accurate to use t top view to mark the spars slots. Make sure you have t rib center lines accurately drawn on the ribs. Leave t ribs solid; no lightening holes. Heaviest ribs should outboard. Using the #2 and #4 full rib as a pattern, ma and cut 1/16" ply half ribs. Glue the half ribs to the fi ribs. Using the landing gear block as a pattern, cut ou notch in both #3 ribs.

Leading and Trailing Edges

The leading and trailing edges are made from three piec of balsa. Use a good balsa stripper such as the Mas Airscrew stripper. The center piece of both the leadi and trailing edges is 1/16" X 1/2". Mark and cut the 1/1 wide by 1/8" deep rib notches on the center piece. T outer pieces for the leading edge are 1/4" X 1/2"; for t trailing edge 1/4" X 3/32". Glue them, holding th together with masking tape. Weight them down flat.

"Lincoln Log" Wing Jig Blocks

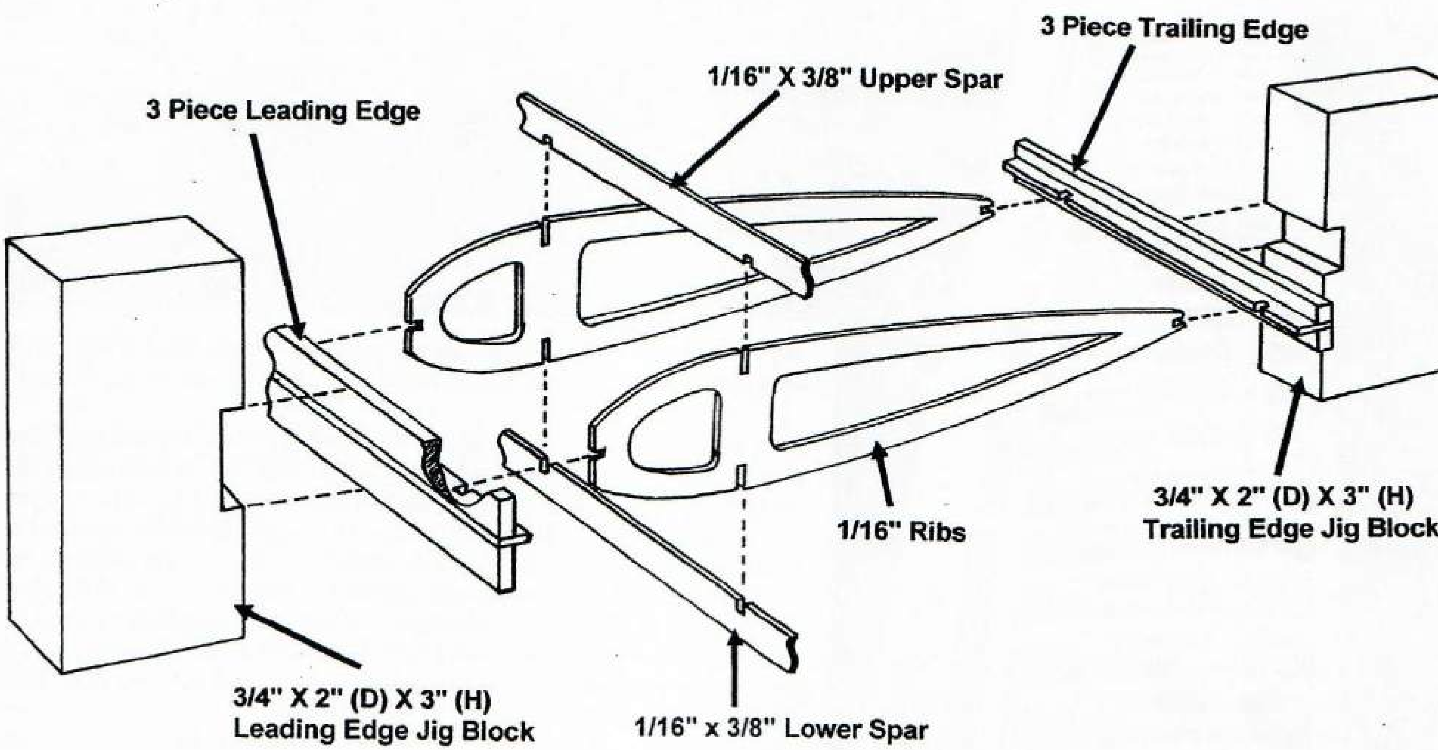


2" X 30"
Angle Iron (2)

Make jig blocks out of 3/4" plywood, board or hard balsa. Blocks are 3" tall by 2" wide. Make eight jig blocks for the leading edge and eight for the trailing edge. Slots in jig blocks are exactly centered, 1/4" deep and fit tightly over leading and trailing edges.

2" X 60 "
Angle Iron
Straight Edge (1)

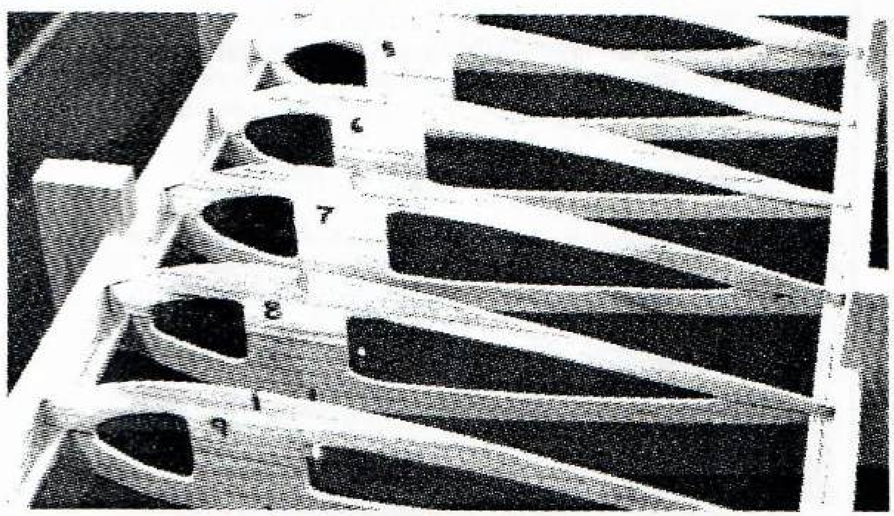
The "Lincoln Log" Wing



Note: Notches in leading edge, trailing edge, spars, and ends of ribs are 1/16" wide by 1/8" deep. Notches in ribs for spars are 1/16" wide by 1/4" deep.

Spars

While the leading and trailing edges are drying cut the 1/16" X 3/8" spars. Use fairly hard balsa. Or, better yet, use spruce. It's only twice as heavy as balsa but five time stronger! Mark the rib locations and cut the 1/16" wide by 1/8" deep notches.



Jig Blocks

The jig blocks can be made out of any kind of 3/4" board lumber, pine, spruce, poplar, oak, basswood, or 3/4 ply, or hard balsa. The main thing is to be dead accurate on these. I cut mine on a radial arm saw using a stop block. That way I know they are exactly the same height and the all the edges are absolutely square. You'll need 16 blocks 3" tall by 2" deep. The notches for the leading and trailing edges should be 1/4" deep and a snug fit. Make eight for the trailing edge and eight for the leading edge. I used a band saw to cut my notches.

Jigging Up !!!

Jigging Up

Now comes the real fun! Place the leading and trailing edges in the jig blocks and start adding the ribs. If a jig block does not fit tightly, you can tack glue it to the leading or trailing edge at the outer corners which will later be removed

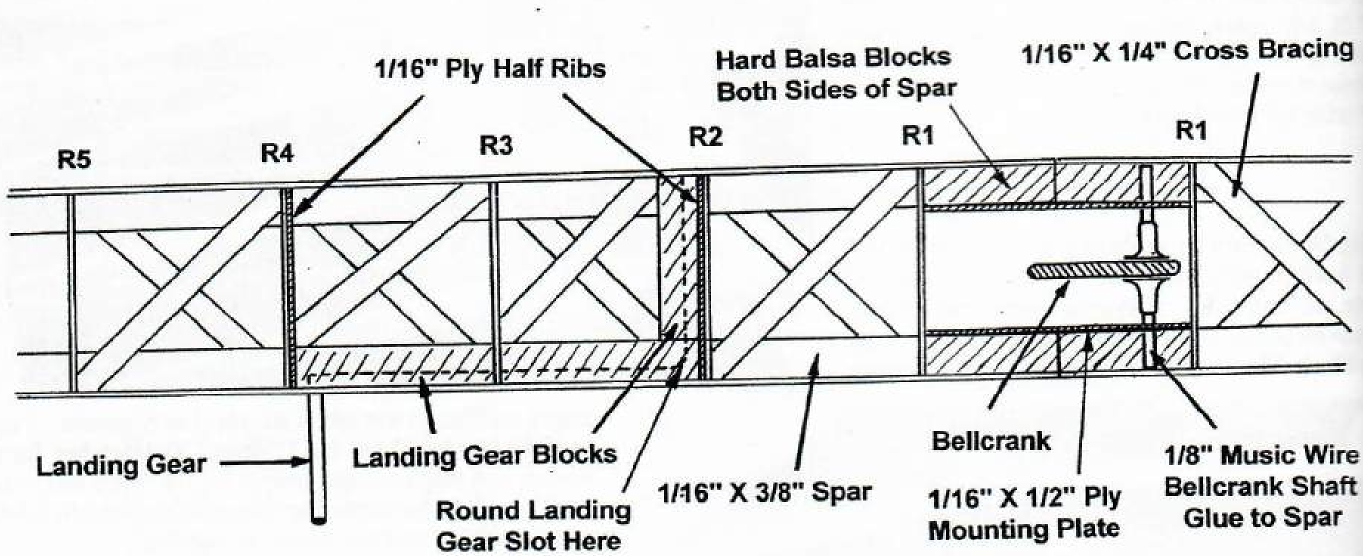
anyway. Everything locks together and is perfectly centered automatically! Join the leading edges and trailing edges together with little strips of balsa.

Drop in the top spar and lightly tack glue it in a few places. We'll thoroughly glue both spars in later as we add the cross bracing. Put a little pressure on the leading edge jig blocks with the trailing edge jig blocks against a straight edge such as a piece of angle iron. In fact, as you glue in all the steps below keep some weights on top of the wing and always keep a little pressure on the front jig blocks against a straight edge along the trailing edge. Now glue the ribs to the leading and trailing edges. Isn't all that gluing area wonderful! And, look ma, no pins! Wow this is really great! After a few minutes to dry, turn the wing over. Drop in and tack glue the bottom spar. Boy, that didn't take very long!

Landing Gear Blocks

You already have your 1/16" ply half ribs in place, so just carefully measure and fit the bottom landing gear blocks. I make my blocks 3/8" X 5/8" from

Landing Gear and Bellcrank Mounting



Landing gear blocks and bellcrank mounting plates are glued to both spars and to the leading edge sheeting. This gives great leverage and strength. Also, since the bellcrank is suspended in open space, the pushrod is free from interference. The cross bracing makes the wing torsionally rigid. The wing is very stiff before covering.

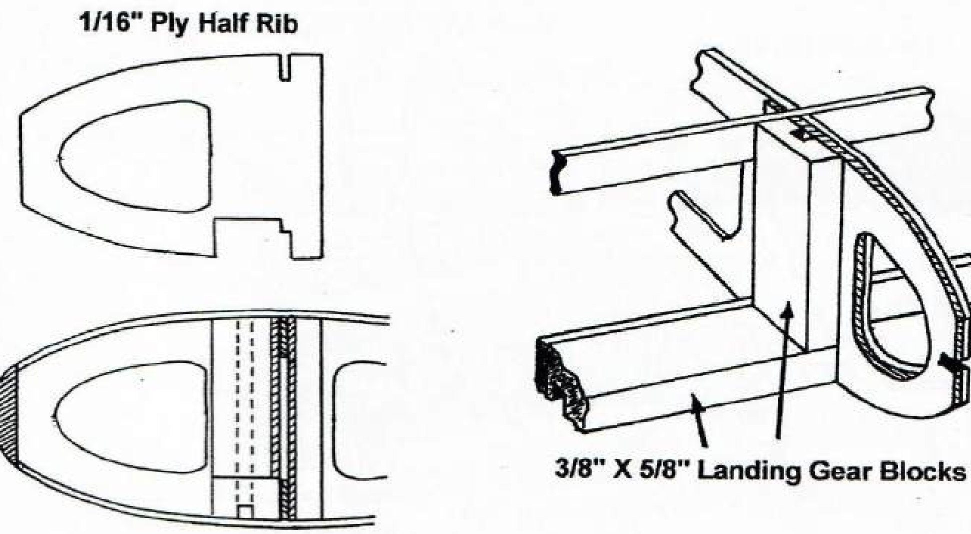
spruce. Make sure the block is flat against the bottom spar. If necessary shape the block to the contour of the ribs. It's important that the leading edge sheeting glue firmly to the landing gear blocks, both the horizontal and the top of the vertical block. Glue the horizontal block to the spar and half ribs. Turn the wing back over. Next fit the vertical landing gear block. It should be full depth and contour with the #2 rib. Glue the vertical block to the 1/16" ply half rib and to the upper spar. After its dried a little, from the top of the vertical block, drill a 1/8" hole through the bottom block. And while you've got the drill out, round off the corner in the bottom block where the landing gears turns up.

Bellcrank

Dry fit two 1/16" x 1/2" plywood plates between the #1 ribs and under the spars. The bellcrank should be positioned to where the pushrod averages being centered through its arc. The 1/8" bellcrank shaft should be tight against the spars and should be full depth of the wing. You could extend the shaft outside the wing for reinforcing against the inside of the fuselage, but that's overkill. When we get through, this bellcrank ain't goin' nowhere! Mark and drill the two plates for the shaft. Add leadouts to the bellcrank. I suggest 1/32" solids. They're so much simpler than stranded. I'll go into detail in a future issue on leadouts. Add the 3/32" flap horn push rod. I suggest a "Z" bend.

Now place the ply plates over the shaft and fit and glue the entire bellcrank assembly in place. Fill in around the bellcrank shaft with hard balsa blocks, both behind and in front of the spars, top and

Mounting Landing Gear Blocks

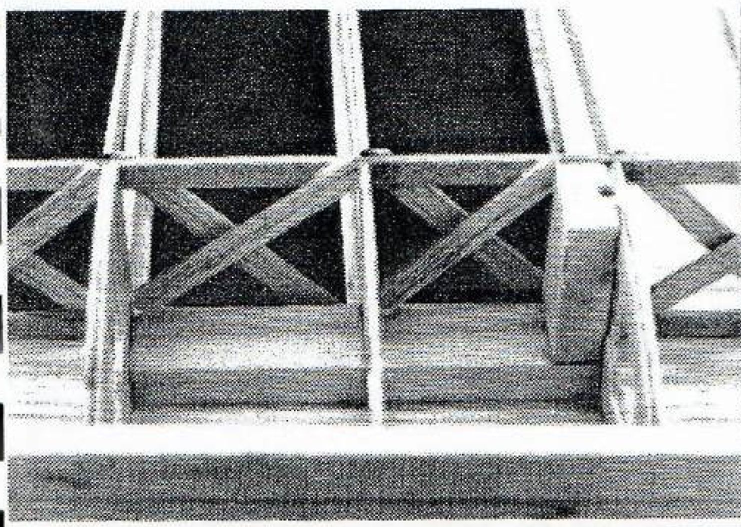


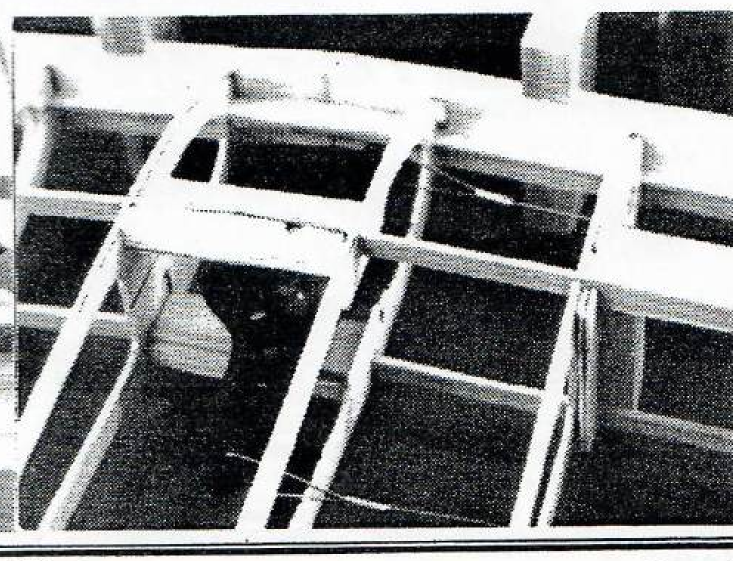
Use rib #2 and rib #4 to draw 1/16" ply half ribs. Glue half ribs to their respective full rib before assembling the wing. The horizontal landing gear block is glued squarely to the lower spar and the vertical block is glued to the upper spar and to the ply half rib. Also shape the landing gear blocks so that they will be glued firmly to the leading edge sheeting. The landing gear wire should extend almost the entire depth of the wing. This gives great leverage and strength.

bottom. Make sure these blocks are contoured to the ribs so that they will glue firmly to the leading edge and center wing sheeting. This mounting method will firmly lock the bellcrank shaft to the spars and to the sheeting. Also, the bellcrank is suspended mid-wing, so there will be no interference with the pushrod or leadouts. You'll have to remove the center web in the inboard #1 rib to make clearance for the bellcrank to fully rotate.

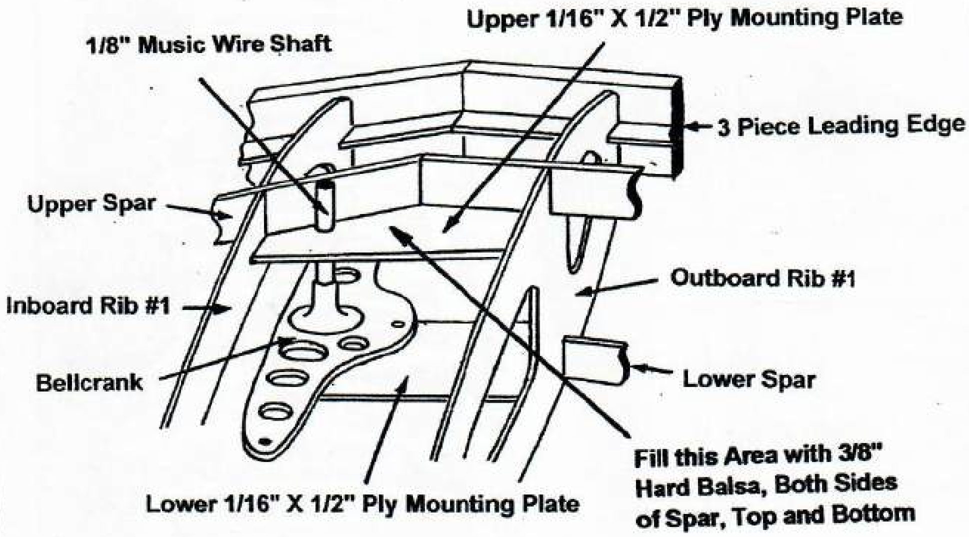
Cross Bracing

Make up 24 1/16" X 1/4" hard balsa cross braces. Trial fit one cross brace and then use it as a pattern for the rest. A perfect fit is not necessary since you've got lots of gluing area on the spars. Cross brace between all ribs out to #7 and #8. No need to cross brace the outer four bays. Put all the cross braces behind the spars going one way and then do all the cross braces in front of the spars. The aliphatic resin glue is tacky enough to hold the cross braces in place while the glue

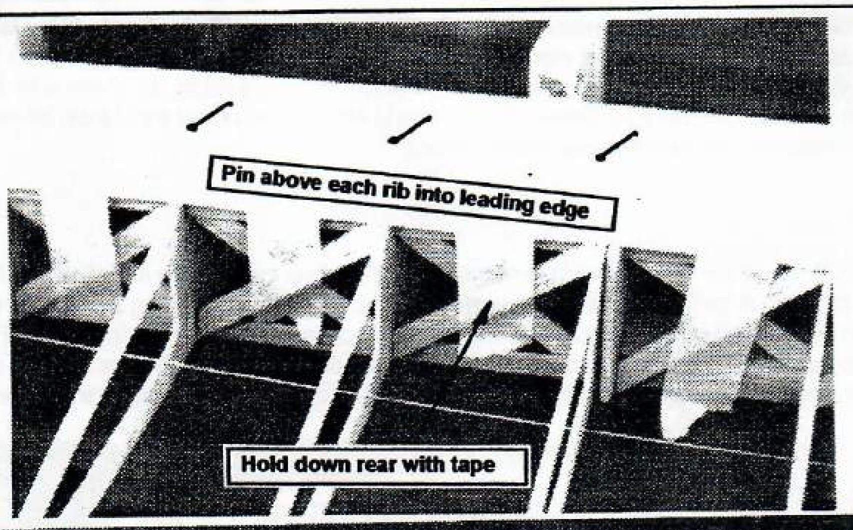


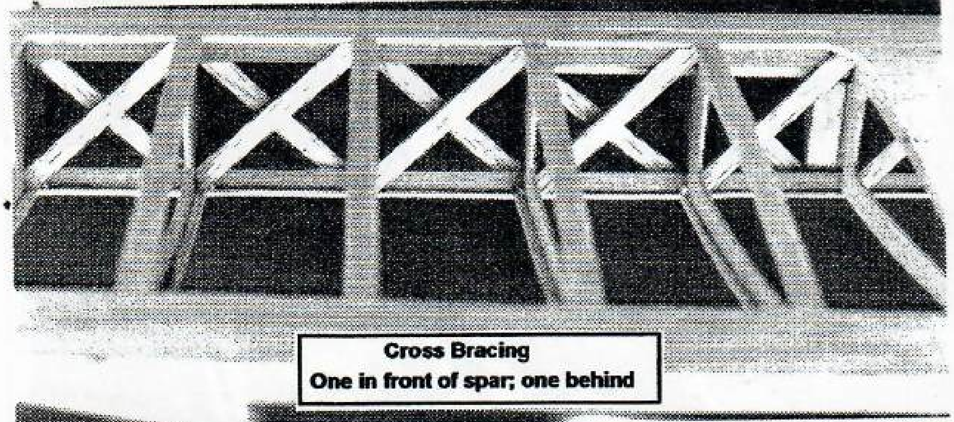


Mounting Bellcrank



Dry fit 1/16" ply mounting plates between #1 ribs. Bellcrank shaft should be tight against both spars. Mark location of 1/8" bellcrank shaft and drill 1/8" holes in mounting plates. Cut bellcrank shaft to full depth of wing. Connect leadouts and pushrod to bellcrank. Slip ply mounting plates over bellcrank shaft. Glue entire bellcrank assembly in place. With 3/8" hard balsa blocks, fill in around bellcrank shaft on both sides of spars. Make sure 3/8" block is contoured to ribs so that it will be firmly glued to sheeting, spar and ply mounting plates. Mounting a bellcrank this way ensures it's there to stay.





dries. Simple. It's this separation of the cross braces that gives the wing so much torsional rigidity. Finish gluing on both sides of both the upper and lower spars. After the cross braces have dried, sand off the little points sticking out above the spars. Also, cut out the center web of all the inboard ribs except #2 and #4 where the ply half ribs are. This keeps the leadouts from rubbing on the web and squeaking!

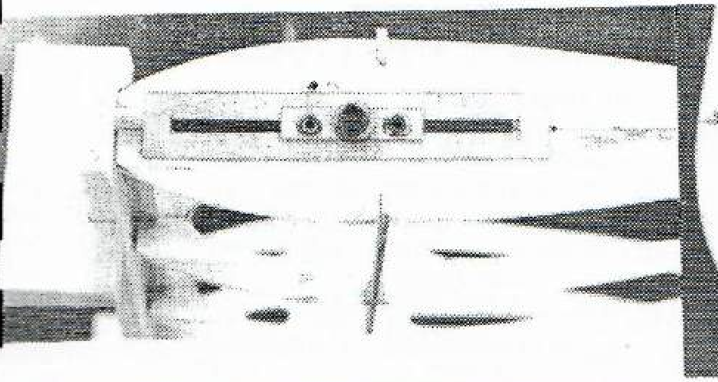
Sheeting

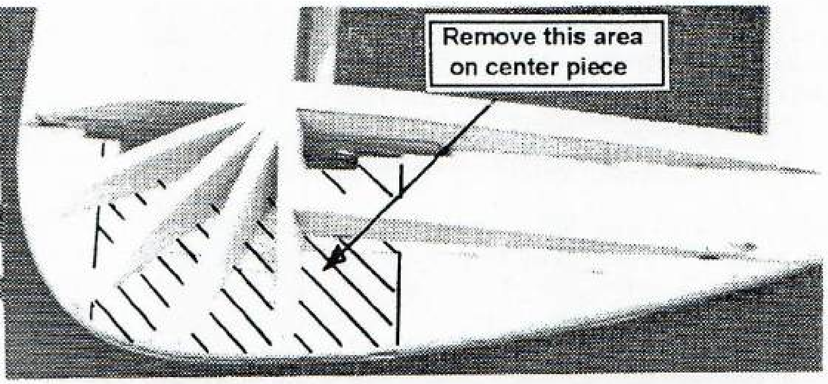
Cut and trial fit leading edge and trailing edge sheeting. No gaps allowed! Make the leading edge sheeting wide enough to cover the ends of the rear cross braces. Here's another place where the aliphatic resin glue is really nice to work with. Put glue on all the upper surfaces that will be glued to the sheeting. Use very little glue on the edge of the sheeting that will mate with the leading and trailing edges. In fact, little dots of glue about every 1/8" works well. You don't want a heavy glue seam there when you are shaping the leading and trailing edges. Here's the only time you'll need to pin anything. Push a small pin just above each rib tightly against the sheeting to hold that edge down against the rib. Don't put a pin between the ribs or you'll have scalloped sheeting. Hold down the inside edge of the sheeting with masking tape.

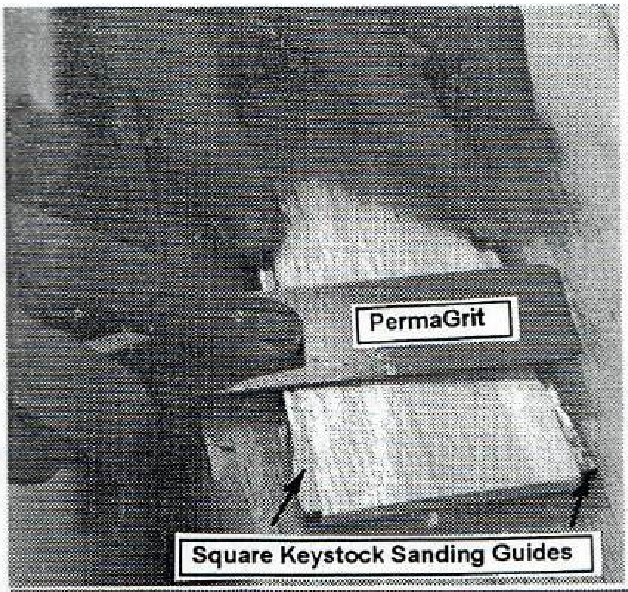
Turn the wing over and do the sheeting on the other side. You'll find that there will be no gaps between the ribs and the sheeting with this system. Add center sheeting. Add 1/16" x 5/16" cap stips. These should be snug fits between sheeting, not tight or you may cause warps. Looking good! Okay, remove the jig blocks from you your work of art. Give it a twist to see how ridgid it is. Nice! Real nice. Sight down the trailing edge and see how dead straight it is. Perfect! What a wonderful way to build!

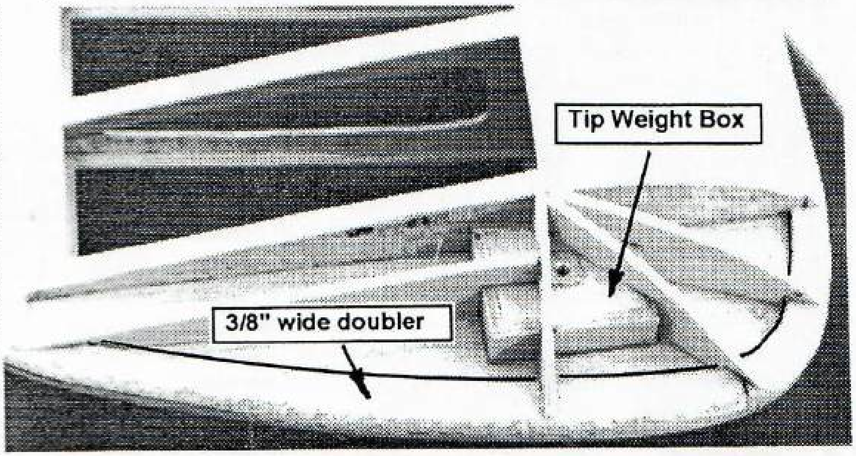
Shaping Leading and Trailing Edges

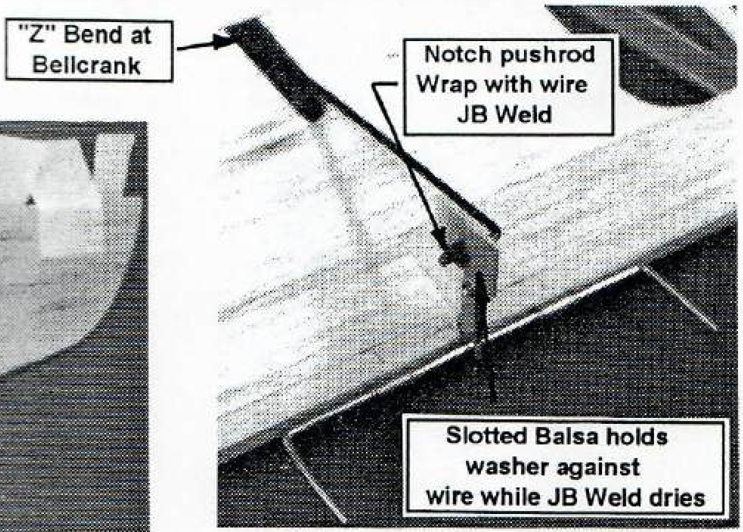
The center piece of the three piece leading and trailing edges gives a nice center line reference. A razor plane is very handy for shaping. While shaping the trailing edge, I put masking tape on the trailing edge sheeting to protect it from getting gouged. Works great! It's easy to get a perfect trailing edge by leaving 3/32" on each of the outer pieces at the rear edge. that gives you a dead straight 1/4" trailing edge. Use a long block sander or a bar sander to sand the cap strips even with the sheeting. Re-

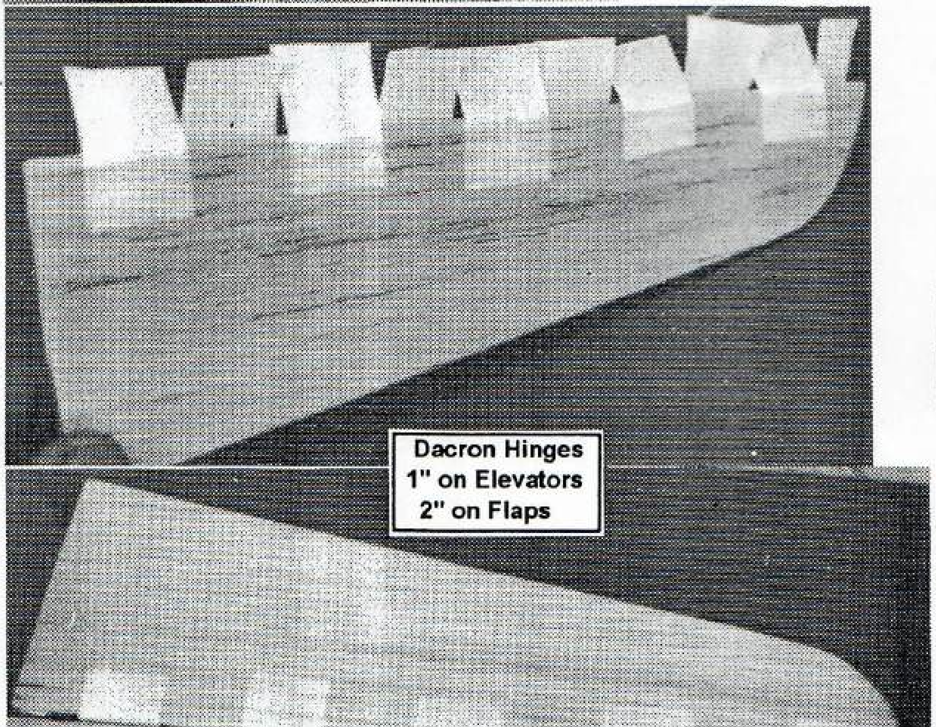












ally take your time doing this as this is one of the most difficult things to do correctly.

Wing Tips

Cut a slot in the inboard tip rib for the adjustable leadout. Glue the adjustable leadout guide in place. Glue an adjustable tip weight box on the outboard tip rib. Make six 1/8" balsa wing tips. Cut away the area for the leadouts on one and glue the three layers together. This leaves a 1/8" slot for the leadouts and for adjusting the guide. Glue it on. Cut an opening in one of the 1/8" wing tips to fit around the weight box. Glue it on. From the remaining two wing tips, cut out a

3/8" wide outer doubler and glue these to the outboard tip. This gives you 3/8" thick tips to shape and enough gluing area to adhere the covering. Add filler pieces per the photos or hollowed out blocks if you prefer. Add the inboard flap filler piece.

Flaps, Horns and Hinges

Shape your flaps using a flat Perma-Grit file/sander and square key stock as thickness guides. This really works well. I like papa Jim's (CSC) horns. The brass clips are especially nice holding the horn firmly in place. Fit the flaps to the horn and make sure the flaps are dead even. I secure the pushrod to the horn by cutting a shallow groove very near the end of the pushrod, slip on a washer, then wrap wire in the grove and around the end. Then I slip a piece of slotted 1/16" balsa between the washer and the horn to hold the washer tight against the wire (see photo). I coat the wire and outside of the washer with a few drops of JB Weld and the pushrod is there to stay. No corrosion; no mess; permanent!

I'm using dacron hinges. I ordered two yards (a life time supply) of the dacron used to cover ultra light aircraft from Aircraft Spruce. I brushed on three coats of nitrate dope on the wing and flaps, lightly sanding between coats. Then I doped the alternating hinges onto the flaps and elevators. After that dried a while, I slid the flaps onto the horns, taped the flap in neutral to the inboard flap filler piece, and doped the hinges to the wing. You need to hold the wing with leading edge down as you do this to insure you're getting the flap exactly centered on the trailing edge of the wing. This is such an easy way to do hinges and the hinge line is completely sealed. The covering then goes on top of the hinges, further securing them as well as hiding them.

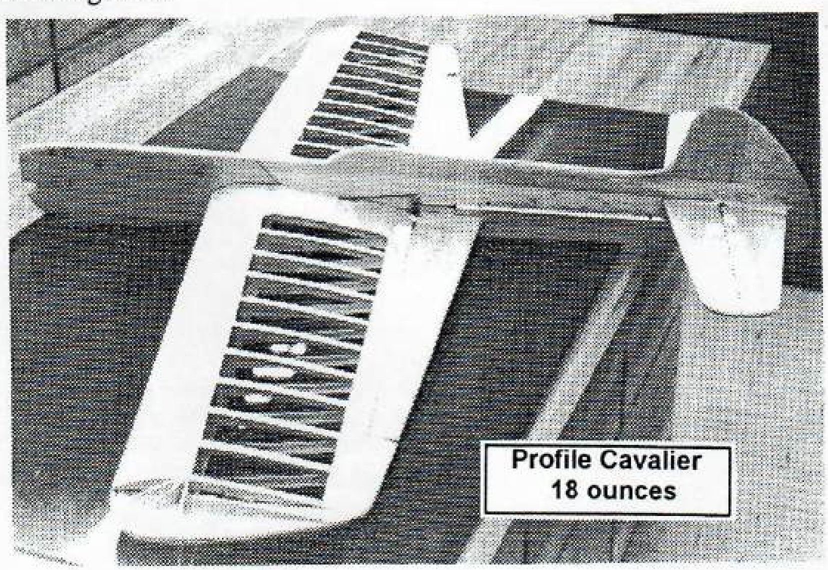
Covering

I used Polyspan to cover these two wings because it's tough, light and heat shrinkable. Overlap the covering about two inches on each side so that there is a double layer across the entire center for extra strength. Starting with this issue we have a new covering advertised called ThermalSpan. I'm trying it on my next ship. In the instructions, he suggests that we use a thinned downed, heat activated glue such as Balsarite to adhere the ThermalSpan. Sounds like a good way to eliminate the problem of getting the covering to stay down around tight corners. I'll let you know how it works.

Conclusion

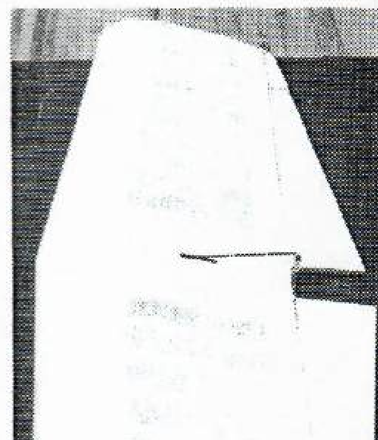
I hope you will try this wing building method. It is truly a pleasure to build a wing this way. I saved so much weight on the wings with these latest two ships that I'm actually going to have to put on extra paint just to bring the weight up to an acceptable minimum. What a nice problem to have! I believe this article is sufficiently illustrated and has enough photos to make everything perfectly clear. However, if you have any problems or questions,give me a call (205-820-1983). And, if you have a better way of doing something, publish it in Stunt News.

This one thing I promise, if you'll build one wing using this method I sincerely believe you'll never build a wing any other way!



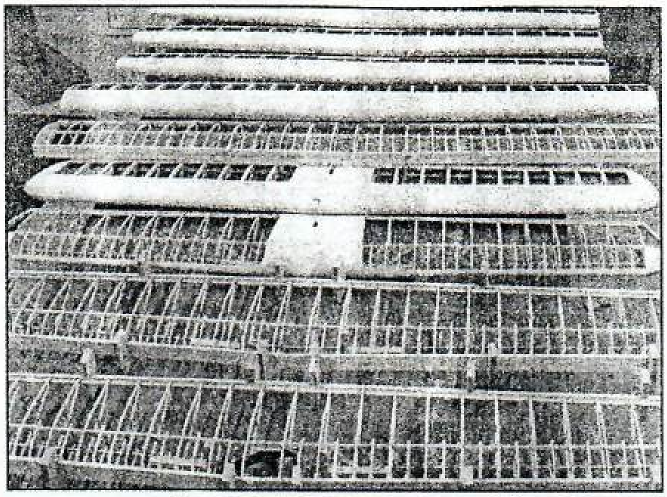
Profile Cavalier
18 ounces

One of the biggest advantages of the "Lincoln Log" wing is that it is self-centering and self-aligning. It makes building a wing such a pleasure. Both wings I built were dead straight and had great torsional rigidity. Another big advantage is saving weight. The complete wing for the profile weighed 8.9 ounces ready to cover. I would recommend that you always cover the wing before mounting it in the fuselage. Generously overlap the coving in the middle for extra strength. The wing to the right is for the full-bodied ship. Covered, ready for mounting it weighed 9.1 ounces! A third big advantage is cost. This is a very inexpensive wing! The jig blocks can be made from scrap pieces of board. There's only about $8 of balsa in the wing. The angle iron straight edges cost about $10. And, you'll use them over and over again. Try this wing building method. You'll love it! Thank you Jim Pollock and Hoyt Hawkins for your help and encouragement.

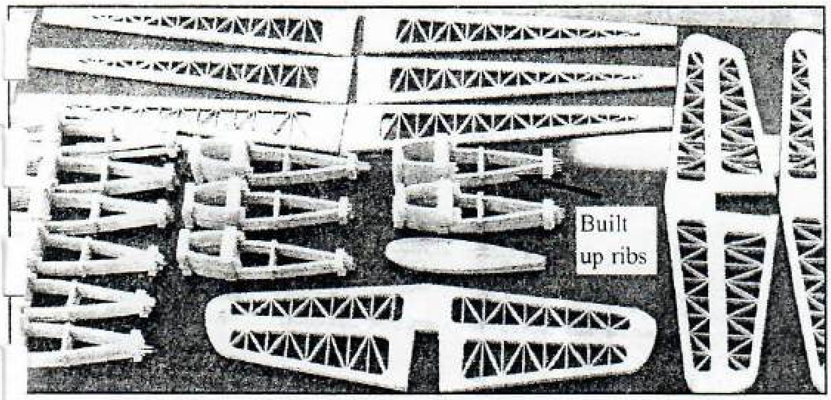


540 in^2; 9.1 ounces!

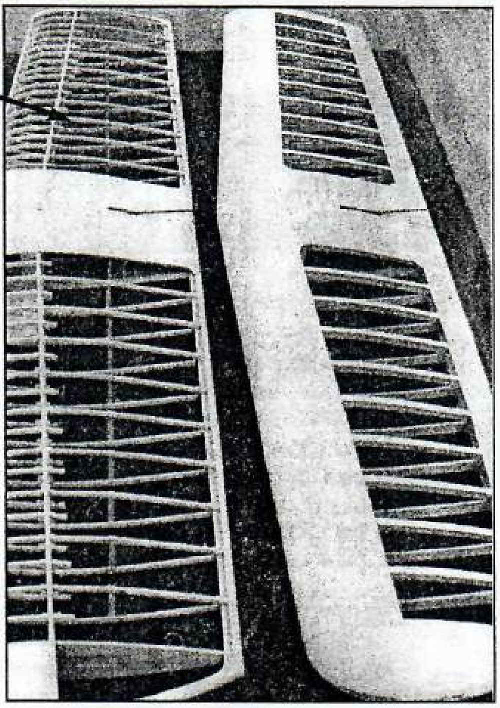
Around Tom's Shop



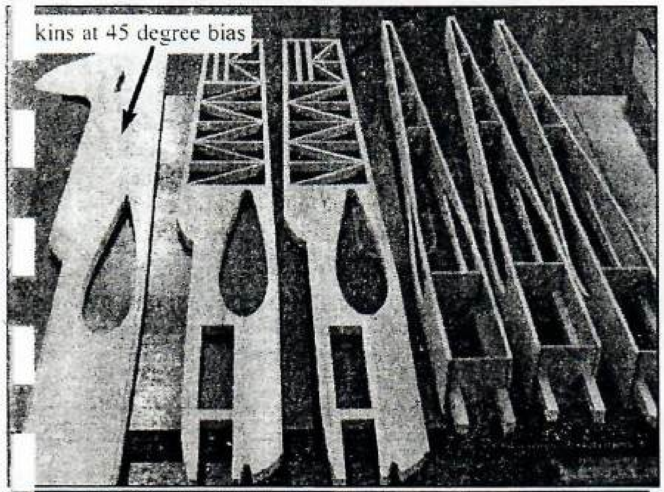
…here were these last two pages to fill to make the pages come out the right number, so I thought I might show a little of what's going on in my "marginally ade-…uate" workshop. In the May/Jun '97 issue of Stunt News on page 54, I had an …ticle on the "Lincoln Log" method of building wings. This method has been …oroughly validated since then with over 20 wings built! They are so easy and



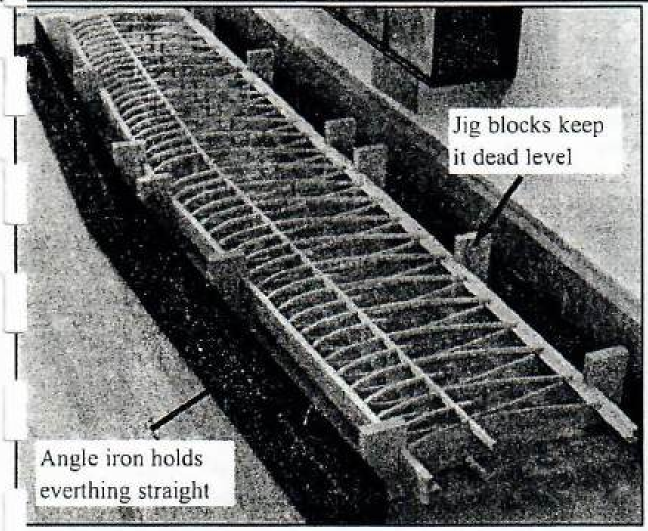
Pseudo I-beam wing



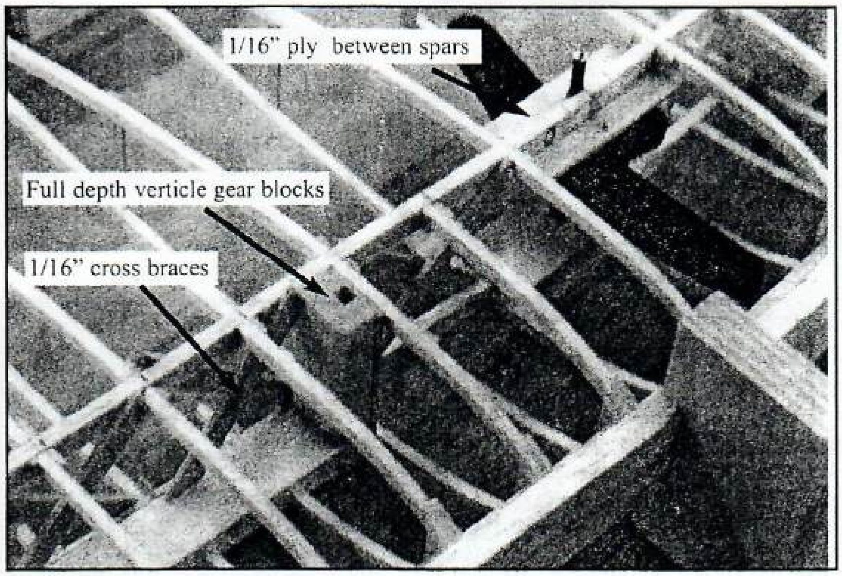
fun to build this way and everyone had been dead straight, very rigid and light. Currently I am building nine stunters, three Cavalier profiles, one regular built-up Cavalier, three advanced train-…s with wing based on Tutor ribs and two Super Cavaliers which will be about 630 squares and …owered by PA40's. As you can see from the photos, I'm using Larry Cunningham's construction method for the profile fuselages. Thanks Larry, it is a great way to make a light, stiff fuselage.



If you look closely you'll see that the ribs are built up rather than cut whole. With this method I can get an entire rib set out of one sheet of balsa and it's quick and light. I'm also experimenting with a pseudo I-beam which uses the built up ribs with normal Lincoln Log spars and cross bracing and no sheeting on leading and trailing edges. The first profile with this wing, ready to cover weighed 12.9 ounces!



The Lincoln Log method is simple an inexpensive. The jig blocks are very easy to make and can be used over and over. It makes it …ery easy to turn the wing over to work on the other side.



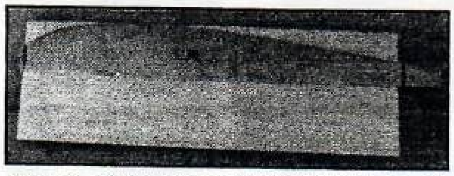
The Lincoln Log method of mounting the bellcrank and landing gear blocks has been well tested and has proven to be bullet proof. The real strength comes from tying everything to the skins and using mounting leverage. The cross bracing is easy to do and provides maximum torsional strength at minimum weight. Try it!

Built Up Ribs Pictorial

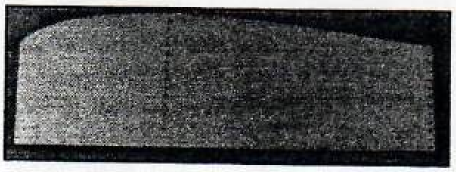
By Tom Morris

In the last issue of Stunt News I showed a few photos of what I was building. It sparked a lot of questions. I got a several letters and phone calls wanting more information. This issue I'll detail how I build ribs. Building ribs rather than cutting them out of a sheet saves a lot of balsa and results in a very accurate, strong and light rib.

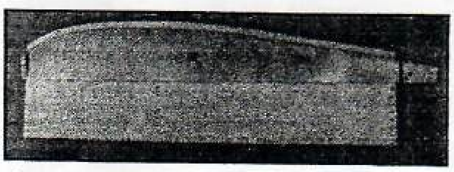
Step 1. Make a rib template. I use thin aluminum



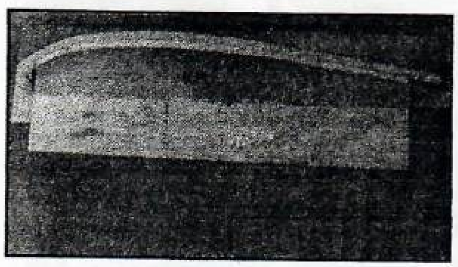
Step 2. Cut a piece of 3/32" light balsa to length and slice off the top rib contour.



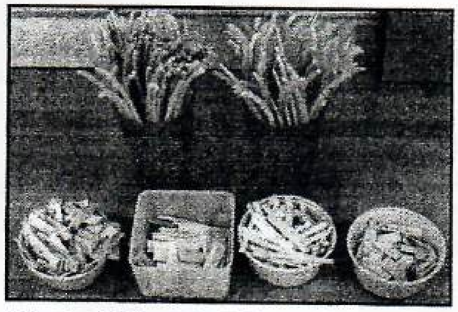
Step 3. Mark 3/16" intervals at the trailing edge and at high point.



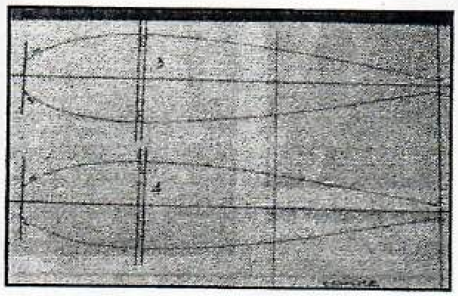
Step 4. Cut off rib slices.



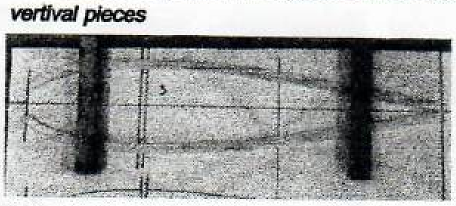
An alternative to measuing and marking each slice is to make a simple jig. I used 1/4" ply as a base, 1/16" ply as an upper stop, drilled a small hole at each end of the aluminum template, cut the head off two small nails, positioned the template 3/16" below the stop, and drove in the nails to position the template. With this jig you can cut identical ribs without measuring. It really speeds things up. I cut enough full ribs and half ribs to build about six wings in about one hour.



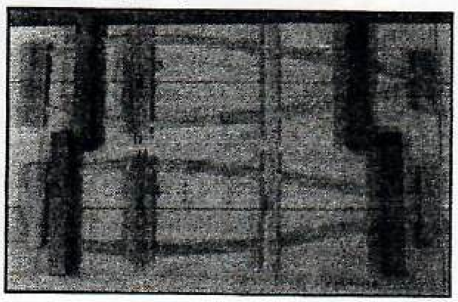
Step 5. Cut the vertcal pieces. Use a stripper to cut 1/16" by 1/2" balsa strips and then gang saw the pieces to length. The verticlal piece in the middle rear is 1/16" by 1/4" wide.



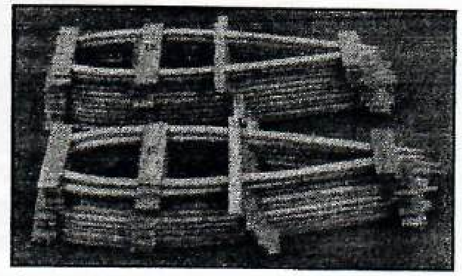
Step 6. Mark the rib plans by extending the vertical lines to guide in the placement of the vertival pieces



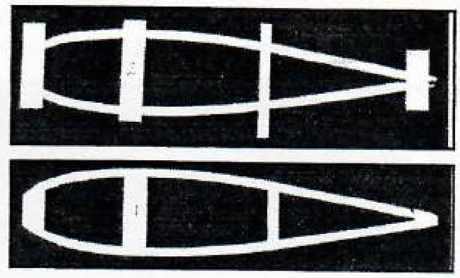
Step 6. Place the rib slices on the plan and weight them down. I use pieces of square key stock and square punches.



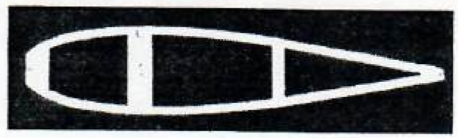
Step 7. Put Elmer's Glue-All on the slices and place down the verticles. You can glue up two or three ribs at a time.



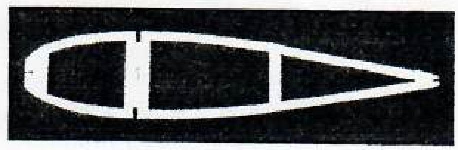
In about 30 minutes you'll have all the ribs assembled.



Step 7. Cut off the excess



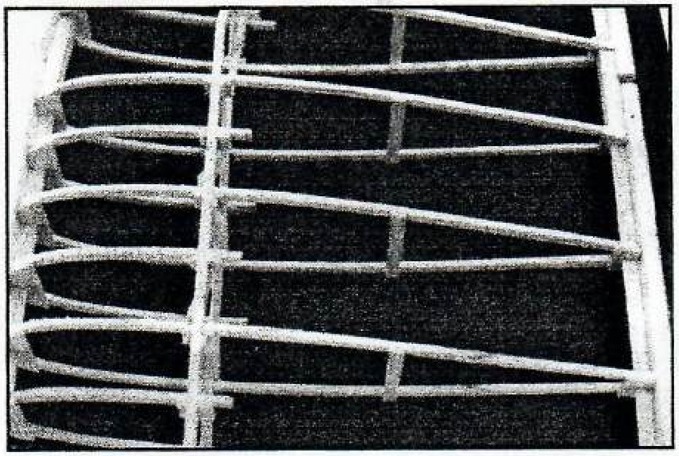
Step 8. Mark the centerline and the spar slots



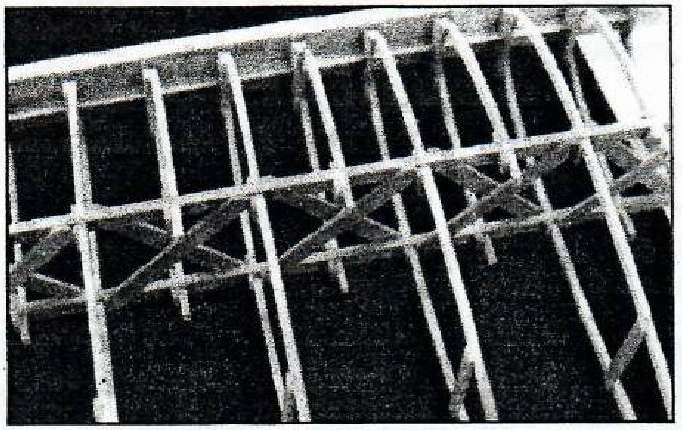
Step 9. Cut the 1/16" by 1/8"slots in the leading and trailing edge and cut the 1/8" by 1/4" spar slots.

Step 10. Build the half ribs

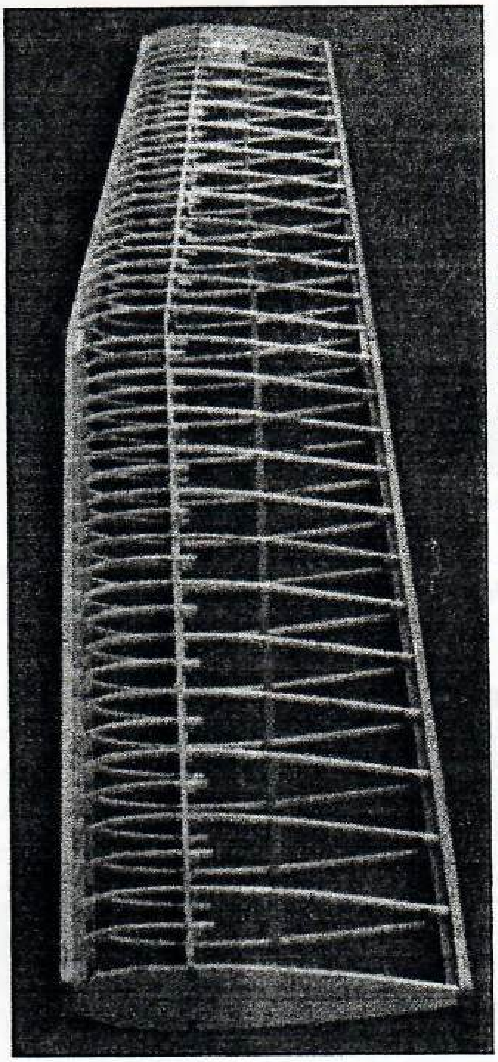
After using this method a couple of times I'm able to make a set of built-up ribs in about the same time it takes to cut them out of a sheet. You can get a complete set of ribs out of one sheet of 3/32" balsa versus nine sheets for a 670 square inch wing.



Once the ribs are built, build and notch the three piece leading and trailing edges, secure them in the jib blocks, snap the ribs in place, and glue it up. No pins required!

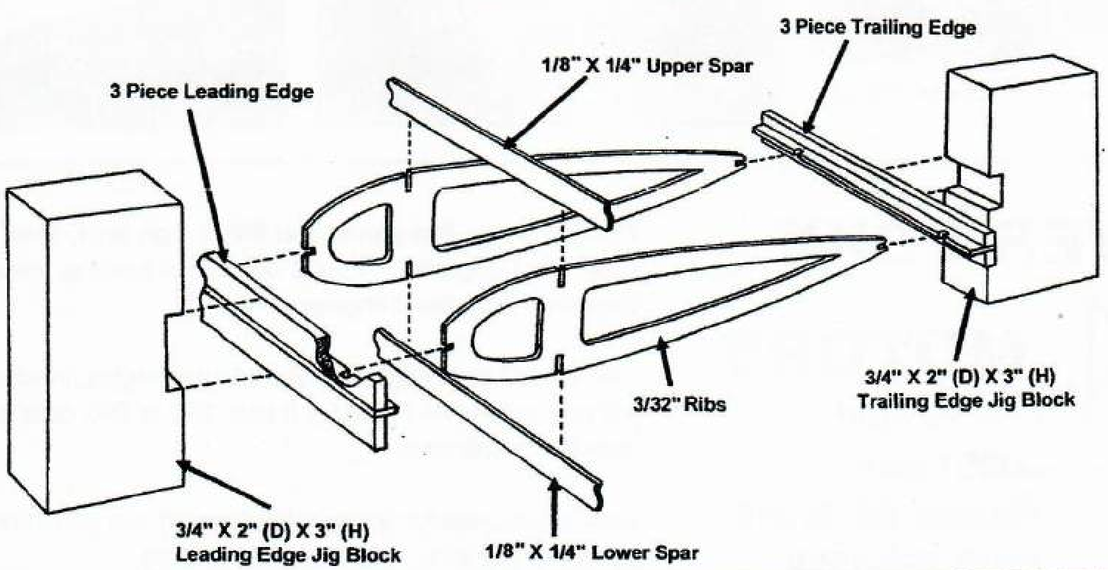


After the spars are glued in, cut away the verticle piece between the spars and glue in the cross bracing.



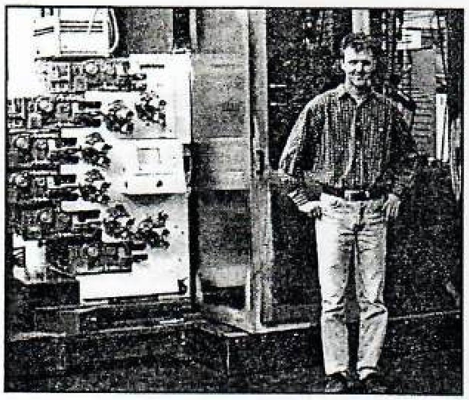
This 670 square inch wing weighs 2.6 ounces at this point. Once the leading and trailing edges are shaped it should be below two ounces. Only the center section will be sheeted. The wing will be covered with polyspan using nitrate dope. The polyspan will be overlapped in the center. There will be no leading or trailing edge sheeting. This "pseudo I-Beam" wing building method saves about three ounces on this size wing and, once covered with polyspan, is plenty stiff.

The "Lincoln Log" Wing



The "Lincoln Log Wing" article was in the May/Jun '97 Stunt News. The only changes I've made after building over 20 wings with this method is that I use the built-up ribs, use 1/8" by 1/4" unnotched spars and the outer pieces of the leading edge are 1/4" by 1/2" and the outer pieces of the trailing edge are 1/4" by 1/4". I'll supply jig blocks if you can't make them yourself. Call me.

SWITZERLAND

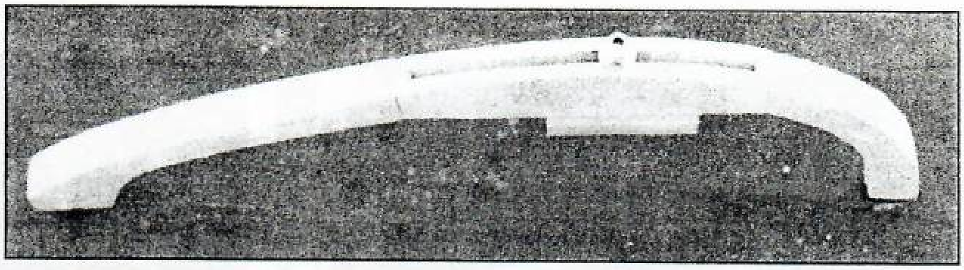


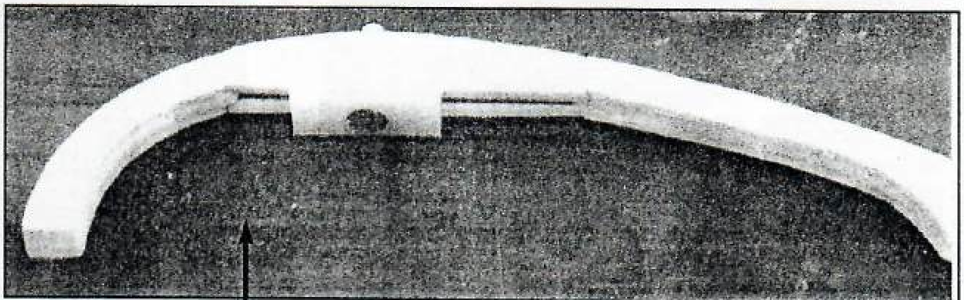
I have only been 1 1/2 years in this hobby and began slowly. But I love flying and dreaming about stunt planes…

Felix Straessler
Switzerland

UKRAINE

Sergy Belko with his ship "Blackwind" Photo by Jim Dunkin



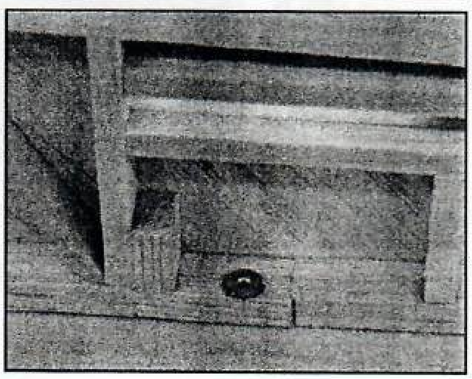


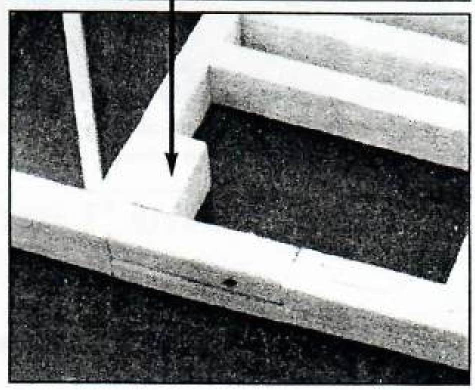
A Simple Wing Tip with Adjustable Leadout

Three pieces of 1/8" balsa are glued together with the center piece being a little longer. Two pieces of 1/8" ply are glued in the center with the back side being straight. Weighs six grams.

A Simple Tail Wheel Block

Two short pieces of motor mount material are glued into the frame after a shallow groove has been cut in the bottom and a blind nut installed on the top. The tail wheel wire runs up the vertical block and is held in place by a single 4-40 bolt and washer. Very light and simple.





How to Draw Perfect Rib Patterns

By Tom Morris

Last issue I showed how I build ribs. I got about 20 calls and letters with more questions. One of the questions was how do I draw the rib patterns. Well, I use the same method as used in building I-beam wings.

Step 1. On the top view of the wing plan draw a line 1/4" back of and parallel to the leading edge. That is your leading edge. Do the same for the trailing edge.

Step 2. Make an aluminum template of the root rib pattern. Use this template to draw the airfoil patterns for each rib and use it in the simple jig I showed last issue to cut the outer rib pieces. Make the template about 1/2" longer than the root rib and mark the leading edge on the template. All ribs are referenced off the leading edge mark. In other words, as the ribs get shorter you will cut off the excess from the rear edge.

Step 3. Decide on the width of the leading edge (5/8" for the Cavalier) and trailing edge (1/4" for the Cavalier).

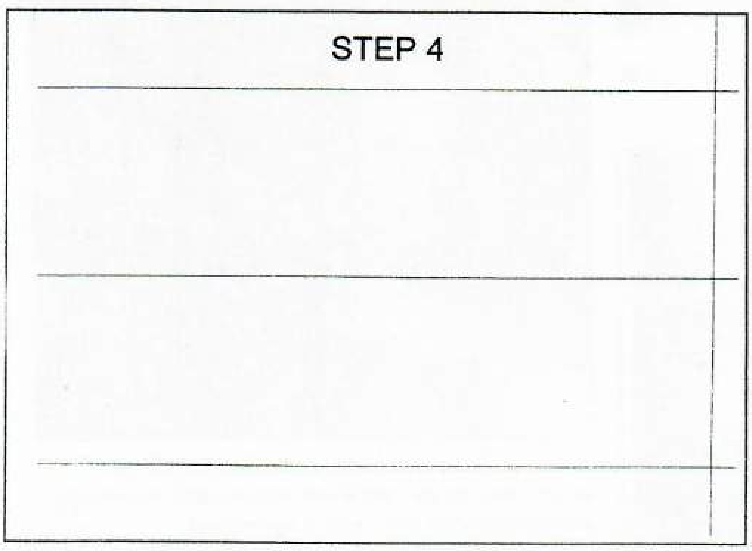
Step 4. On an 8 1/2" X 11" sheet of quad paper draw a line representing the trailing edge of the rib. The draw three center lines.

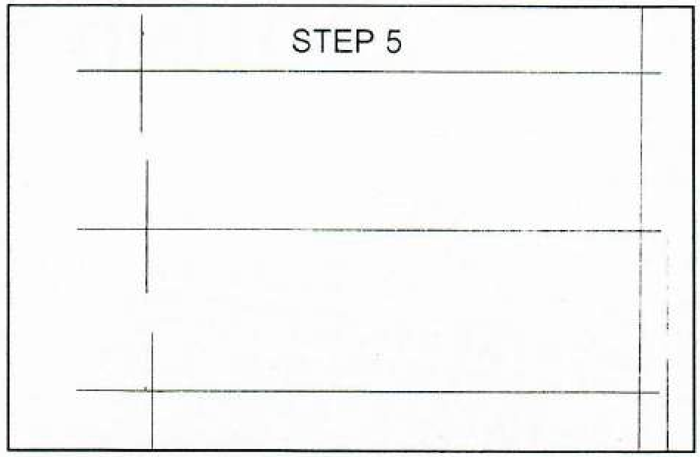
Step 5. Measure the length of each rib from the top view of the plan and mark it off on each centerline. All measurements are from the trailing edge.

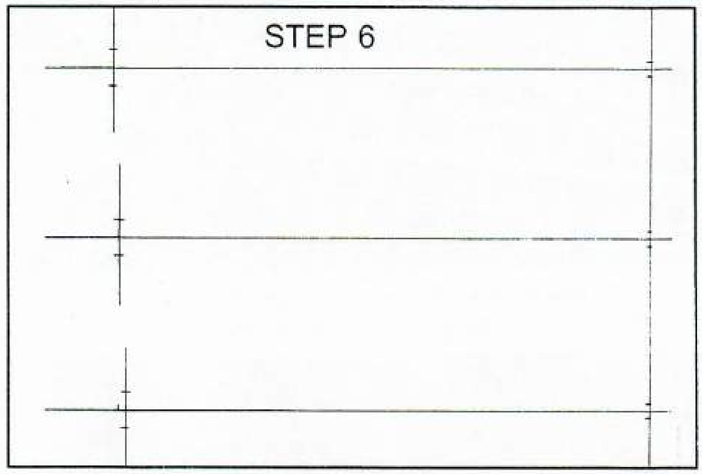
Step 6. Mark the leading and trailing edge widths on the leading and trailing edge lines.

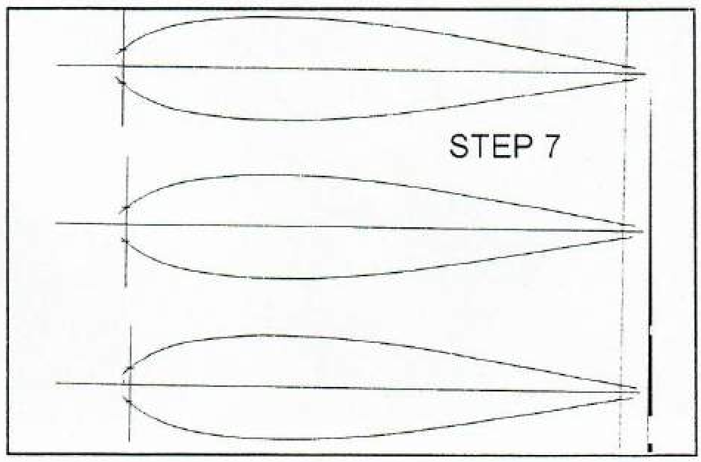
Step 7. Align the airfoil template on these marks and trace the airfoil. Register off the leading edge mark on the airfoil.

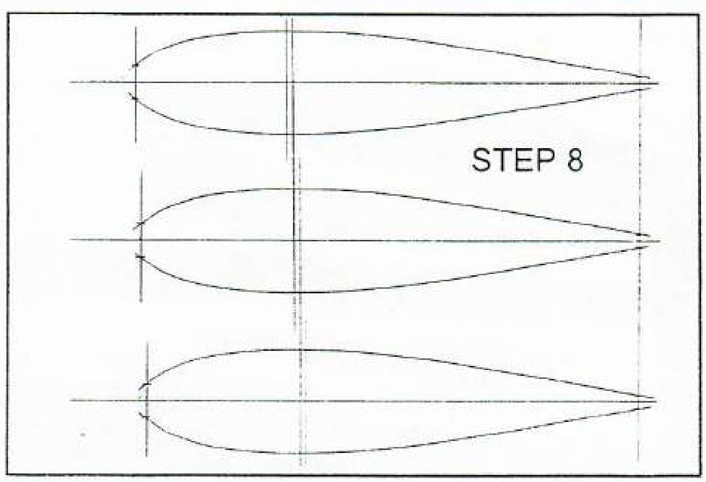
Step 8. Measure and mark the spar locations.





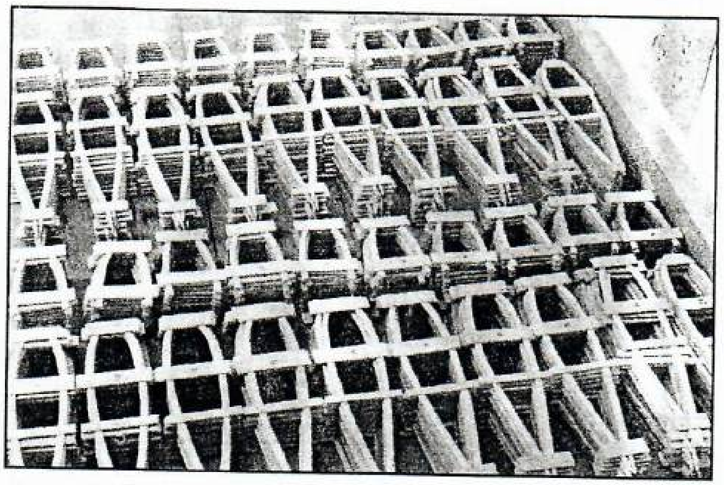




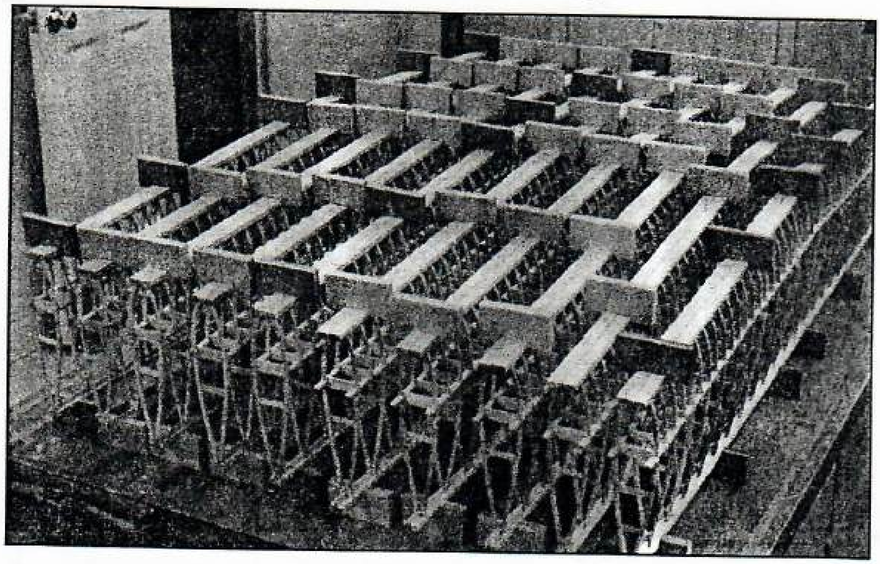


AROUND TOM'S SHOP

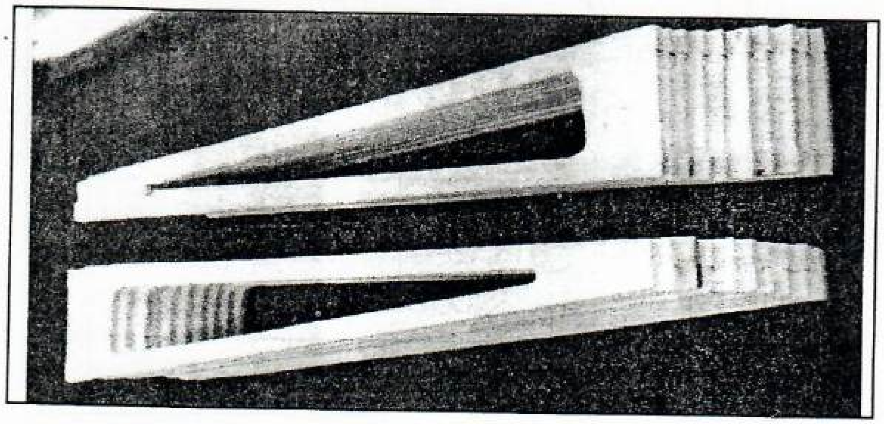
November was a good month for building. Few other distractions. Many people have asked me to build them a Cavalier profile so I decided to build ten at one time. This pictorial is to be sung to the tune of the twelve days of Christmas.

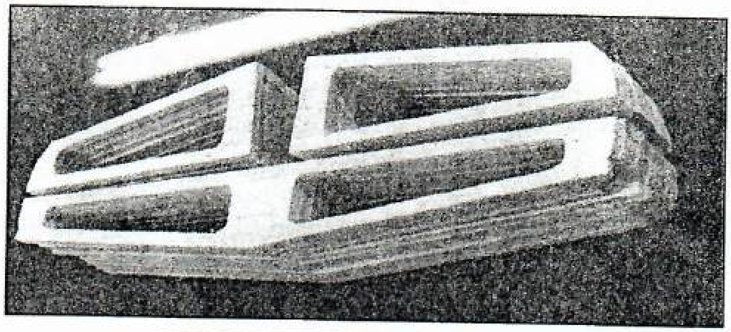


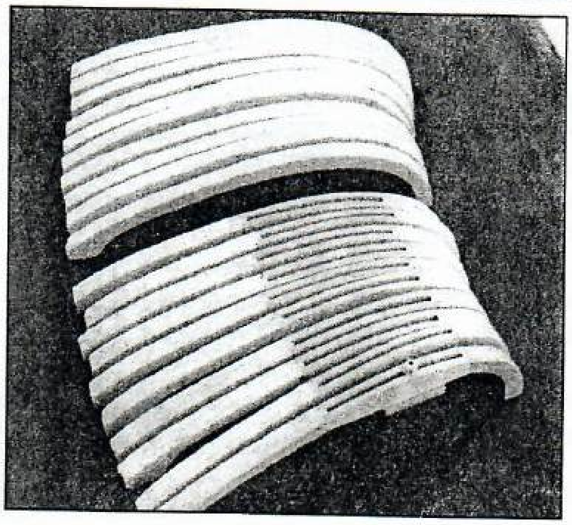
So, on the first week of November my good shop gave to me. 240 full ribs, 220 half ribs, and good start to ten Cavaliers.



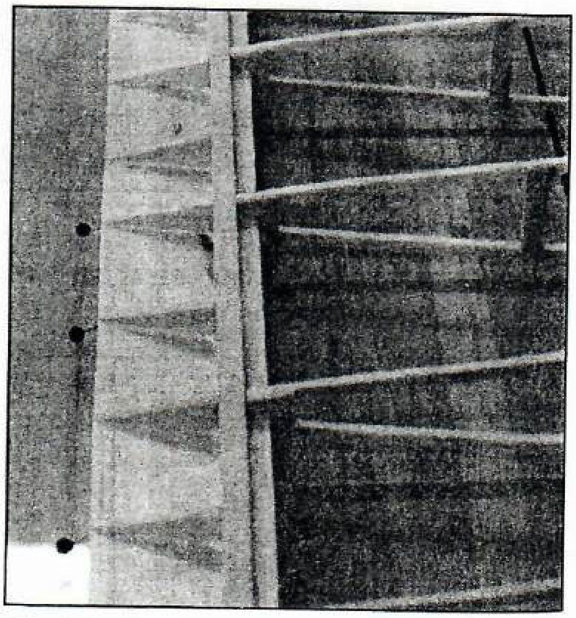
So, on the second week of November my good shop gave to me. 10 wings in framing, 240 full ribs, 220 half ribs, and good start to ten Cavaliers.



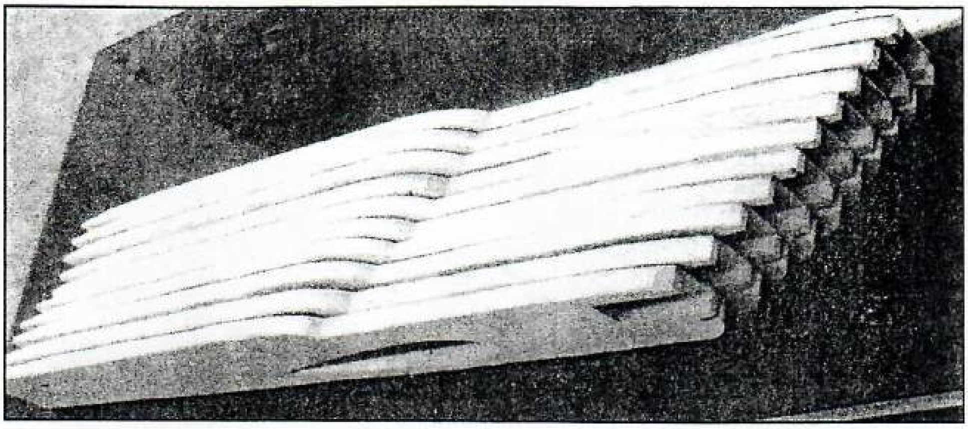




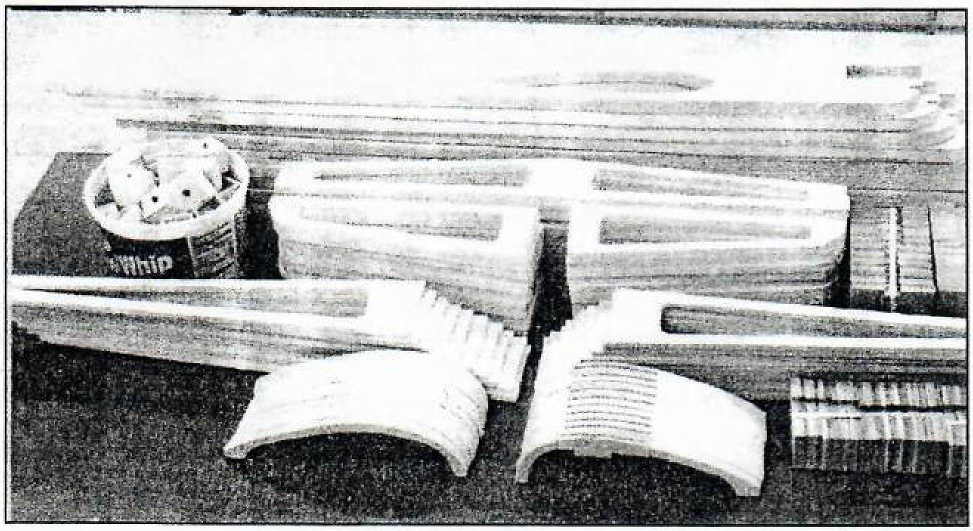
So, on the third week of November my good shop gave to me, 10 inboard flaps, ten outboard flaps, 10 horizontal stabs, 20 elevators, 10 inboard wing tips, 10 outboard wing tips, 10 wings in framing, 240 full ribs, 220 half ribs, and good start to ten Cavaliers.



"Lincoln Log" wing method can be used on flapless wings. Just cut the ribs where rear sheeting begins and add sheeted area after wing is assembled.



So, on the forth week of November my good shop gave to me, 10 fuselages, 10 inboard flaps, ten outboard flaps, 10 horizontal stabs, 20 elevators, 10 inboard wing tips, 10 outboard wing tips, 10 wings in framing, 240 full ribs, 220 half ribs, and good start to ten Cavaliers.



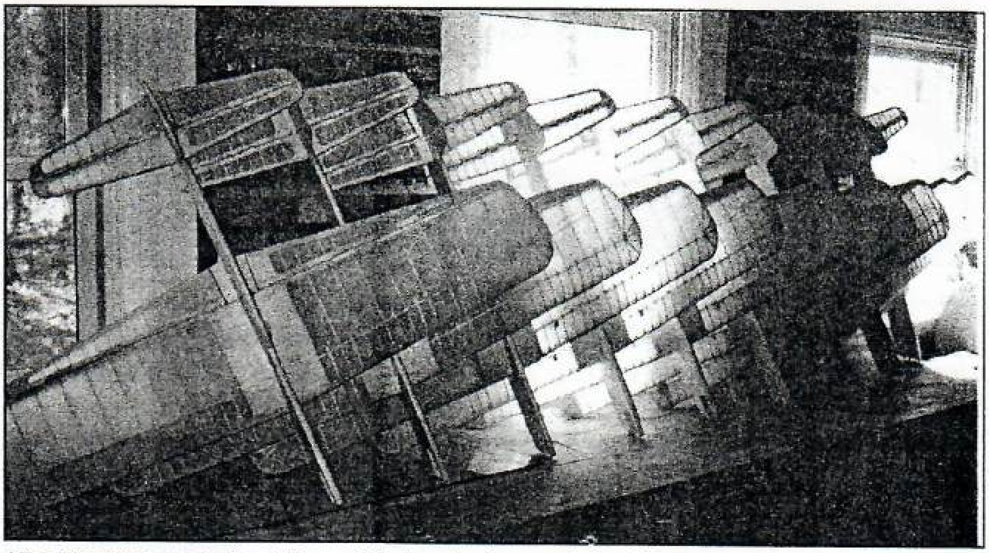
Also in there were 50 weight tip boxes and 25 sets of "Lincoln Log" wing jig blocks. Then came 10 days of Stunt News making! But we got a great start on the Cavaliers.

Index to Advertisers

BUILD 'EM LIGHT

By Tom Morris

How many times have you heard someone say build 'em light and straight? Well. I am on a journey to figure out just how to do that. Here's what I've learned so far.



Nine Cavaliers ready for delivery. The heaviest one weighs 12.6 ounces. That's a complete 540 square inch airplane, covered, with five coats of dope. One was delivered Christmas eve.

Two years ago I learned the pattern with a 43 ounce Cavalier profile. It flew okay. Didn't know any difference. Then last year I flew a 38 ounce Cavalier. Wow! What a difference five ounces makes! So, this year I'm determined to cut at least another five ounces off.

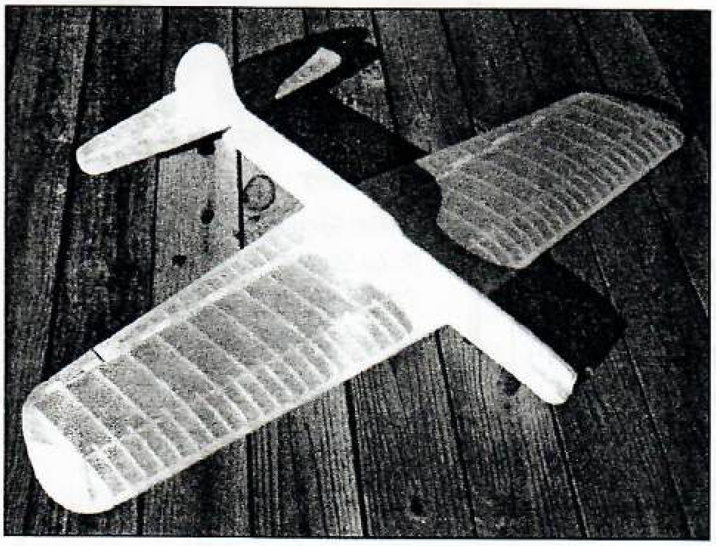
Last year someone asked me to build them a Cavalier. I said yes. After 21 more yeses I said no! I've delivered seven so far and now have nine more ready for delivery. You saw all the parts in the last issue.

So how do you build light?

First start with 4-6 pound per cubic foot density contest grade balsa. I get mine from Lonestar Models, Lubbock, TX, 1-800-687-5555.

Second, use as little balsa as possible. This is where the "Lincoln Log Pseudo I-Beam" wing design really shines. It weighs about five ounces completely covered. The ribs are built up using very little balsa, but are plenty strong. The spars are 1/8" by 1/4" with 1/16" by 1/4" cross bracing between each rib. It weighs only a fraction of an I-beam spar, yet is very strong.

Third, use all built up parts, fuselage, stab, elevators, flaps, wing tips. You also save a lot of money this way. There's less than $8 worth of wood in this model.



Build light by using methods that minimize the amount of wood in plane. the "Pseudo I-Beam Lincoln Log" wing is one way to do this. All components should be built up, so that when you hold it up to the light, you see mostly air. Even the fuselage is built up.

Fourth, use spruce instead of maple for your motor mounts and tail wheel mount. As long as you use aluminum pads under your engine to spread the load, you won't crush the spruce.

Fifth, mount the bell crank with two small pieces of 2" by 7/8" by 1/16" ply between the spars.

Sixth, make the adjustable leadouts as part of the wing tip instead of another piece. Photos to follow.

Seventh, get strength in the middle of the wing by covering your wing with Polyspan which is 50 time stronger than silkspan, before you put it in the fuselage and overlapping (doubling) the polyspan over the center sheeting. Polyspan absorbs less dope than silkspan and finishes lighter.

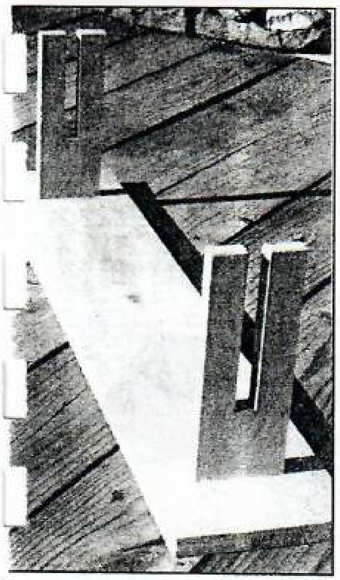
Eighth, use ligth, but strong bellcrank and horns. I use a 1/8" phenolic bell crank and 3/32" horns on this size model.

Ninth, use cloth hinges. I use 1.8 ounce per square yard dacron from Aircraft Spruce, 1-800-824-1930, part number 09-00100, 60" wide, $3.15 per lineal yard. One yard is a life-time supply. You just dope 'em on, couldn't be any easier or lighter. Seals the hingeline too!

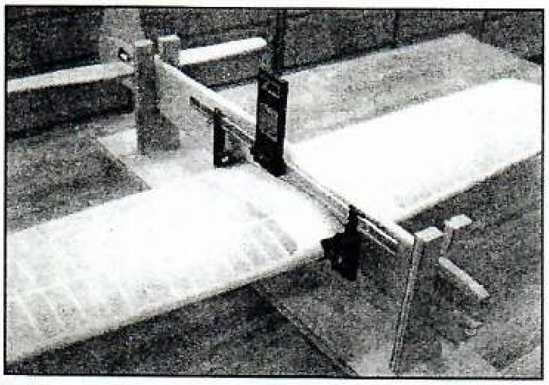
Tenth, don't ruin it with a lot of paint. Put a little stain in the clear coats to have a beautiful semi-transparent base coat and go easy on the trim colors and you should not add more than four ounces of paint.

Eleventh, since you'll have less weight behind the CG, shorten the nose a little to keep the CG right, so you don't have to add a ton of tail weight. I cut off 1 1/8th inch which should be about right for the Cavalier.

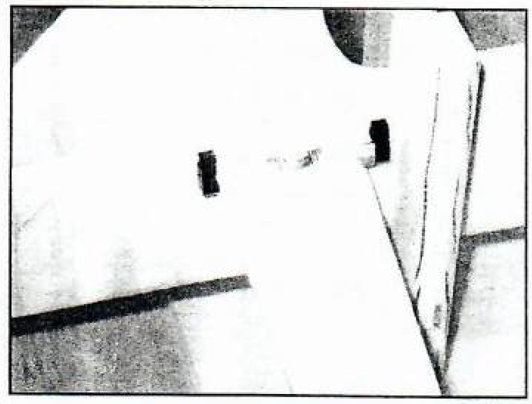
BUILD 'EM STRAIGHT



· Make a simple jig that will ɟ the fuselage verticle and vel.

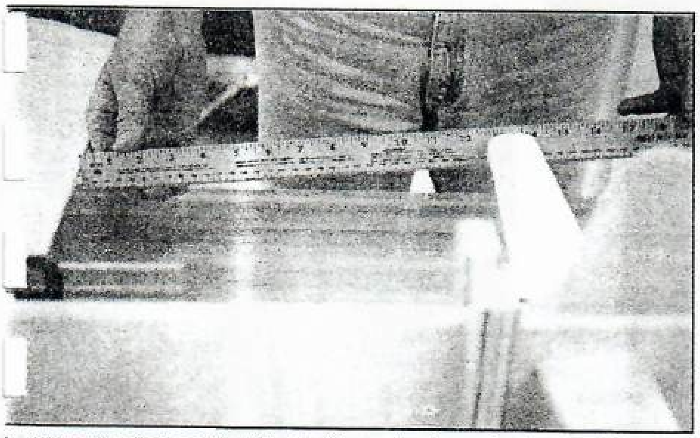


#2 - Secure the wing squarely in the fuselage. I use a framing square to square the trailing edge to the fuselage and level the wing with the bench. After the glue dries, using an incidence meter, shim the fuselage until the wing is dead level.

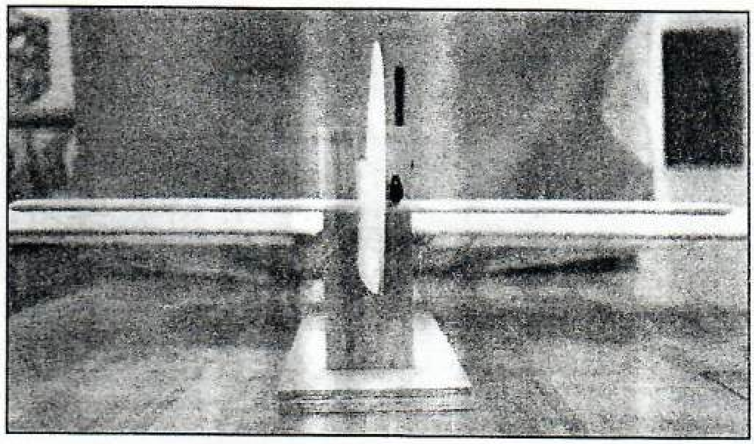


#3 - Place a small line level on the stab and file the opening until the stab is dead level, or if you prefer, slightly positive!

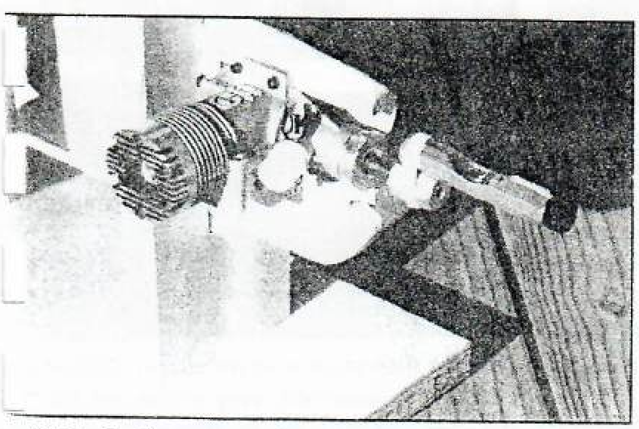
Side Note: This fuselage holding jig also came in handy while fitting flaps, elevators and fillets.



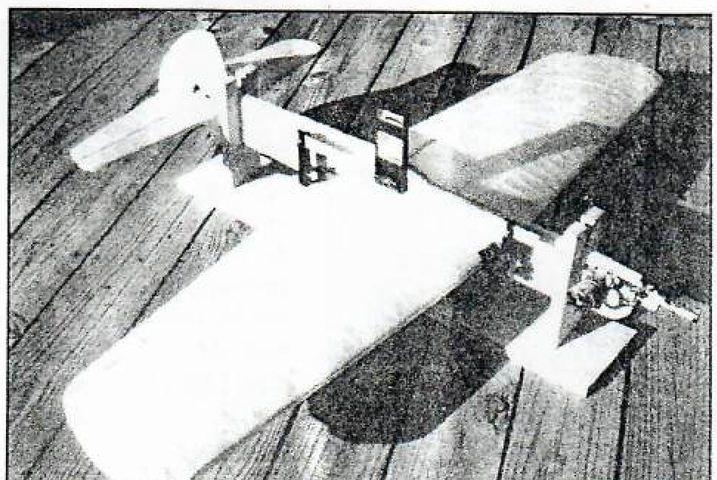
Align the stab so that the trailing edge is perfectly parallel to the ɿ trailing edge. I use a yard stick. Check this several times as adjust the horizontal stab in all three directions.



#5 - The third direction is to level the horizontal stab with the wing. I sight from the rear until I line up the two trailing edges. Again, check all three directions several times. Sometimes I'll place a small weight out on the high side to hold it level while the glue dries. Take your time. Do it Right.



- When the time comes to mount your motor, screw on a ner adapter and attach a second line level with a rubber l or tape to the adaptor.



#7 - With the wing and horizontal stab level, level your engine. Recheck it anytime you remove your engine and anytime your plane starts turning easier in one direction. What a great feeling to know that you are zero, zero, zero for the first launch! This was adapted from what I observed when visiting Byron Barker last Fall. Thanks, Byron.

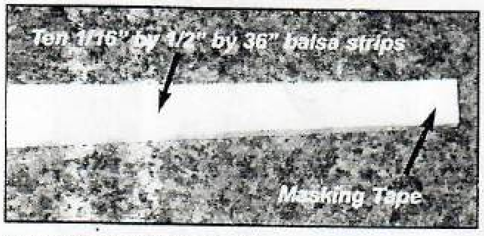
18

BUILD 'EM FAST

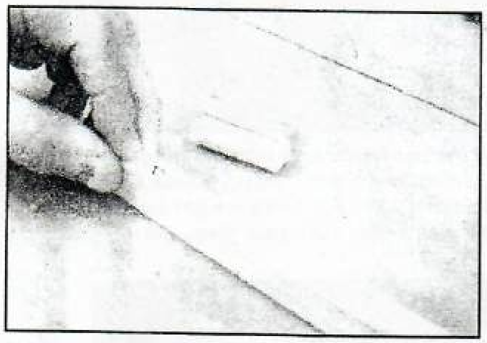
Even if you are building only one at a time here are some tips that will speed things up.

Building ten at a time forced me to come up with ways of working faster. To build the 240 full ribs and 220 half ribs, that meant 920 ribs slices. Those were done with the simple jig I mentioned in the last issue. It works great.

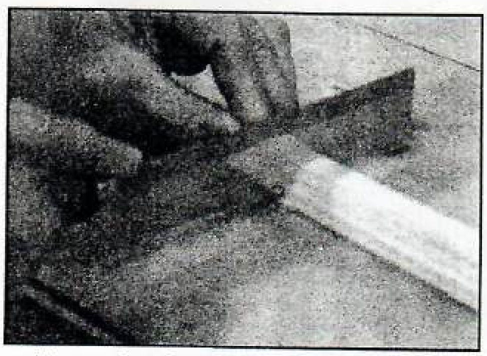
Next I had to cut all the rib verticle pieces. With the balsa stripper I cut ten 1/16" by 1/2" by 36" strips. I then taped them together with masking tape at one end to hold them together while I cut.



Next place a pattern piece on top.



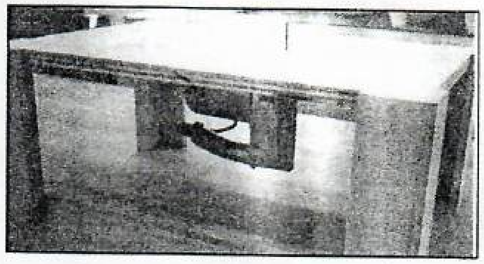
With a saw cut straight down and you've cut ten pieces at one time. This makes



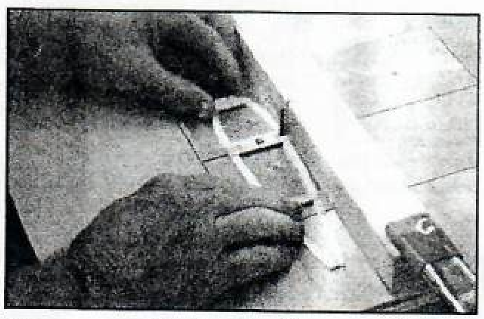
quick work of cutting rib verticles, spar-cross braces, and the ribs for the flaps, elevators and horizontal stab. Works great!

The next major task was cutting the spar nothches and leading edge and trailing edge notches in the ribs. With 24 full wing ribs and 22 half ribs, that's 162 notches times 10, equals 1620 notches in ribs. I also had to cut rib notches in the flaps, elevators and horizontal stab. That's another 960 notches!

A few months ago I built a router table out of scrap pieces of plywood to round over my handle bases. The router is attached to a 1/8" aluminum plate which is set flush into the plywood. So, I made another 1/8" plate to attach my sabre saw. I simply drilled and tapped the sabre saw base to use the same three screws that secure the router.



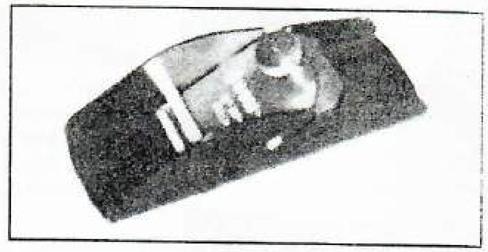
This allows me to cut the flap, elevator and stab notches which are on the inside. Just a quick touch and it makes a perfect notch for the 1/16" ribs.



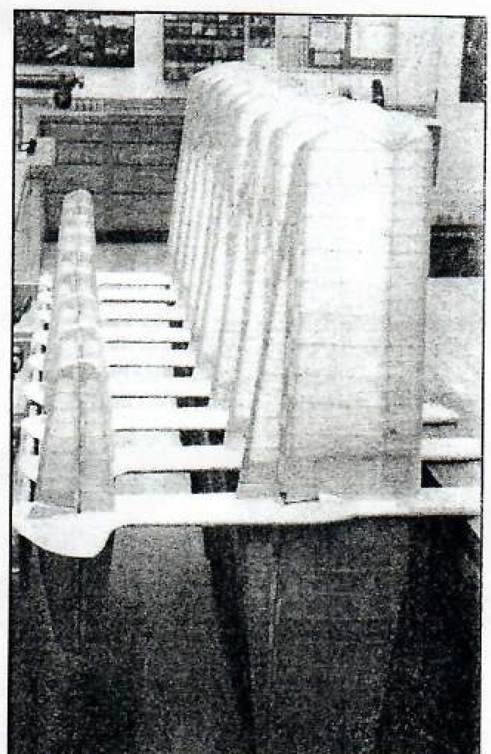
The spar notches in the ribs are 1/4" deep, so I clamped a fence to act as a stop to give the 1/4" depth of cut. Two quick passes and you've got a perfect 1/8" by 1/4" notch. For the leading and trailing edge notches I reset the stop fence to 1/8" depth. One quick plunge and you've got a perfect notch for the 1/16" leading and trailing edge shelves.

These gang and notch cutting techniques allowed me to cut and fit all 960 flap, elevator and stab ribs in three hours! What a great time saver.

After the wing was assembled I had to shape the leading and trailing edges. I used a Stanley 12-101 razor plane to shape the trailing edges. By placing the heel of the plane over the ribs, I found that I could plane the trailing edge to the exact shape in just a few passes. Very little sanding was required. It made quick work of the 2[illegible] trailing edges.



I used my electric finishing sander to taper the solid portions of the flaps and elevators. I found you can't use it over the open ribs. It will cup them out. But, it save a ton of time in the tapering job. The open rib areas were done with the flat Perma Grit file.



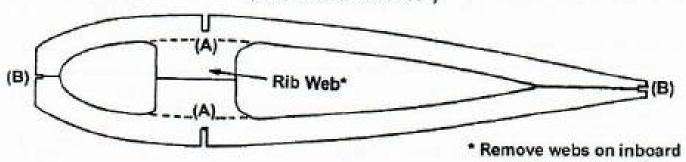
Another time saver was the use of Elmers Glue All to secure the wing and do the fillets. Yes, I used Elmers for fillets. They are small, but just as smooth as glass! Note how the planes were positioned while putting on the Elmers fillets.

Building

A "Lincoln Log" Wing Update By Tom Morris

Typical Rib in "Lincoln Log" Wing

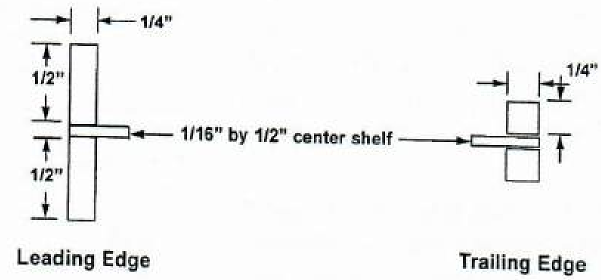
All ribs are 1/16" Balsa
(Over 700 in² use 3/32")



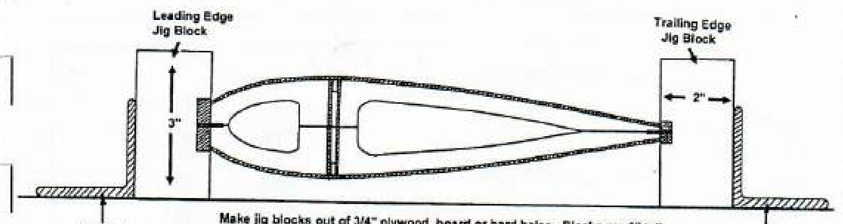
* Remove webs on inboard ribs after cross bracing is installed

(A) Spar Slots are 1/8" (W) X 1/4" (D)

(B) End Slots are 1/16" (W) X 1/8" (D)



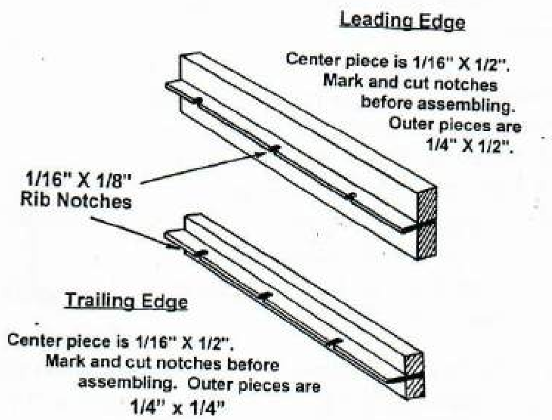
"Lincoln Log" Wing Jig Blocks



Make jig blocks out of 3/4" plywood, board or hard balsa. Blocks are 3" tall by 2" wide. Make eight jig blocks for the leading edge and eight for the trailing edge. Slots in jig blocks are exactly centered, 1/4" deep and fit tightly over leading and trailing edges.

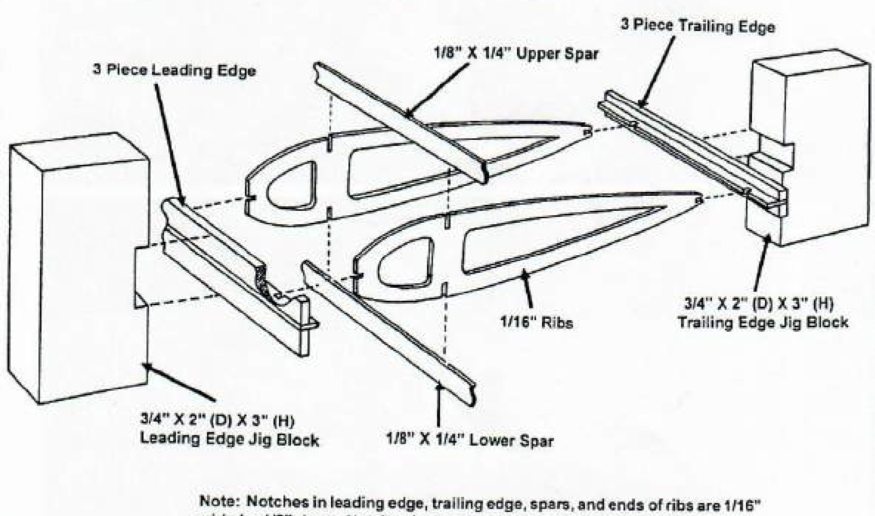
Because we have over 500 new PAMPA members since the first "Lincoln Log" wing article was published in May/Jun 1997, and since I get several calls per week about this building method, I thought it would be appropriate to republish at least an abbreviated version. There have been a few minor changes which are reflected on some of the drawings. I have now built over 100 wings with this method and it is still fun and works great. The pseudo I-beam was a little too light and difficult to finish, so I've gone back to regular sheeting. I still overlap the covering in the middle.

Three Piece Leading and Trailing Edges



Notches are 1/16" wide by 1/8" deep.

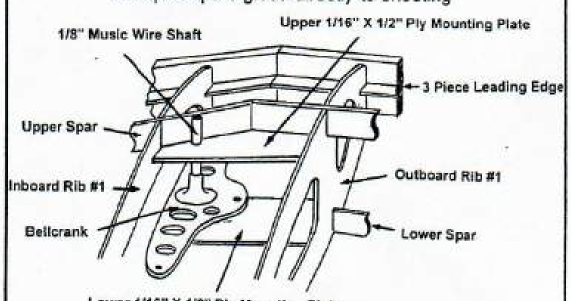
The "Lincoln Log" Wing



Note: Notches in leading edge, trailing edge, spars, and ends of ribs are 1/16" wide by 1/8" deep. Notches in ribs for spars are 1/8" wide by 1/4" deep.

Mounting Bellcrank

I now inset 1/16" ply mounting plates on top of spars glued directly to sheeting



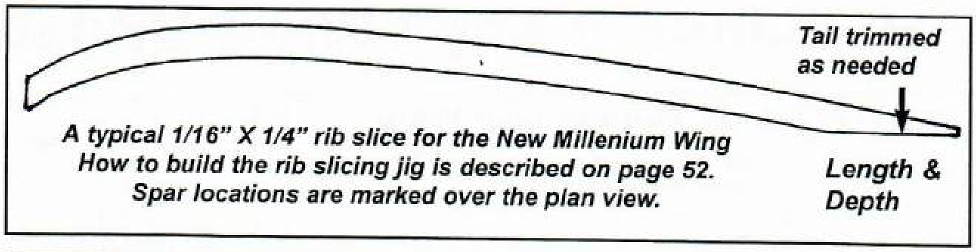
Dry fit 1/16" ply mounting plates between #1 ribs. Bellcrank shaft should be tight against both spars. Mark location of 1/8" bellcrank shaft and drill 1/8" holes in mounting plates. Cut bellcrank shaft to full depth of wing. Connect leadouts and pushrod to bellcrank. Slip ply mounting plates over bellcrank shaft. Glue entire bellcrank assembly in place. With 3/8" hard balsa blocks, fill in around bellcrank shaft on both sides of spars. Make sure 3/8" block is contoured to ribs so that it will be firmly glued to sheeting, spar and ply mounting plates. Mounting a bellcrank this way ensures it's there to stay.

Introducing
The New Millenium Wing

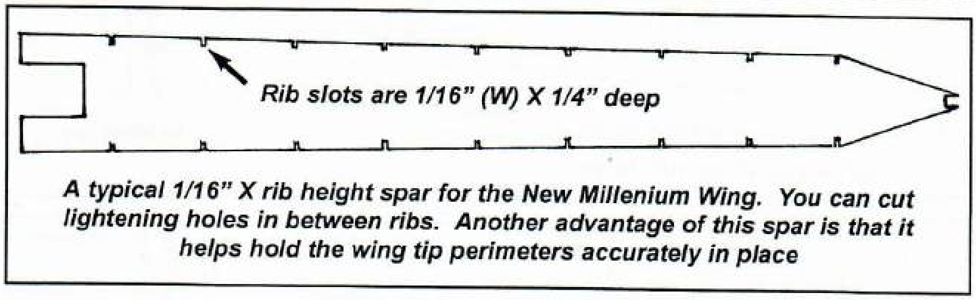
20

By Tom Morris

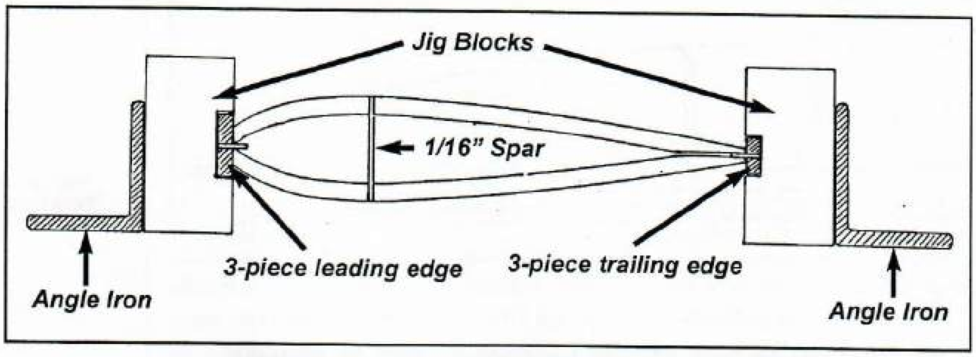
We could have called it the "Lincoln Log", pseudo I-Beam, sheeted D-tube, but that would have been a bit too much, don't you think!



A typical 1/16" X 1/4" rib slice for the New Millenium Wing How to build the rib slicing jig is described on page 52. Spar locations are marked over the plan view.

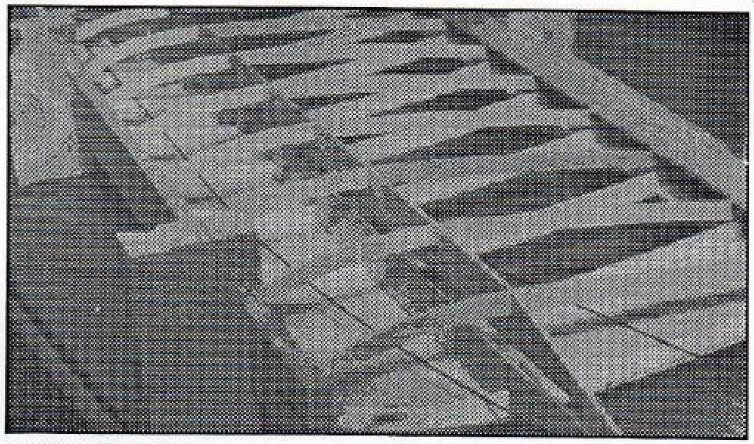


A typical 1/16" X rib height spar for the New Millenium Wing. You can cut lightening holes in between ribs. Another advantage of this spar is that it helps hold the wing tip perimeters accurately in place

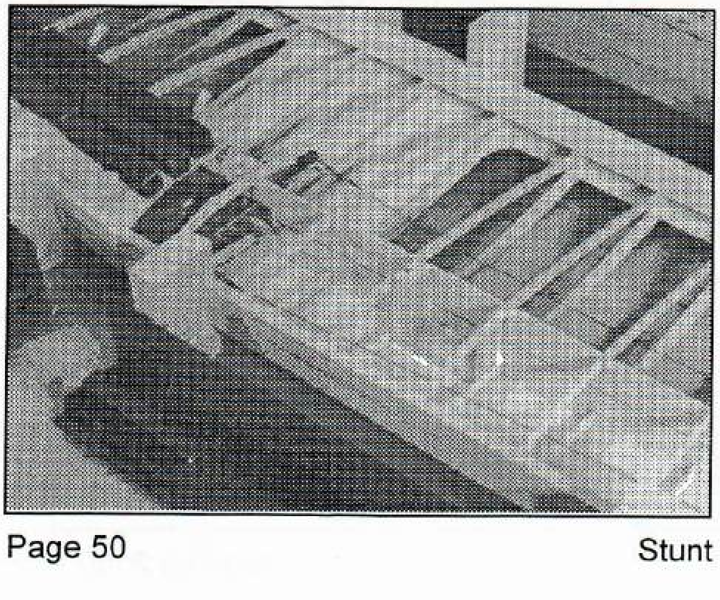


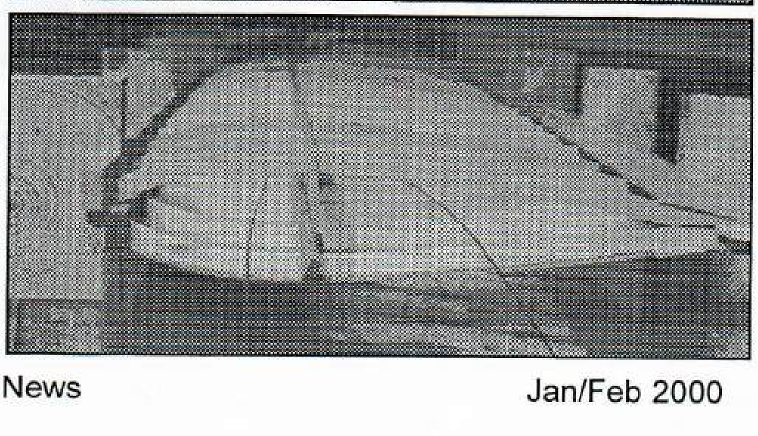
In our never ending search for the perfect wing building method John Simpson and I have tried at least a dozen different configurations. The "Lincoln Log" wing solves a lot of wing building problems and produces a really great wing, but you still have to cut out a set of ribs which takes six sheets of balsa and about 2.5 hours of somewhat tedious work.

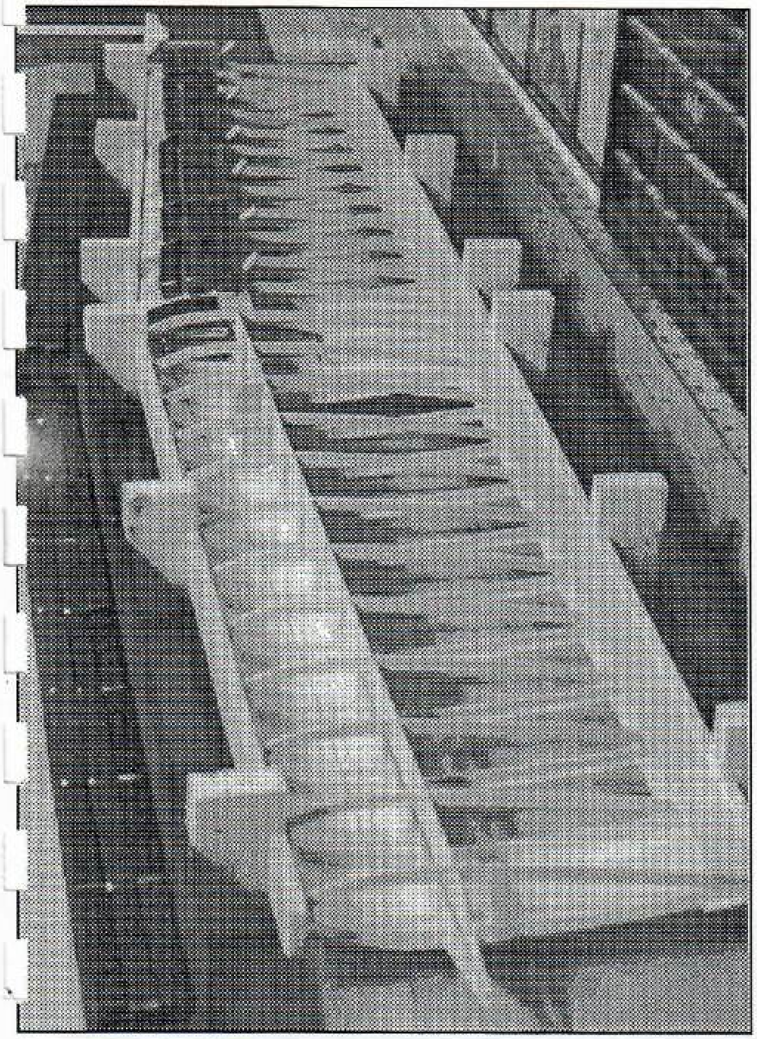
John came up with the idea of combining the "Lincoln Log" three piece leading and trailing edges, the I-beam rib slices and the Nobler type "D" tube sheeted spar construction. Now you can cut out the rib slices with a simple jig (See page 52) in about ten minutes using less than two sheets of balsa. Halleluia!

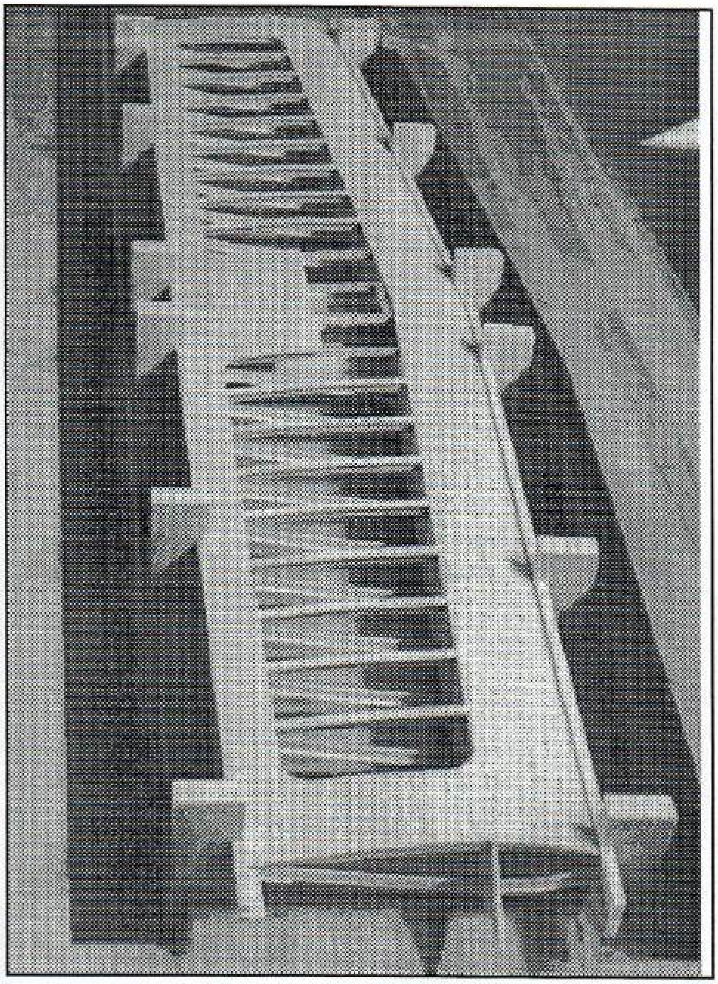


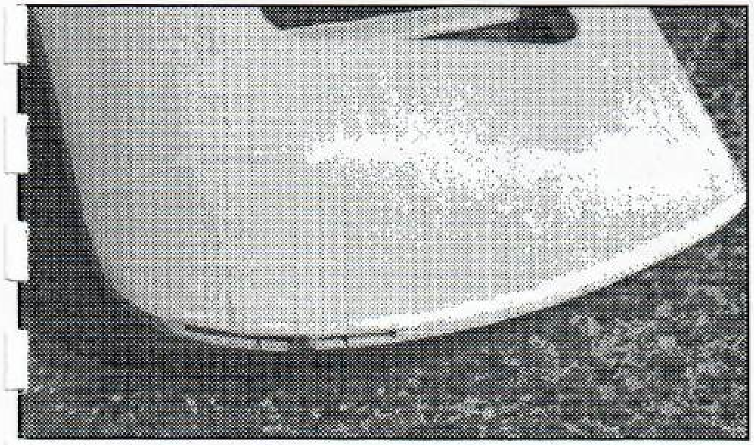
The spar height is determined by the root and tip rib thicknesses. The spar is notched 1/16" X 1/4" at each rib location. The 3-piece leading and trailing edges are not notched, only marked for rib locations. The rib slices are trimmed at the tail for length and depth. The plywood bellcrank mounting plates are inset on the top of the spars directly under the sheeting. The landing gear blocks are supported on each side by a 1/32" ply rectangle. The verticle landing gear blocks are full depth and glued firmly to the











sheeting, both top and bottom, ply doublers and against the spar. The inboard spar is notched to allow the leadouts to clear when the adjustable is all the way forward. Leading edge, trailing edge and center sheeting is as usual for a sheeted "D" tube wing.

The wing tips are fully sheeted with 1/16" balsa. Also notice that the outer bay is sheated per Robin Sizemore's suggestion. This increases rigidity at almost no extra weight. Also, it allows symmetry in the open bay with round fillets on both ends. It looks better that way. Before the tip is sheeted another rib sliver is placed on the outboard rib to act a shelf for placing the tip sheeting. The sheeting is then glued to this ledge and to the tip perimeter. It makes a very strong, yet light and easy to finish tip. The adjustable is built right into the tip.

The only glue I use is Elmers, period. On any surface joint, instead of a continuous bead of glue, I put small drops about every 3/8". This keeps the joint from being a hard line and showing up later as you sand. Sheeting is pinned in a few places, but primarily it is held down by steel weights. Once you get away from pin based building to weight based building, your speed will pick up significantly. I have all sizes and shapes of weights in my shop. I use them in nearly every building step. Another plus is that weights hold everything dead flat on the bench. These wings come out of the jig blocks dead straight everytime.

Once the sheeting is completed except one piece of the center sheeting to allow access to install the pushrod, it's time to shape the leading and trailing edges. This is done with a Stanley 12-101 razor plane. Pull the razor plane toward you instead of pushing it away from you. You rest the heal of the plane on the sheeting and plane until you hear the clicking of the glue joint from the dots of glue. With just a little practice you can shape the leading and trailing edges in just a few minutes so accurately that you can go straight to sandpaper bypassing the Permagrit file.

I am currently building these wings for 50 Cavalier "Quick-Build" kits to be available in March. I will try to make a video available in late January.

I truly believe our search for the best wing building method is over. We've found it! This is it!

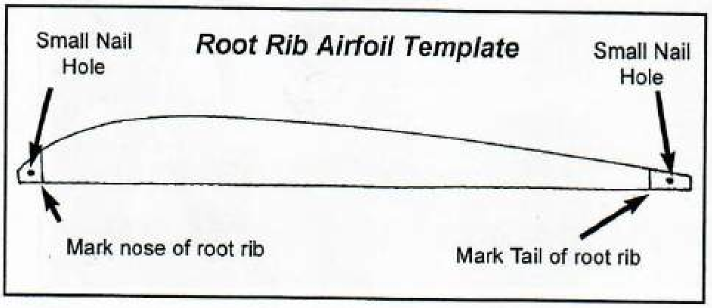
how to build a rib slicing Jig

By Tom Morris

This jig is composed of three parts:

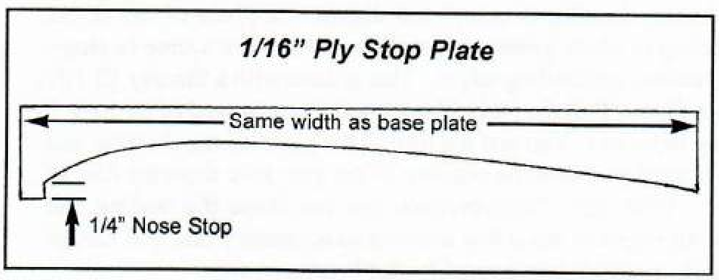
Root Rib Airfoil Template
1/4" Ply Rectangular Base Plate
1/16" Ply Stop Plate

Step One - Cut a **root rib airfoil template.** I use 3/32" phenolic or 1/16" aluminum. You could use 1/16" plywood or formica. Make the template about 3/8" longer than the root rib on both ends. Mark the nose and tail of the root rib. Drill a small hole on the extra length at both ends.

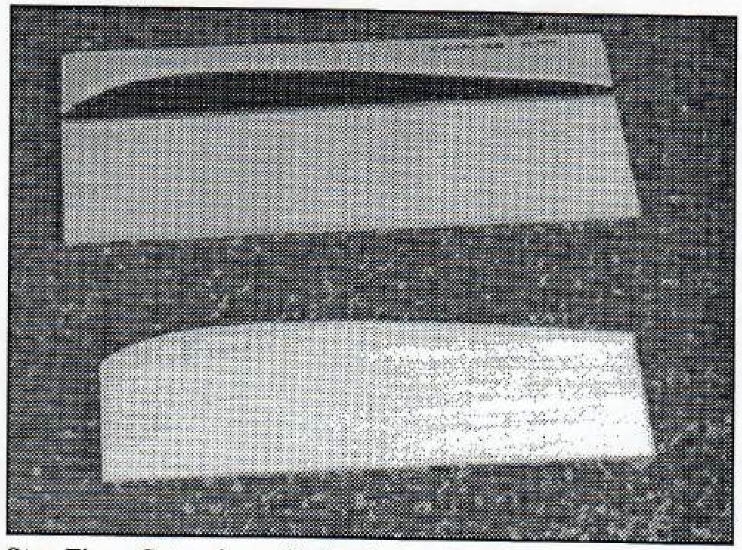


Step Two - Cut a retangular **base plate** from 1/4" ply. Make it the width of the template and about six inches high.

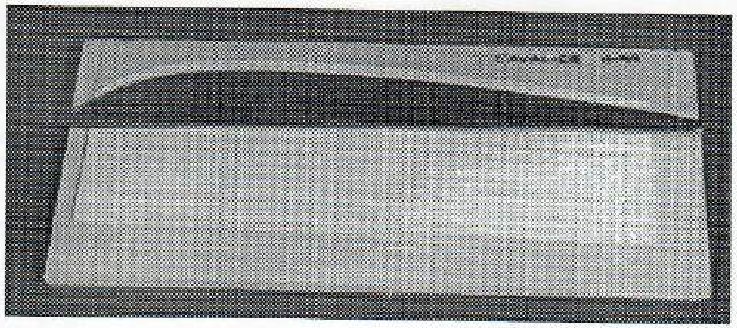
Step Three - Using your root rib airfoil template, cut a **stop plate** out of 1/16" ply. Cut the width the same as the base plate. Trace on the airfoil using your root rib airfoil template. Put a nose stop 1/4" long. All rib slices will be referenced off this stop. Glue the stop plate onto the top of the base plate.



Step Four - Measure down from the stop plate 1/4" at the high point of the airfoil and at the tail of the airfoil and mark the base plate. Place the root rib airfoil template on these marks with the nose mark lined up with the nose stop and seat small brads in the two nail holes. You may have to cut the heads off the nails to remove the template. You should now have your template 1/4" away from the stop plate. You're now ready to cut rib slices.



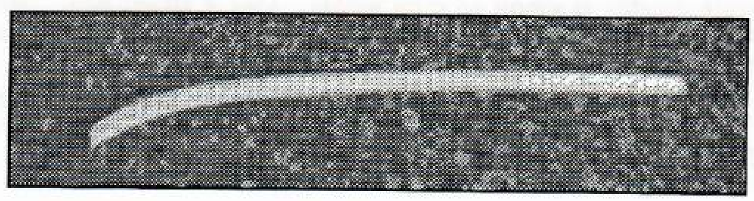
Step Five - Cut a piece of balsa the length of the root rib. Remove the template from the jig. Using the root rib airfoil template, slice the airfoil off the top as you see the lower piece in the photo above.



Step Six - Now place that piece of balsa on the base plate up against the stop. Make sure the the nose is against the nose stop. Replace the root rib airfoil template over the nails on top of the balsa as you see in the photo above.

Step Seven - Start slicing off rib slices. You could edge glue several sheets together before you started slicing so you wouldn't have to stop slicing with each individual sheet. After you remove each slice, just slide the sheet up against the stop and keep on slicing. Make sure the nose of the sheet is against the nose stop.

With a little practice you can slice all the rib slices you'll need for a wing in about ten minutes. Remember for tapered wings as the ribs get shorter to remove the length from the tail of the rib slices.



How To Cut Ten Sets of "Lincoln Log" Jig Blocks In 30 minutes

By Tom Morris

With a table saw you can cut enough dead accurate jig blocks for your whole club in about 30 minutes

Step 1 - Buy a straight, light spruce 1 1/2" X 3 1/2" X 8' (2 X 4 X 8').

Step 2 - Cross cut it in half. You now have two pieces of 1 1/2" X 3 1/2" X 4'.

Step 3 - Rip saw 1/4" off one edge.

Step 4 - Set the fence at 3" and rip saw both pieces to 3" wide. You now have two pieces 1 1/2" X 3" X 4'.

Step 5 - Set the fence at 1 1/2" and the blade 1/4" high. Saw a blade width by 1/4" kerf down the middle of each piece. Revese ends and saw again.

Step 6 - Move the fence out 1/8". Saw another kerf. Reverse ends and saw again.

Step 7 - Repeat step six until the opening is approximately 9/16". Lay one piece aside. This will make your trailing edge blocks.

Step 8 - With the remaining piece continue Step 6 until the opening is approximately 1 1/16". This will be your leading edge blocks.

Step 9 - Clean out the openings with a chisel.

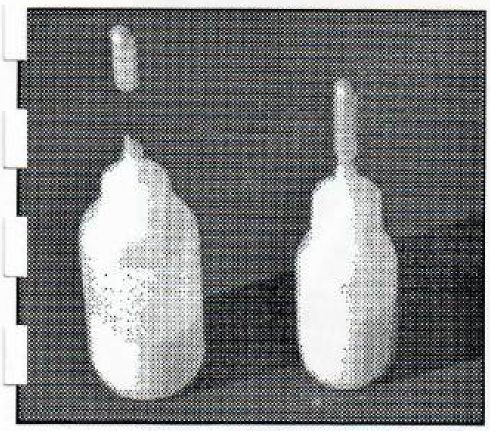
Step 10 - Set up for crossing cutting 5/8" slices off the 1 1/2" X 3" X 4' pieces. As you slice off the 5/8" pieces keep them in order and rubber band them together in packs of six (a complete set). Write the date on them. That way if you cut other sets later, you will use only matched sets and not mix them up with other sets.

You will get ten complete sets of leading edge blocks and ten complete sets of trailing edge blocks out of one two by four. And they will build you a perfect wing every time!

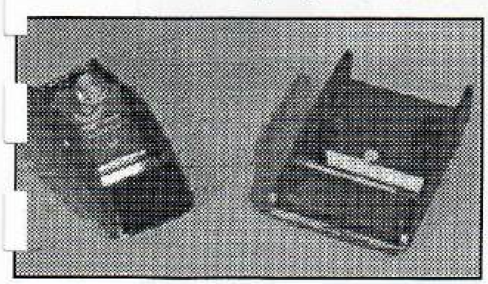
Set your balsa stripper at a little over 1/4" and cut two pieces of scrap 1/4" balsa. Put a scrap piece of 1/16" balsa in between and test fit in the jig block. Adjust your balsa stripper until you get a snug fit of the three piece trailing edge in the jig block. Now cut your full length pieces. Repeat for the leading edge. In just a minute or two you should be able to get a nice press fit between the jig block and the three piece leading and trailing edges.

Happy building! If you or any of your building buddies don't have a table saw, I'll be glad to supply the jig blocks.

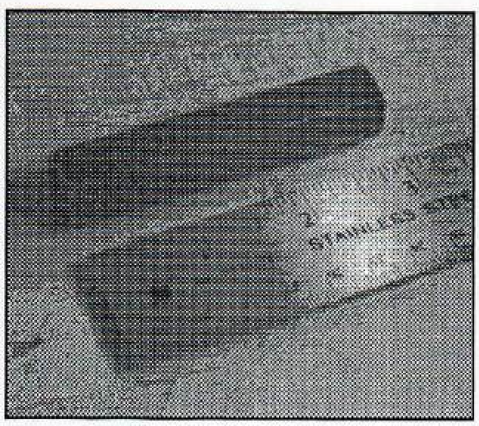
Four Very Useful Tools



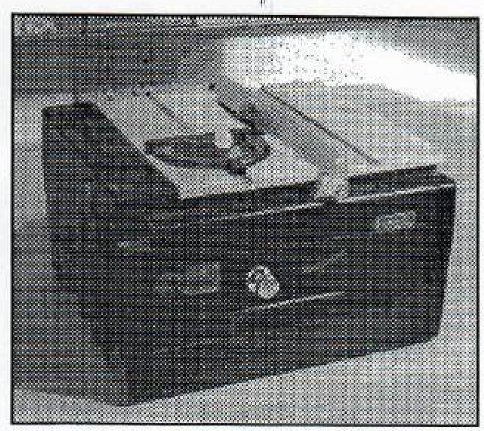
Glue bottles from Small Parts (800) 220-4242. 1 oz R-SQB-01, 2 oz RB-SQB-02. Needles 1" R-NE-142PL, 2" R-NE-143PL. 30 different size needles. These work best with undiluted Elmers. And that's the glue you should use!



Razor Planes. Stanley 12-101, Lowes, Home Depot, etc. Master Airscrew, hobby shops, Brodaks, Tower. Need both. Stanley for shaping LE & TE. Master Airscrew for flaps.

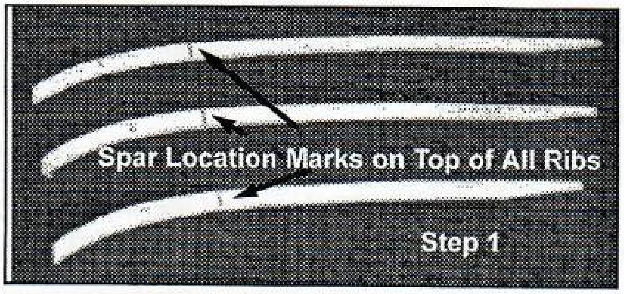


1/2" X 1/2" X 3" steel weights. You need 50. Get then from your local steel yard. About $8 here. While you're there, pick up several other sizes of steel weights. You'll be amazed at how many ways you use them. I hardly ever use pins. It really speeds things up using weights rather than pins.



The MicroLux Tilting Arbor Table Saw from Micro-Mark (800) 225-1066. $300. I don't know how I ever built model airplanes without this jewel. It cuts perfect pieces with minimun effort and will saw all day long without skipping a beat. I use it to cut all balsa strips, leading and trailing edges, spars, plywood bellcrank supports, horn clips, weight tip boxes, etc. It is wonderful!

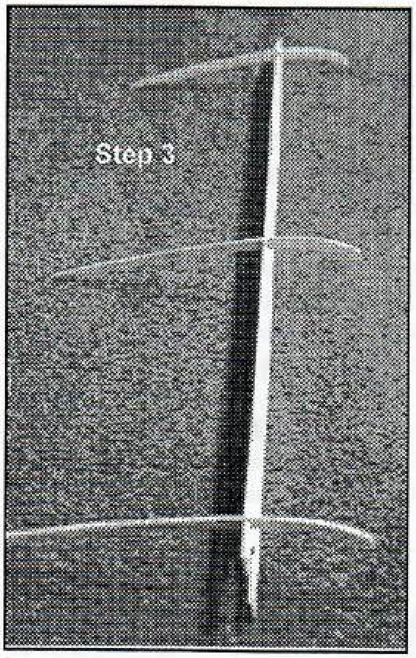
Building the "Lincoln Log New Millennium" Wing



☐ **Step 1** - Locate Rib slices #1 through #12. Note the dot drawn on top of the ribs which is the spar location.

☐ **Step 2** - Fit the 3-piece leading and trailing edges in jig blocks. The trailing edge should be aligned on a straight edge. I use angle iron, but anything straight will do.

☐ **Step 3** - Insert #1, 4, 8 & 12 rib slices into spar making sure the spar is on the marks on the rib slices.

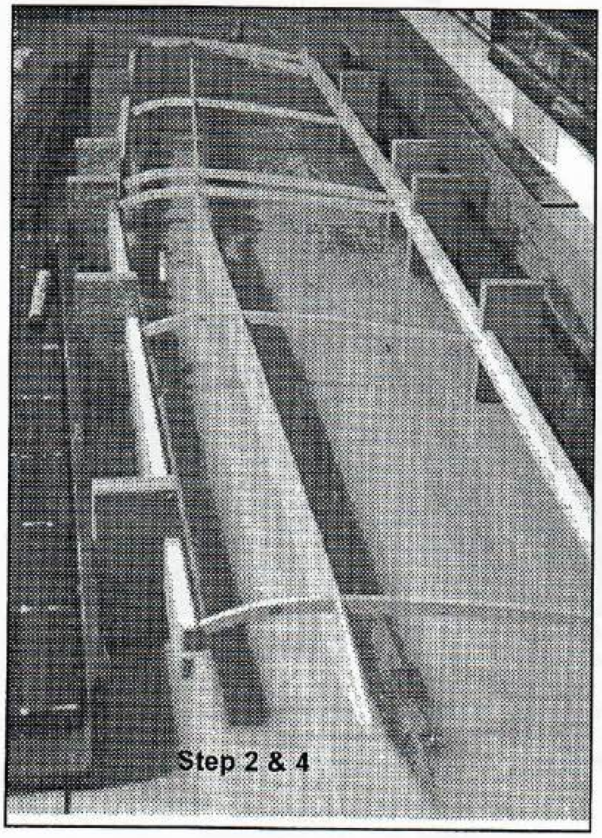


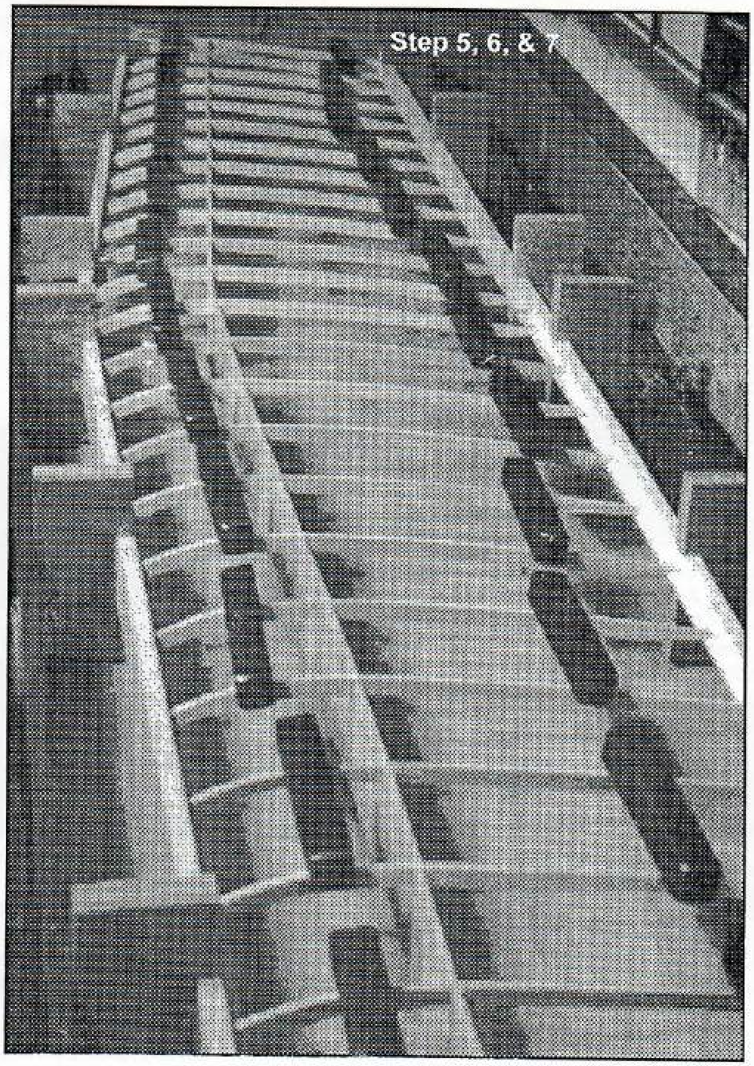
☐ **Step 4** - Place these ribs on the marks on the LE & TE center shelf.

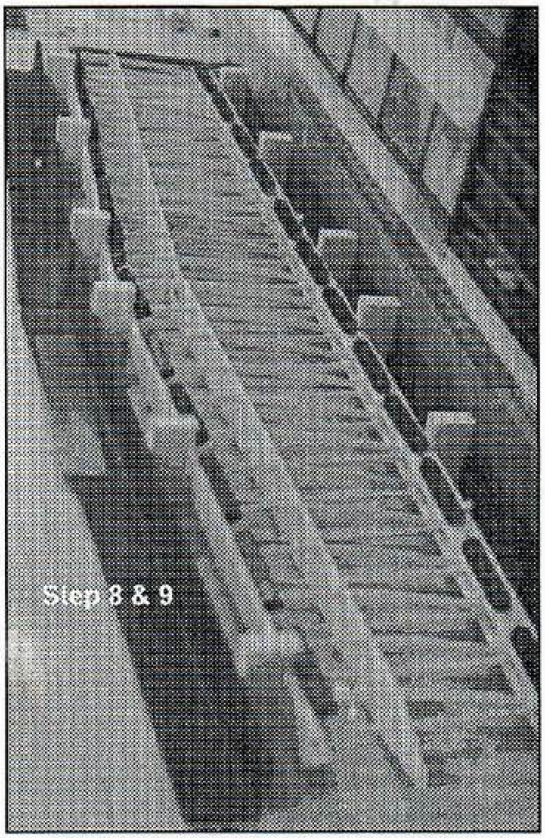
☐ **Step 5** - Add the remaining rib slices. It helps to place small weights on top of rib slices to keep everything in place. I use 1/2" X 1/2" X 3" steel weights.

☐ **Step 6** - Glue all rib slices to TE, LE & spar aligning ribs perpendicular to the trailing edge. Don't glue front inside of spar at #3 & #4 ribs where the landing gear doublers will later be glued.

☐ **Step 7** - Join the two halves by gluing the LE & TE joints and by adding the center 1/16" balsa doublers.



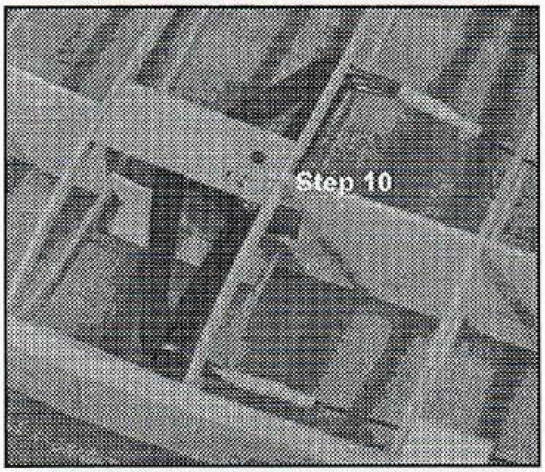




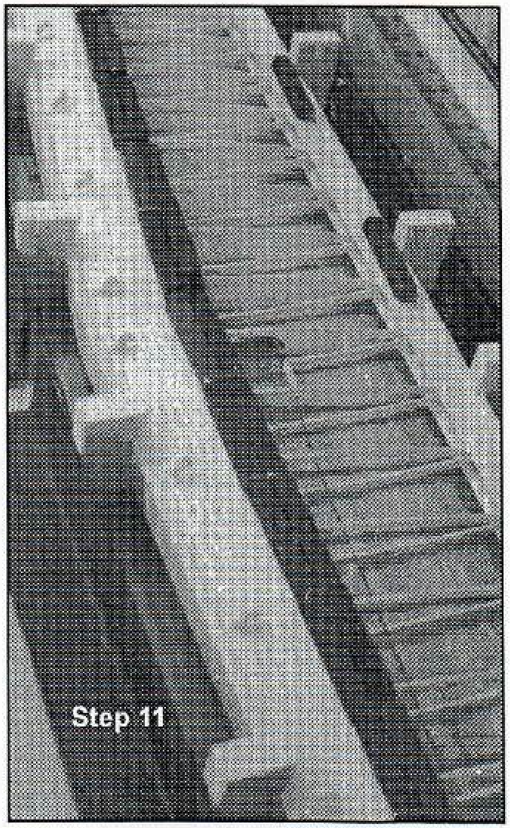
☐ **Step 8** - Add the top TE Sheeting.

☐ **Step 9** - After drying, carefully turn the wing over and glue in the bottom rib slices, center doublers and the bottom TE sheeting on the bottom side. Be sure to align the spar on the spar marks on rib slices #1, #6 & #12, keeping the spar straight and verticle.

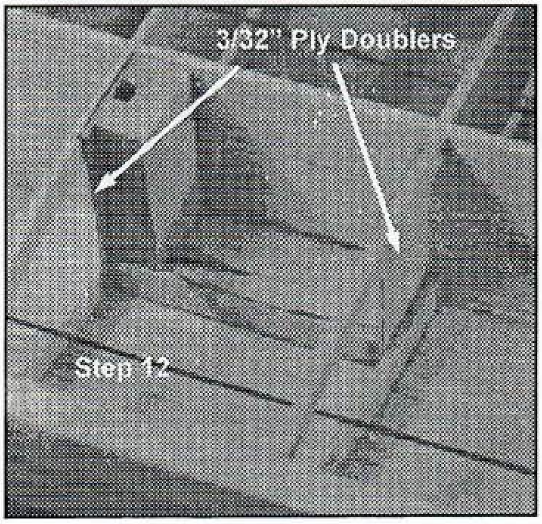
☐ **Step 10** - Turn the wing back over and add the Bellcrank assembly. Place the two ply supports over the shaft and slide the assembly in place and glue. After the top has dried turn the wing over and glue the bottom support. Be sure the supports are flush with the top of the spar so the leading edge and center sheeting will be firmly glued to the supports. With a Dremel tool and cut off wheel, cut off the excess bellcrank shaft, leaving 1/16" above the mounting plates.
Epoxy the bellcrank shaft to the spar to keep it from moving vertically.



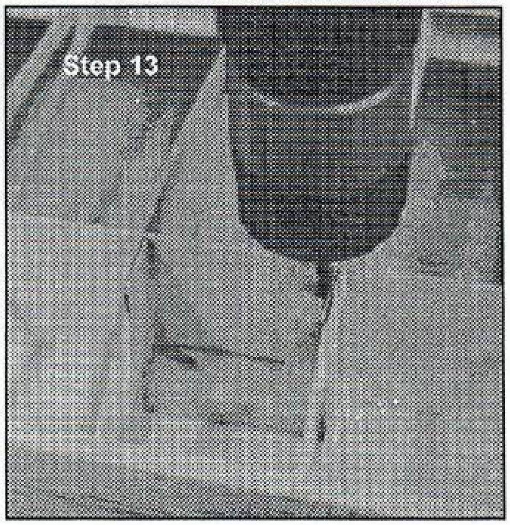
☐ **Step 11** - Turn the wing over and glue on the bottom leading edge sheeting. Pin sheeting to LE above every other rib. On all outer surface glue joints, instead of a continuous bead of glue, put a drop every 3/8". This will eliminate a hard glue line.

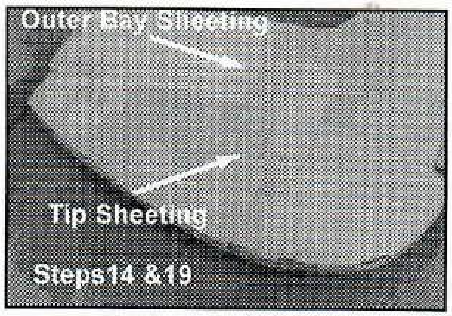


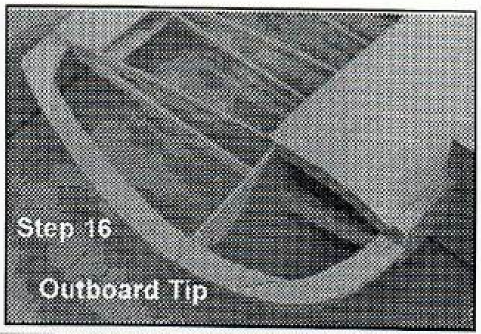
☐ **Step 12** - Turn the wing over and glue in the landing gear block doublers and the landing gear blocks. Make sure you've got a good fit all around and that the top of the doublers and verticle blocks are flush with the top of the ribs and thus will be glued firmly to the top LE sheeting.



☐ **Step 13** - When landing gear blocks are dry, drill out a 1/8" hole for the landing gear wire from the top down the verticle block shaft and through the horizontal block and bottom sheeting.



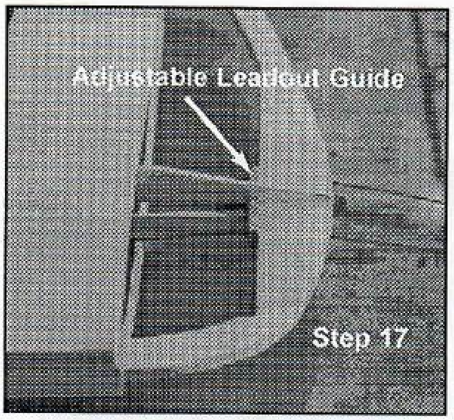


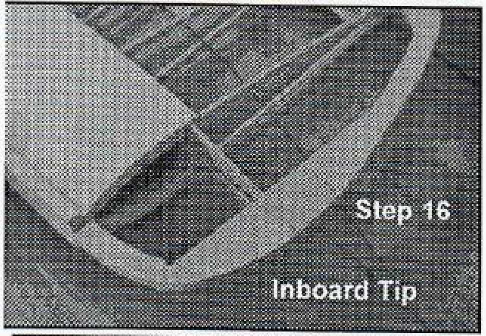


□ **Step 14** - Add outer bay sheeting. Glue back piece first. Trim front piece to fit.

□ **Step 15** - Sand off all excess TE, LE & Outer Bay Sheeting flush with outer rib.

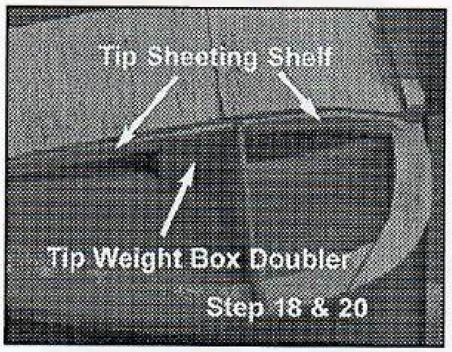
□ **Step 16** - Add the tip perimeters.

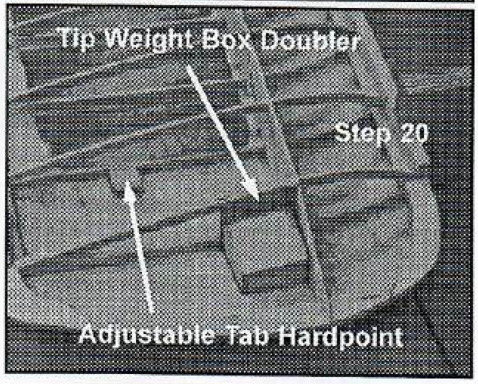




□ **Step 17** - Add the adjustable leadout guide and bolt. Thread leadouts through guide. Cut away some of the spar extension to clear the adjustable leadout guide so it slides freely.

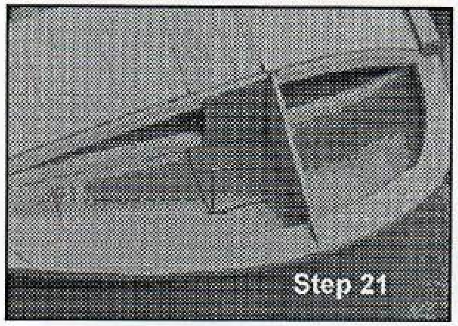
□ **Step 18** - Add the tip sheeting shelf (#12 rib slices) for the tip sheeting top and bottom.

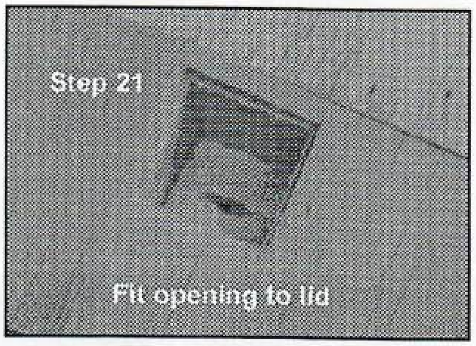




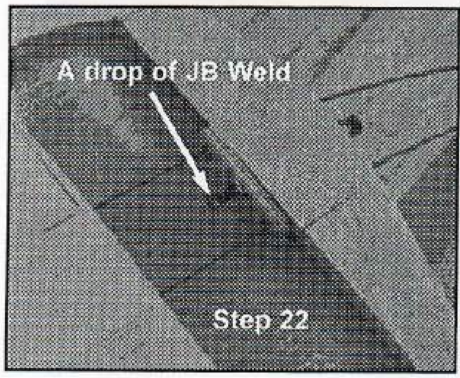
□ **Step 19** - Test fit and add the bottom tip sheeting. Use masking tape and a few pins to hold it in position. You may have to sand off some of the spar to get a good fit.

□ **Step 20** - Add the tip weight box doubler against the spar and the adjustable tab hard point (about 3 " from the trailing edge, on the bottom sheeting, against rib #11). Drill a 5/64" hole through the center of the adjustable tab hard point and bottom sheeting.

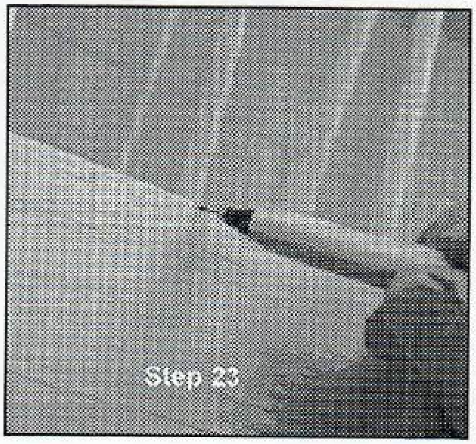




□ **Step 21** - Mark around the tip weight box. Stick a pin through the sheeting about 1/16" in from the out edges of the weight box corners. On the bottom side cut the rough opening for the weight box lid leaving about 1/16" around the edge of the weight box. Trim to fit. Glue in the weight box. Final fit the opening to the lid.



□ **Step 22** - Put a drop of JB Weld on the end of the adjustable leadout bolt so it can't be backed out.

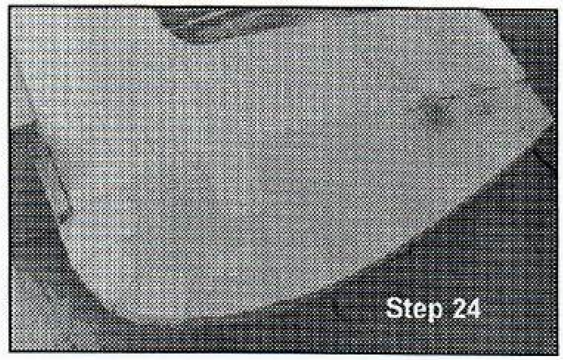


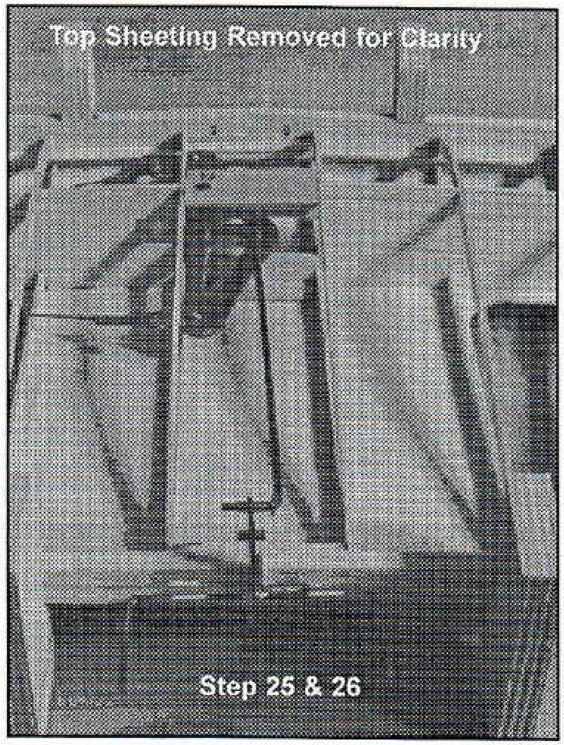
□ **Step 23** - Mark the location of each rib and add cap strips to ribs #3 through #10.

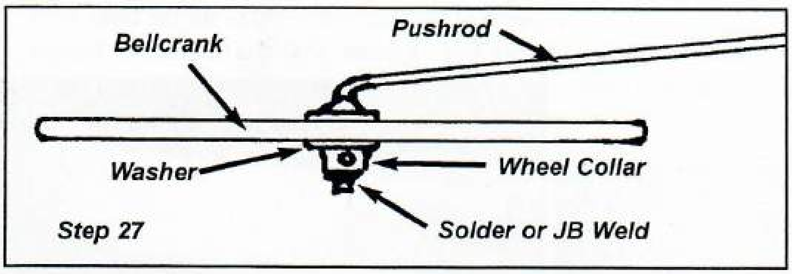
☐ **Step 24** - Add the top tip sheeting

☐ **Step 25** - Fit the control horn to the trailing edge. Make slots for the keepers and file a knotch in the trailing edge to clear for the swing of the horn verticle.

☐ **Step 26** - Place the pushrod into the bellcrank, and with the horn and bellcrank in neutral mark the pushrod. Bend the pushrod. Test fit and adjust. Don't worry about the system being tight at this point. It will loosen up when lubricated.

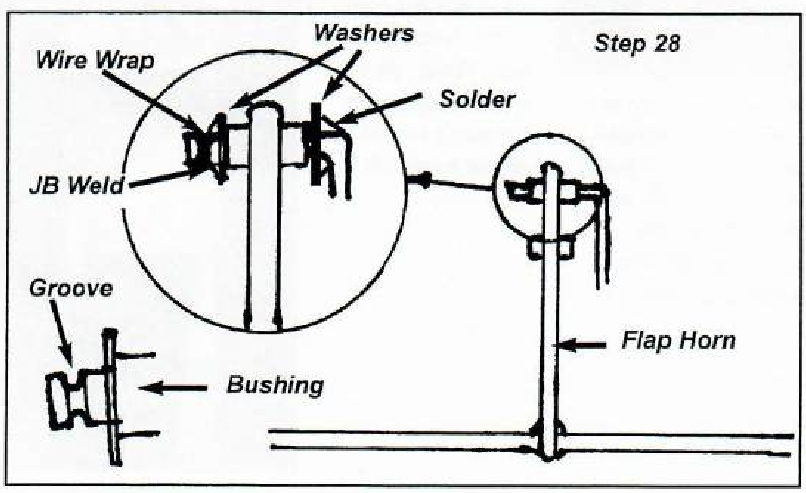






☐ **Step 27** - Permanently mount the pushrod to the bellcrank. Clean the pushrod end. I use Scotch Brite scouring pads. Push the pushrod through the hole in the bellcrank. Put on a #4 washer, then the 3/32" wheel collar. Tighten the wheel collar making sure there is no verticle play in the connection. Either solder or JB Weld thewheel collar to the pushrod. Don't overheat the bellcrank.

☐ **Step 28** - Prepare the end of the pushrod at the horn bearing. Solder a #4 washer perpendicular next to the bend. Cut off excess leaving about 1/8" sticking out the other side of the bushing. with a Dremel cut off wheel cut a shallow groove close to the end. Permanently attach the pushrod to the horn. Put on a #4 washer, then wrap fine copper wire into the groove and against the washer, Put a drop of JB Weld on the wire and against the washer. Don't get any on the horn.

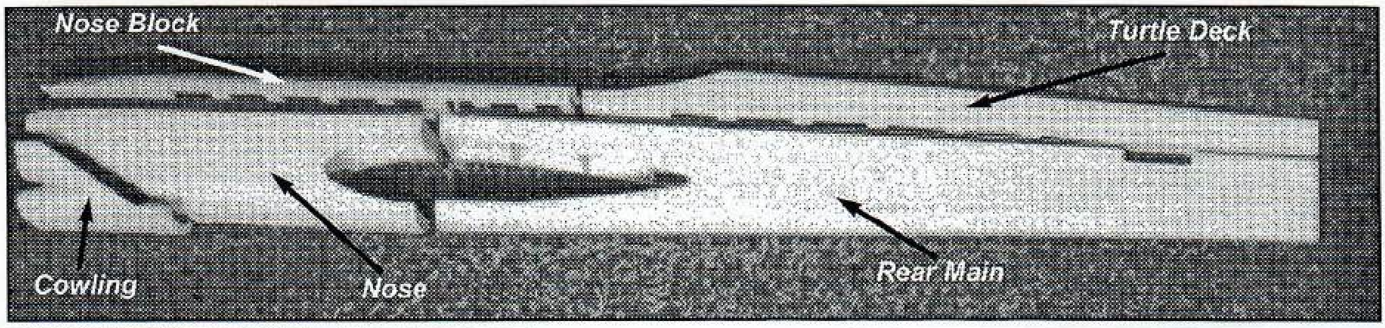


☐ **Step 29** - Put on the remaining center sheeting.

☐ **Step 30** - Remove the wing from the jib blocks and shape the leading and trailing edges. I use the Master Airscrew razor plane to remove most of the excess material. Rest the heal of the pane on the sheeting or cap strips and you can get it to almost the final shape. Finish up with a Perma Grit file and 150 sandpaper.

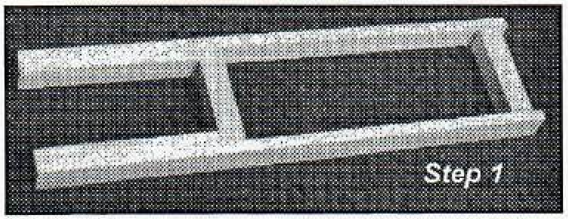
☐ **Step 31** - Sand, shape and fill as necessary.

Building the Ladder Crutch Molded Skin Fuselage

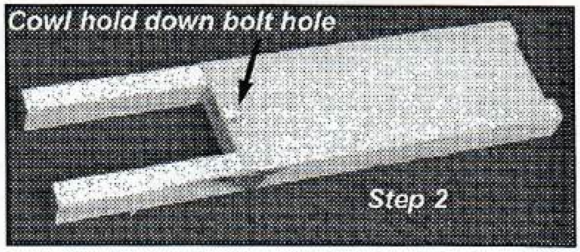


Nose

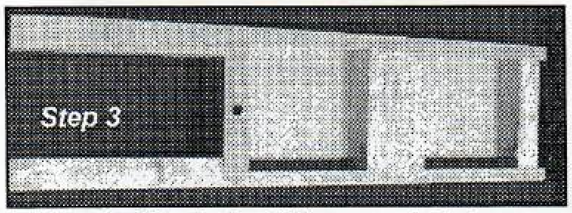
Note: I use Elmer's Glue All for gluing all parts.



☐ **Step 1** - Glue cross braces between motor mounts. Make sure crutch is dead flat and square.

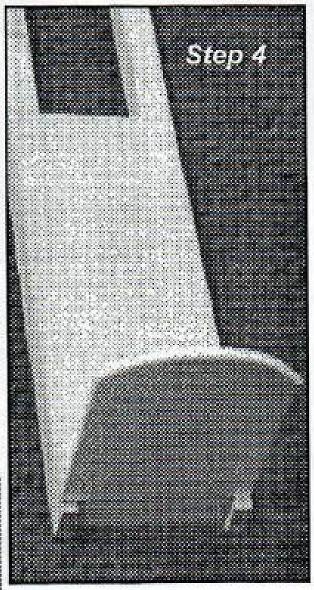


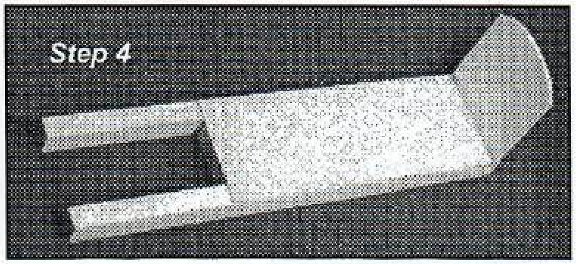
☐ **Step 2** - Glue 1/32" cross-grain ply tank compartment roof across crutch. Drill 5/32" hole in the center of the front cross brace for the cowl hold down bolt. Counter sink the hole on the bottom.

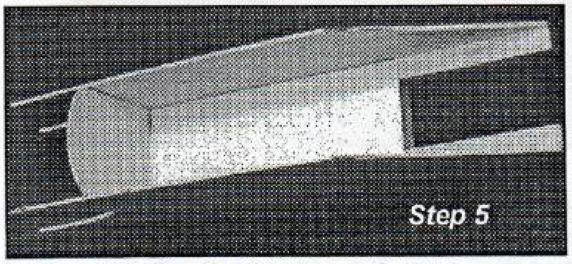


☐ **Step 3** - Glue in the 1/2" cross grain balsa vibration damper between the motor mounts.

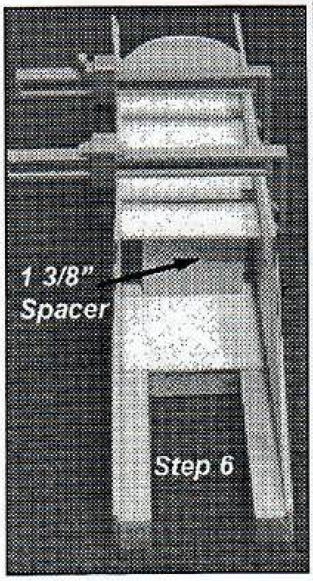
☐ **Step 4** - Glue former F1 perpendicular to rear of crutch.

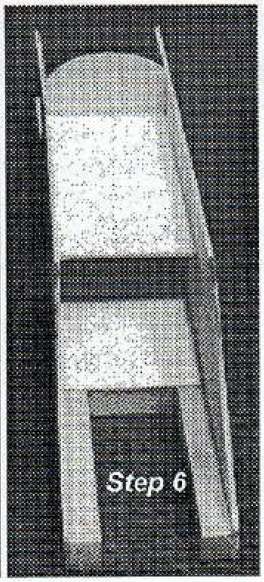






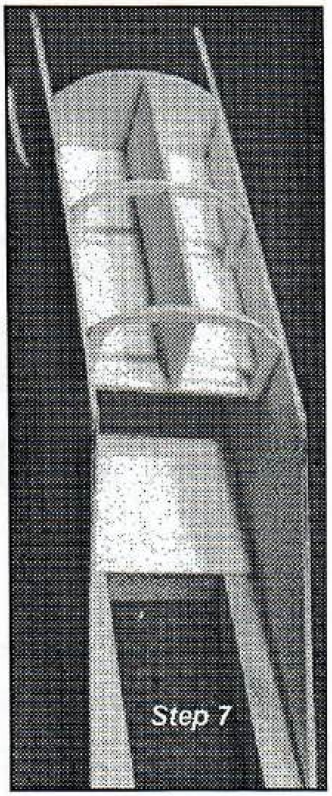
☐ **Step 5** - Glue ply sides to crutch. Make sure they are even with the front and top of the crutch. To insure accuracy, I glue on one side at a time.

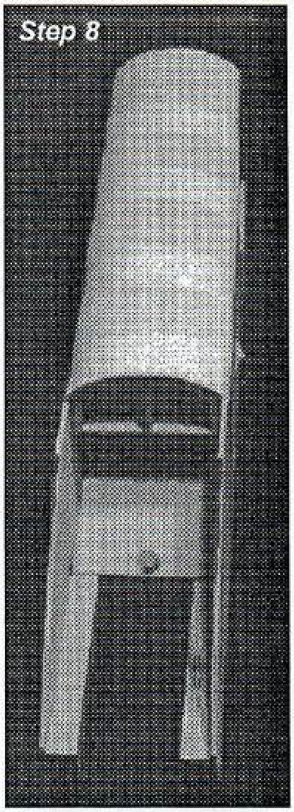




☐ **Step 6** - Glue ply tank shelf between sides and against F1. I use a piece of pine 1 3/8" high as a spacer. This leaves enough room for nearly all engine and tank set ups

☐ **Step 7** - Glue the 1/16" balsa center verticle support and the 1/16" ply horseshoe shaped formers to the bottom of the tank floor. Before gluing adjust the former's height and shape to match former F1.

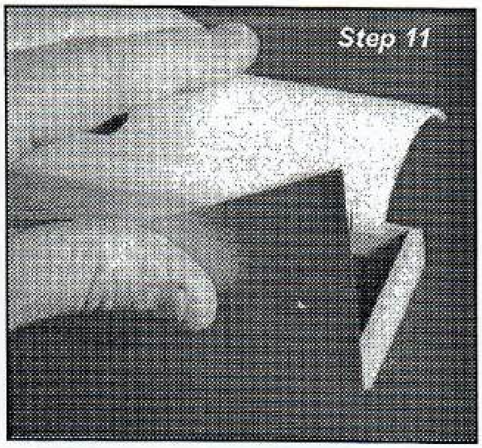




☐ **Step 8** - Fit and glue the molded bottom skin.

☐ **Step 9** - Mark and cut air outlets on bottom skin.

☐ **Step 10** - Glue former F1C inside ply sides. Make sure it is perpendicular to sides.

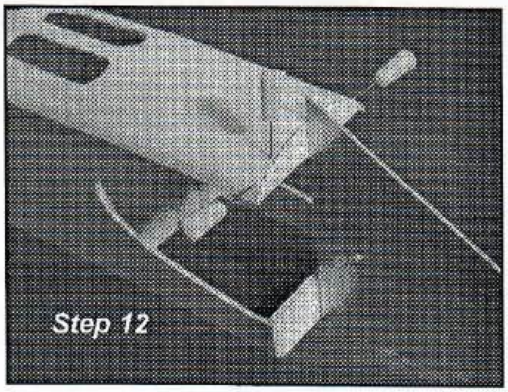


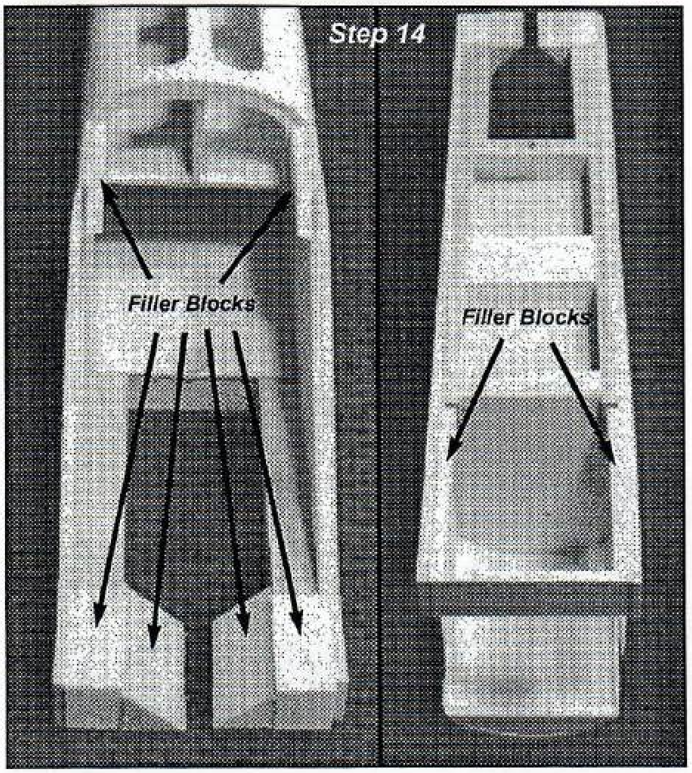
☐ **Step 11** - With a square mark the location of Former F1B. Cut off the excess molded skin.

☐ **Step 12** - Glue former F1B to molded skin. Make sure it is flush with F1C. Glue F2B to F1B.

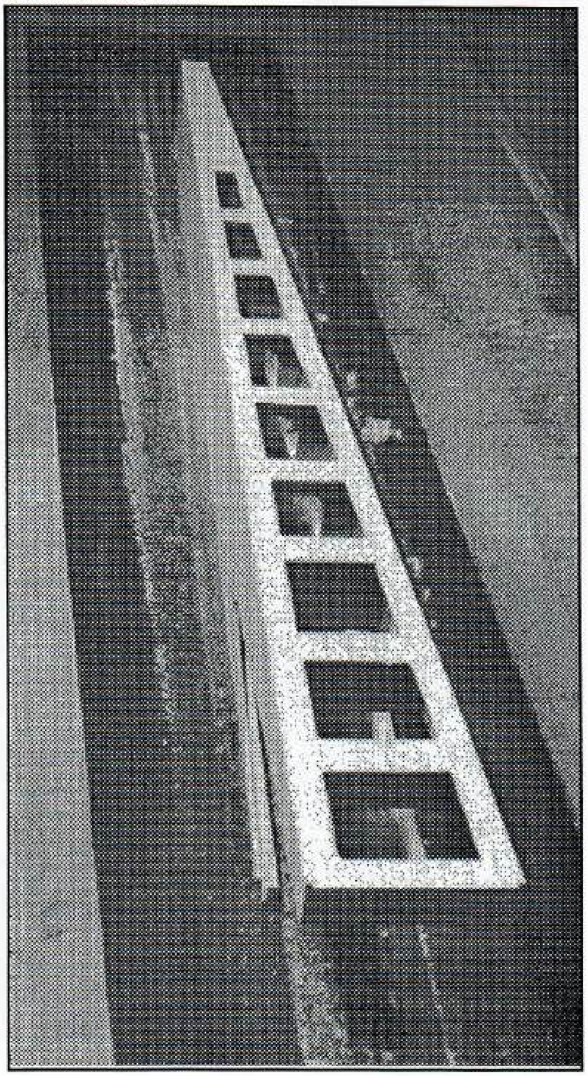
☐ **Step 13** - Glue 3/32" balsa sides to ply sides. To insure accuracy, I glue on one at a time.

☐ **Step 14** - Glue in filler blocks. There are eight.

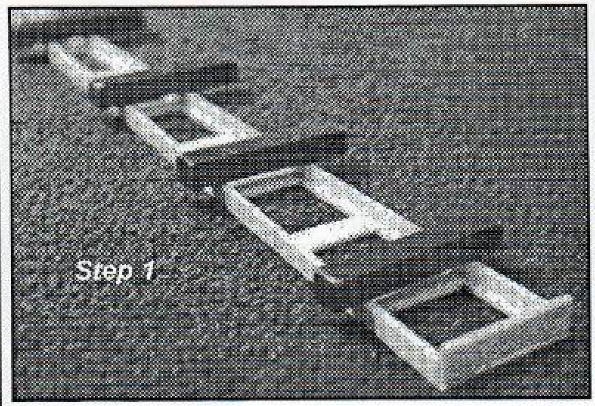


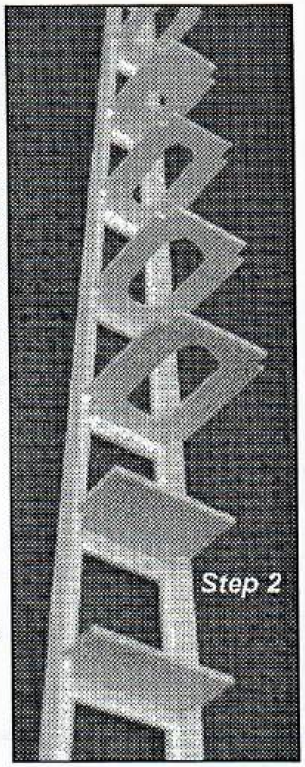


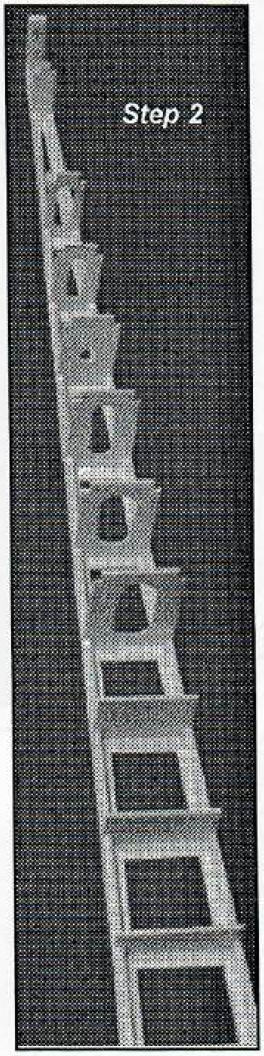
General note: (Photo at right) All mating surfaces need to be perfectly flat. The easiest way to do this is to cut a 100 grit belt sander belt and glue it to a piece of glass. Then move the fuselage part over the flat sanding surface. It works great and you'll find yourself using this tool often.



Rear Main Fuselage





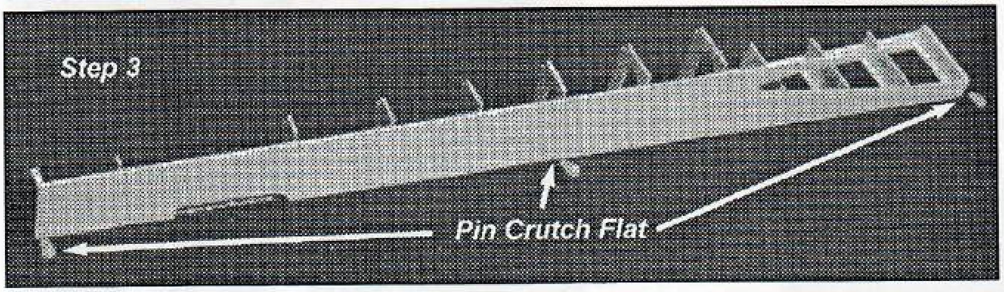


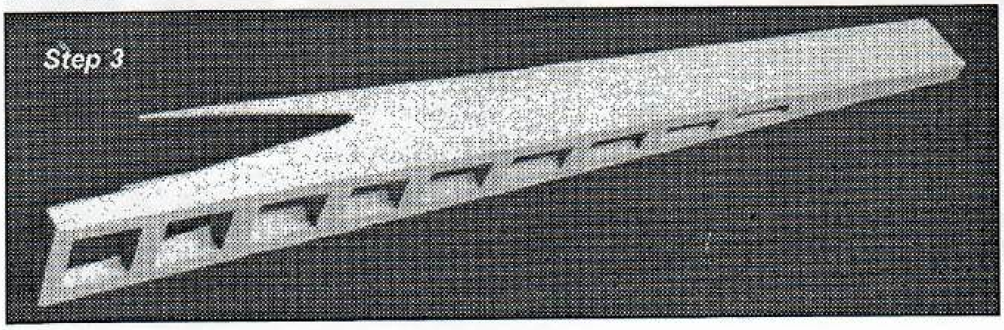
☐ **Step 1 -** On a dead flat surface, fit and glue 1/8" X 1/4" stringers between 1/8" formers F2C and F14C on top of the 1/16" balsa ladder crutch. Make sure the stringers are even with the edges of the crutch all the way down each side.

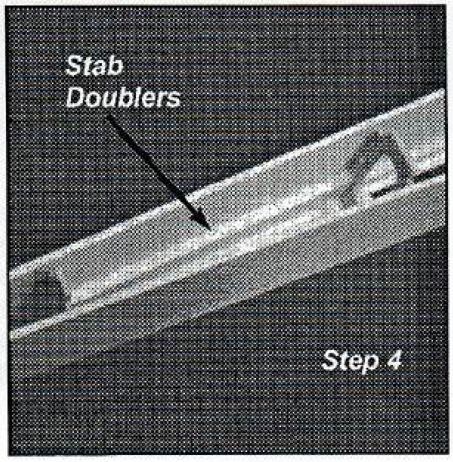
☐ **Step 2 -** Glue in formers F3C to F13C. There is no F12C. Make sure they are verticle. When dry, sand the edges even and flat.

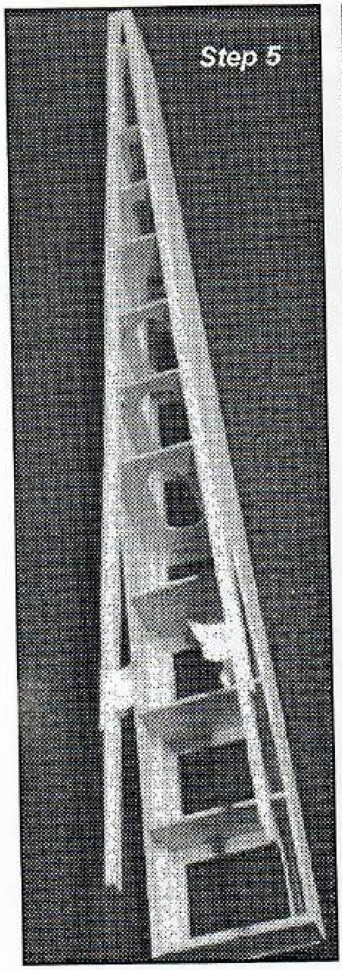
☐ **Step 3 -** Glue on the sides. Make sure the crutch is dead flat. Do this by pinning the side to the stringers at each end and in the center. Then put it on it's side and weight it down until it dries. I glue one side at a time to ensure accuracy. Turn it over and glue on the other side in the same manner. Make sure the crutch is flat.

☐ **Step 4 -** Glue in the 1/8" X 1/4" balsa stab doublers between F11C and F13C.

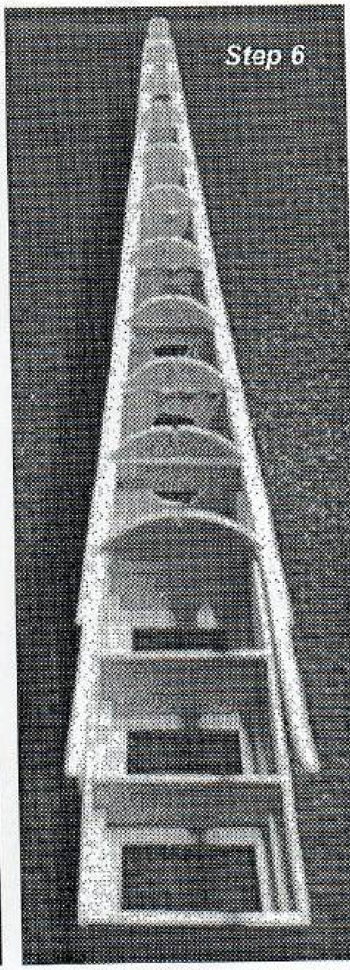






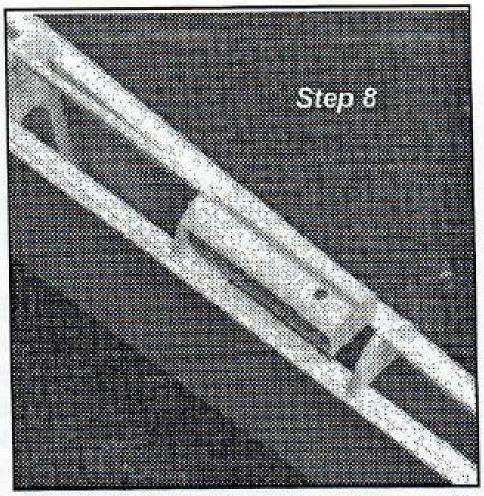


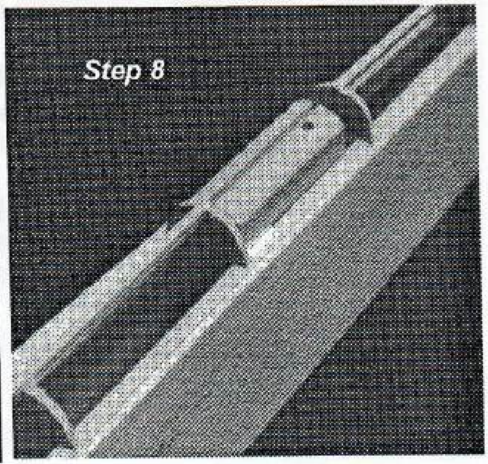
☐ **Step 5** - Glue in bottom stringers. Pin and tape as necessary. Sand Flat.



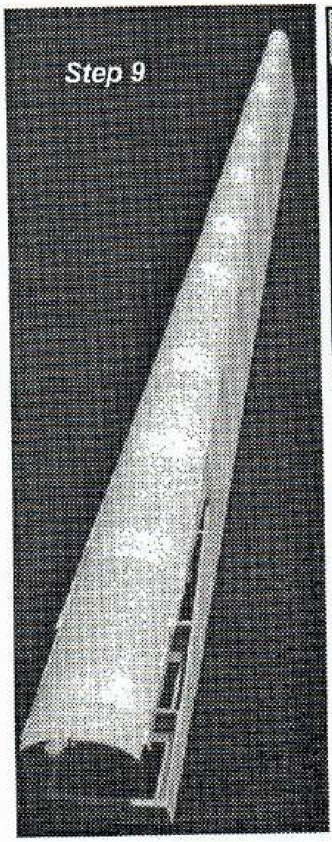
☐ **Step 6** - Glue on formers F5B through F14B. Former F5B is in two pieces.

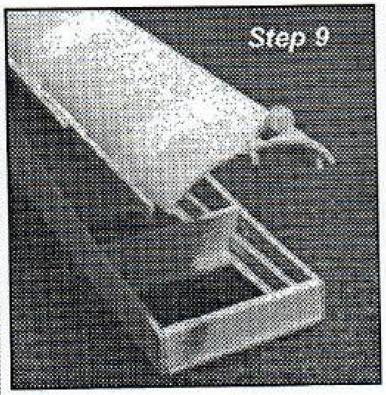
☐ **Step 7** - After formers are dry, glue in center stringer.



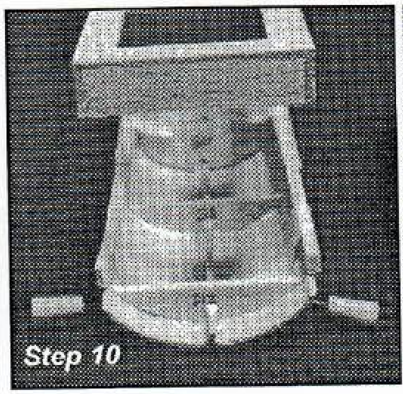


☐ **Step 8** - Glue in tailwheel block behind former F11C. Remove enough of the stringer to fit in the block.

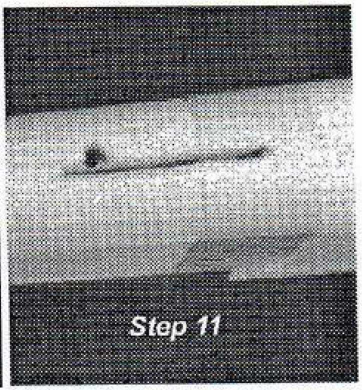




☐ **Step 9** - Fit and glue on the bottom molded skin. File off the sides to match the width of the molded skin. Glue up the stringer, formers and the edges. Tape in place. Pin the center stringer in the middle of the molded skin on the front edge.

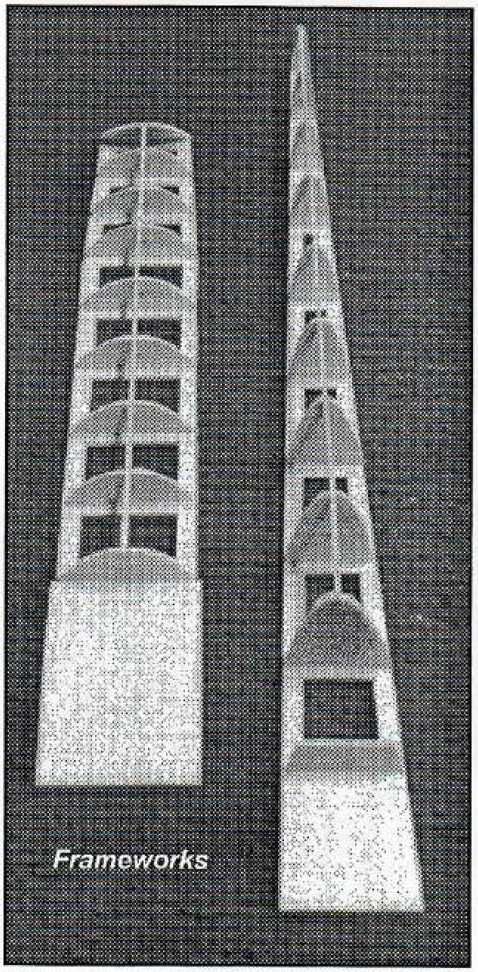


☐ **Step 10** - With a square mark the exact location of former F2B even with F2C. Glue in formers F3B and F4B. When dry cut and sand off the excess sheeting. Do not glue Former F2B to rearmain fuselage as shown in photo. Former F2B is glued to F1B to make a shelf when joining the rear main fuselage to the nose.



☐ **Step 11** - With pin drill, from the top, drill through the blind nut in the tail wheel block. Remove the balsa skin above the tailwheel wire slot in the block and around the blind nut hole.

Nose Block and Turtle Deck



☐ **Step 1 -** Glue formers to ladder crutch.

☐ **Step 2 -**When dry, glue stinger to formers.

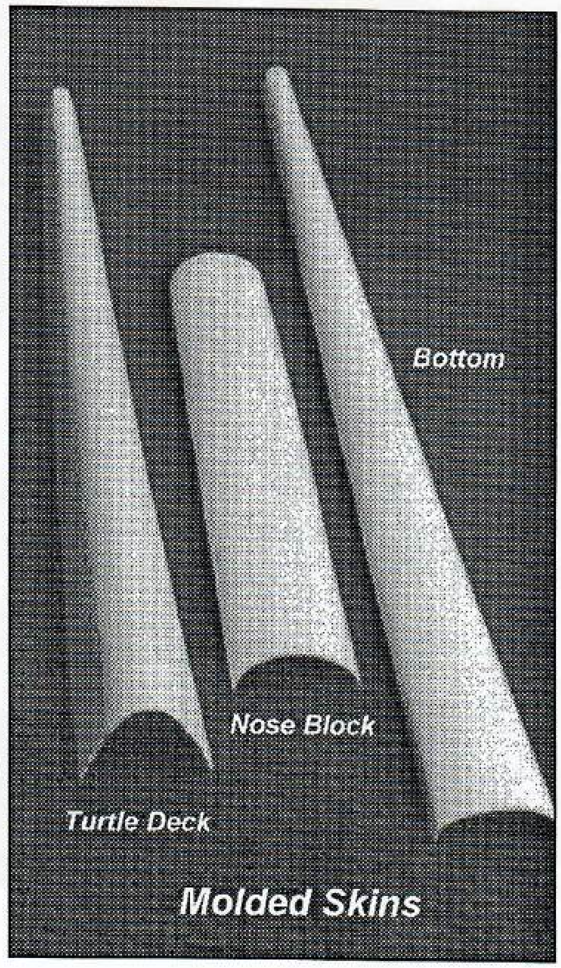
☐ **Step 3 -** Glue molded skins to frames. Center the skins on the frame and tack glue in several places, then finish gluing with Elmer's. Make sure you keep the frame straight. Don't distort it by pressing too hard.

☐ **Step 4 -** Cut and sand off excess sheeting. Sand bottom perfectly flat. See general note on page 2.

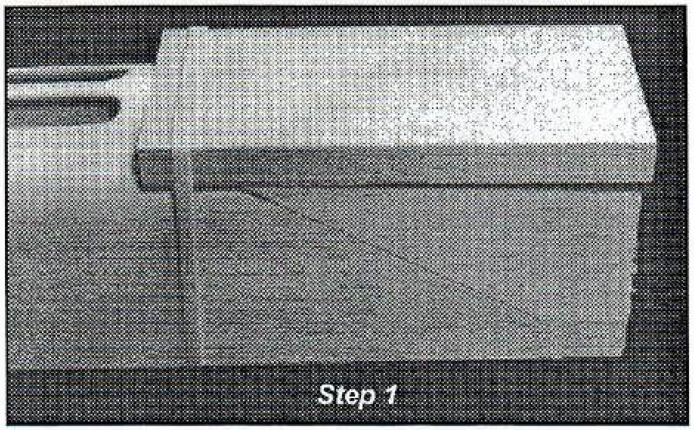
☐ **Step 5 -** Add the three solid 1/4" balsa blocks to the front of the nose block.

☐ **Step 6 -** Shape the solid blocks to conform with the molded skin.

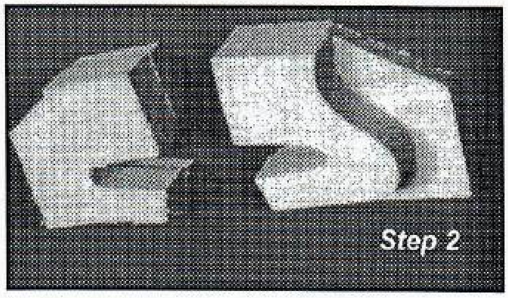
☐ **Step 7 -** Using the canopy plan, remove the excess molded skin around the canopy.



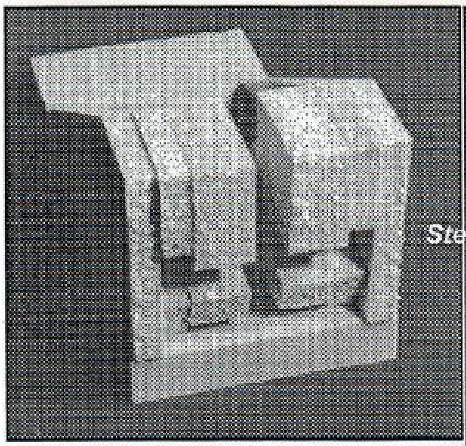
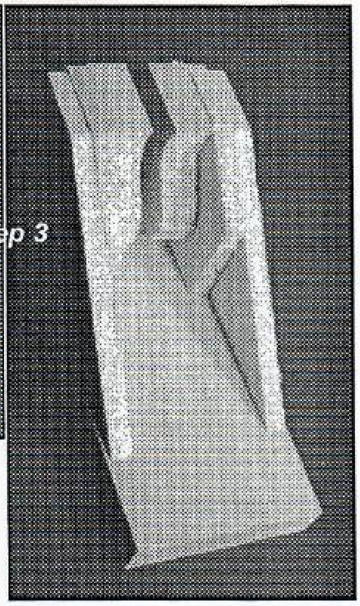
Cowling



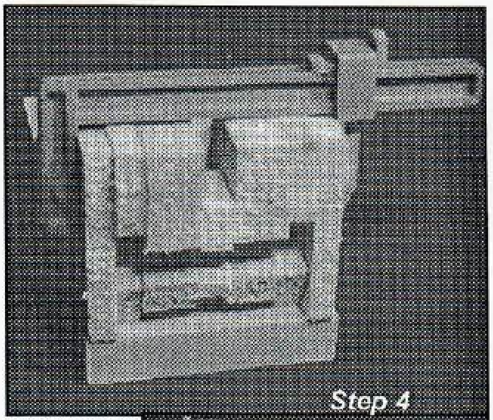
☐ **Step 1** - Rubber band the 1/2" bottom block to the flat portion of the nose. Fit the 1/4" sides to the nose. Glue the 1/4" sides to the bottom block. Keep everything lined up.

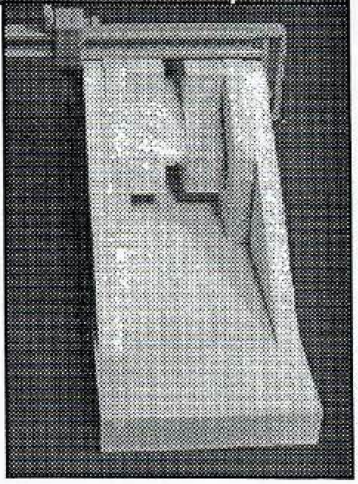


☐ **Step 2** - Glue 1/2" filler blocks to 1/4" filler blocks. Make a left and a right pair. Line up the top and the front.

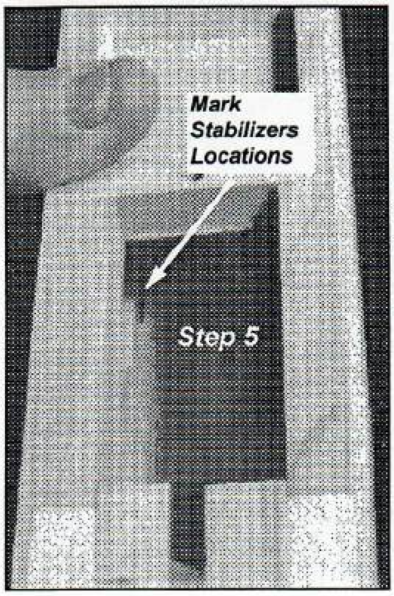



☐ **Step 3** - Glue in the pair of filler blocks keeping the slanted top even with the sides. Don't worry about the keeping the front even for now.

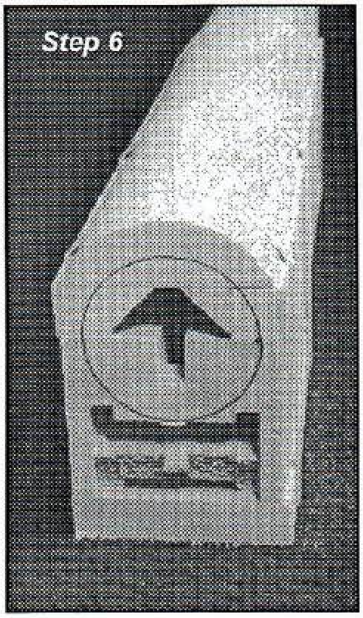




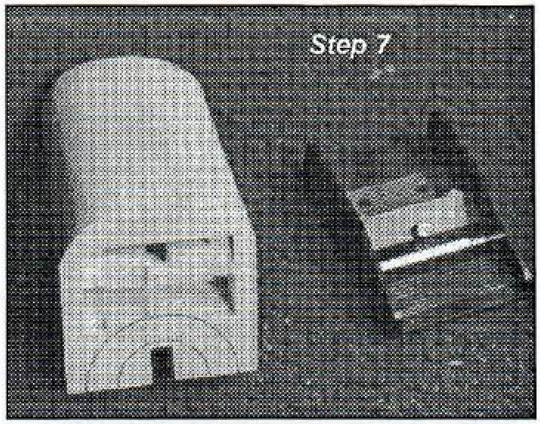
☐ **Step 4** - Cut the 3/8" filler block to fit the remaining space. Glue it in so that it is even with the groove in front that will become the top of the air inlet. Look ahead so you can get the picture.

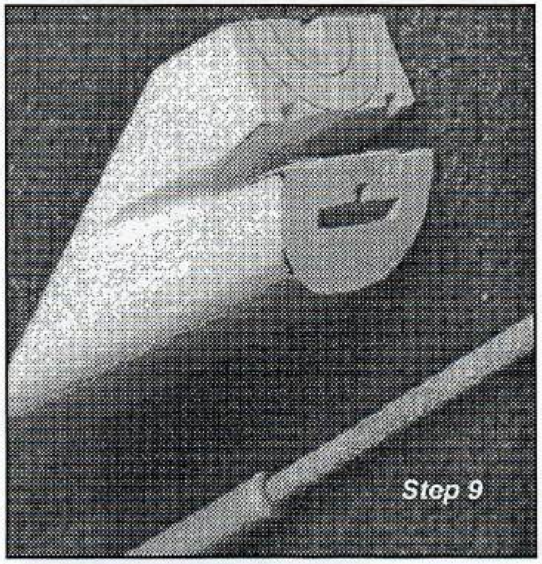


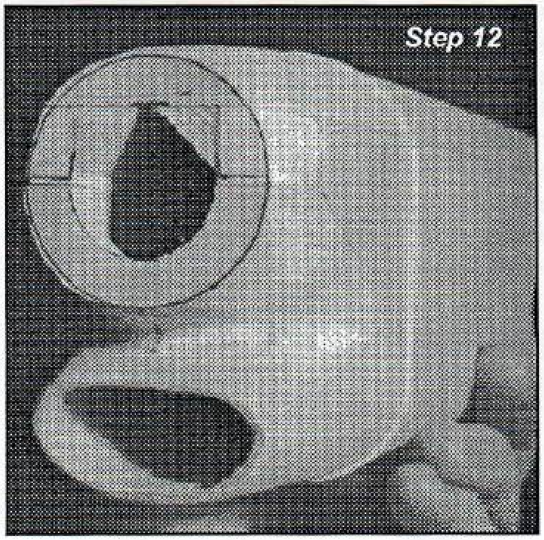
☐ **Step 5** - With the cowling in place, from the top, mark the location of the rear cowling stabilizers. Cut a groove and glue in the stabilizers.



☐ **Step 6** - Rubber band or tape the nose, top block and cowling together. Sand the front flat. I use a disc sander. Be sure you keep things square. Using the nose ring, mark the location of the nose ring. Remember the engine will be seated on 1/16" aluminum pads. The thrust line should be right where the cowling separates from the nose.







☐ **Step 7 -** Shape the outside of the bottom block. I use a Master Airscrew razor plane to shape all blocks.

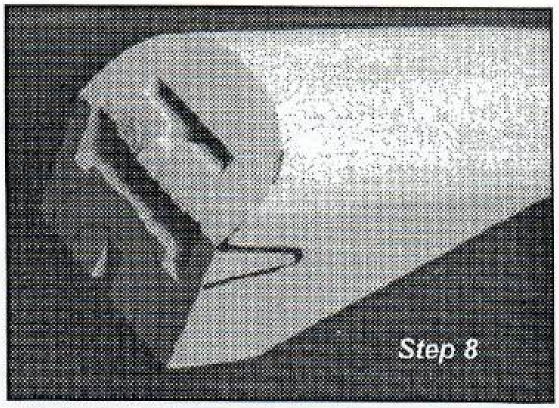
☐ **Step 8 -** Mark the "V" groove between the spinner and the air inlet. Remove that portion of the sides.

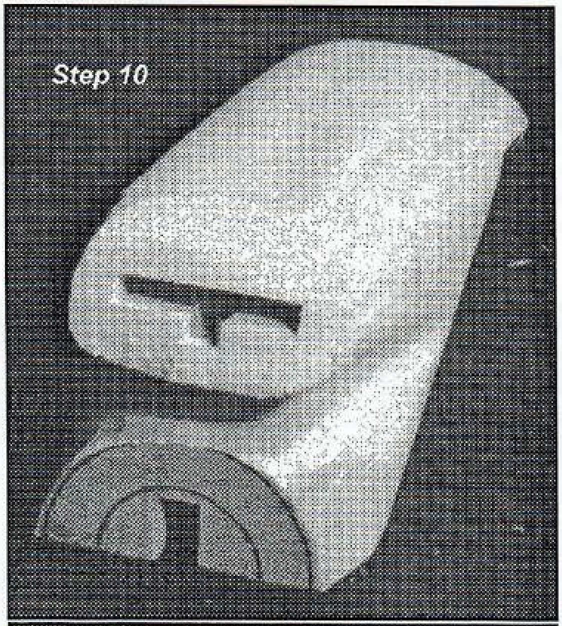
☐ **Step 9 -** With a round Perma Grit file, file the groove deep into the sides and around the front between the spinner and the air inlet.

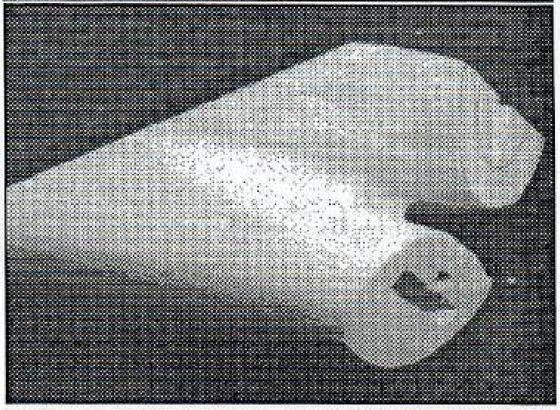
☐ **Step 10 -** Carve material away until you've got a round shape around the spinner and the air inlet. Leave a little excess around the spinner for final fitting after the engine and spinner are mounted.

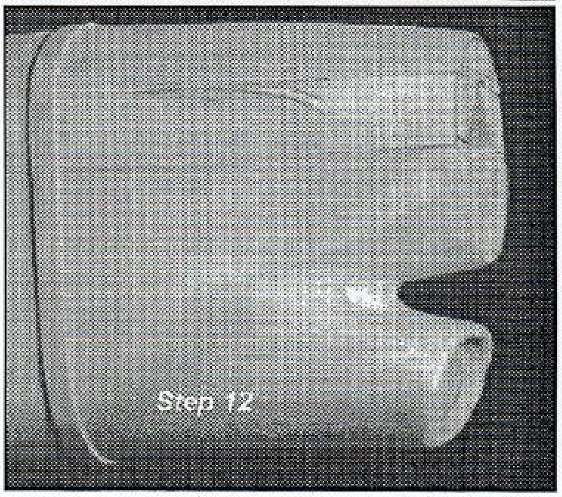
☐ **Step 11 -** Carve the inside of the air inlet and the inside of the cowling. I use a Dremel tool with a carbide ball shaped end. Leave the sides a full 1/4". Don't get too thin in the area where the hold down bolt is located. I leave a balsa "island" around the bolt hole for extra strength.

☐ **Step 12 -** Tape or rubber band the top block, nose and cowling together. Carve the nose block to the spinner shape.

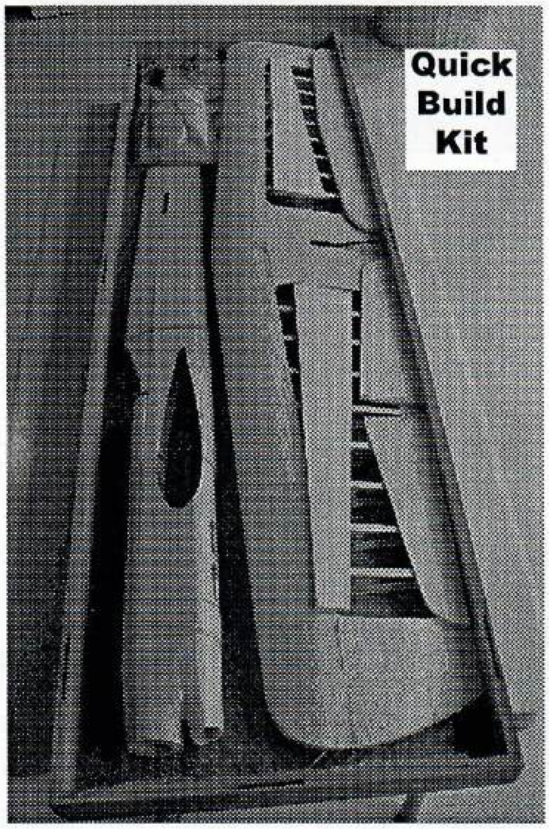




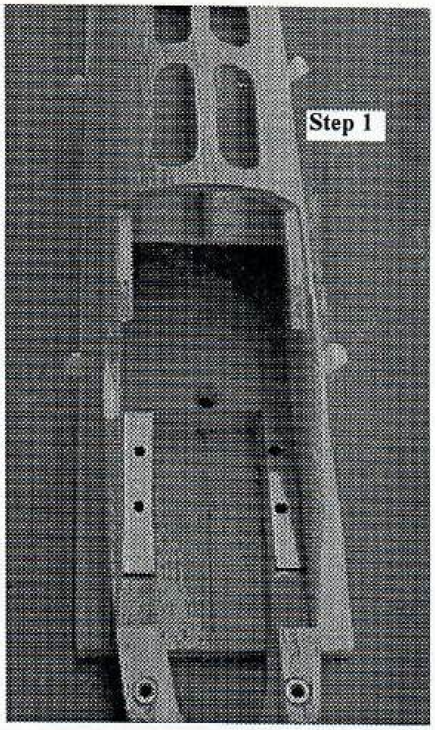




Assembling a Quick Build Kit



Step 1. Mount you engine to the motor mounts. A 3/4" shaft extension is recommended. I also recommend marking and drilling one hole at a time. It takes a little longer, but you can get a perfect fit everytime. I slightly recess my blind nuts into the motor mounts so the fuselage lays flat on the assembly board.

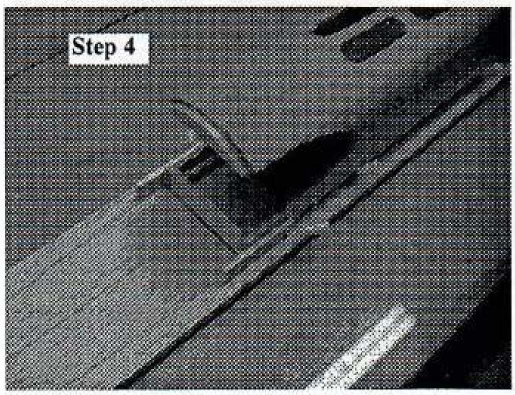


Step 2. Fit your cowling to the engine. Also slightly recess the cowling holedown blind nut.

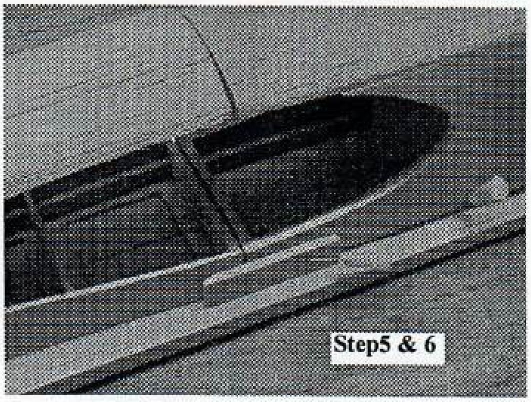
Step 3. Draw a straight line down the center of a hard piece of 1/4" balsa. Mark center lines on the nose and rear main fuselage.

Step 4. Center and Pin the nose to the assembly board.

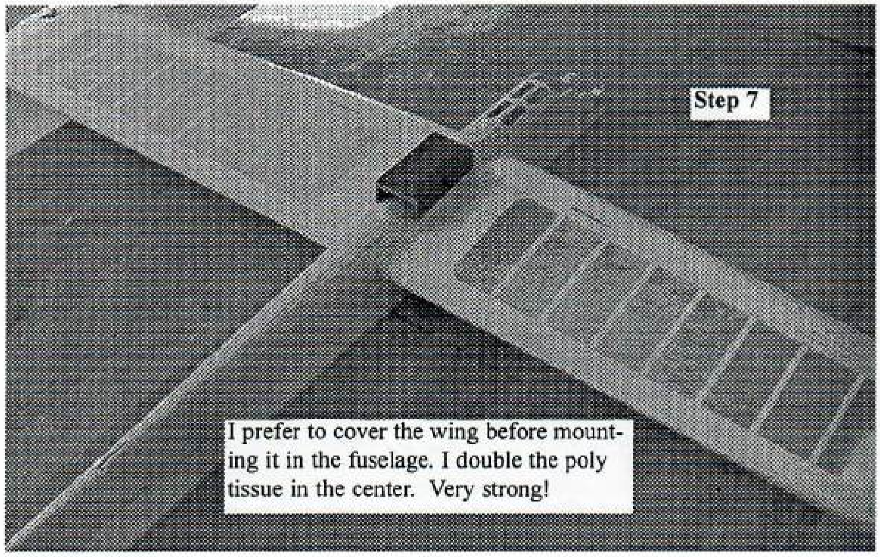
Step 5. Tack glue a couple of 1/8" scrap pieces to each side of the nose to act as guides when the rear main is glued to the wing and nose.



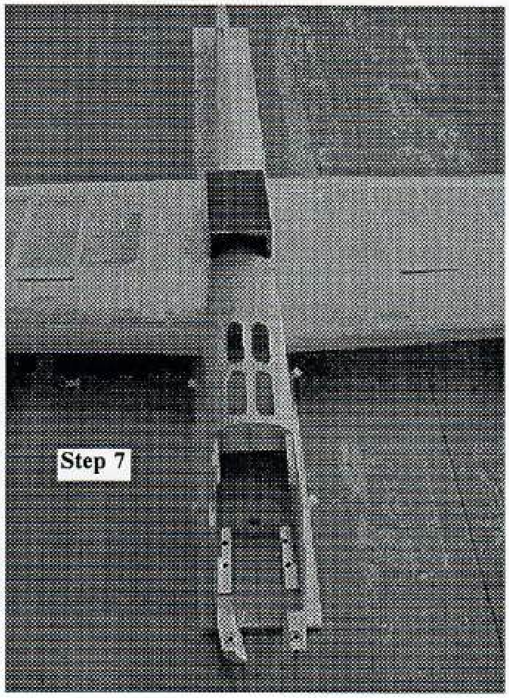
Step 6. Test fit the rear main fuselage to the nose and both parts to the wing. Mark center lines on the fuselage and wing. Make sure your fit is lose enough th line up these marks.

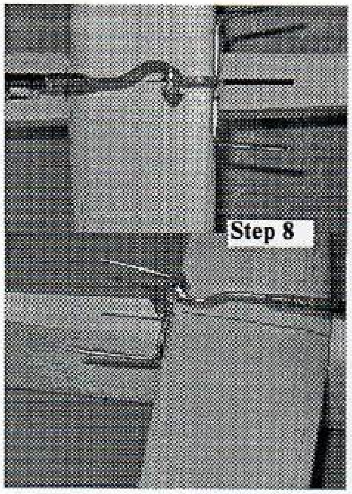


Step 7. Glue the wing into the rear main and nose. Make sure the wing is level in the fuselage and square to the fuselage. I do this by two lines drawn on my workbench at 90 degrees. Don't overdo the amount of glue on the top of the wing. You will have the opportunity to take care of gluing the top when you turn it over.

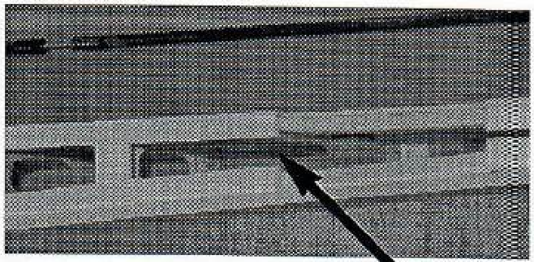


I prefer to cover the wing before mounting it in the fuselage. I double the poly tissue in the center. Very strong!

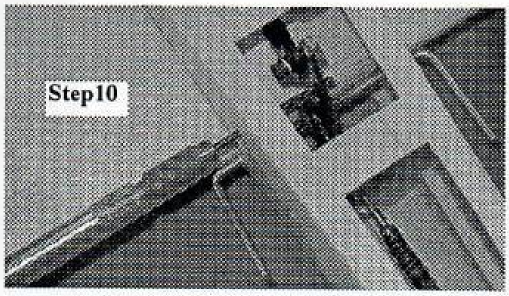




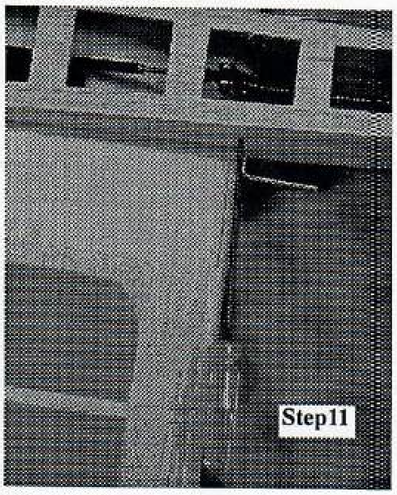
Step 8. Connect the pushrod to the elevator horn. cut a shallow notch in the 3/32" pushrod wire just beyond the brass bushing in the horn. Put on a #3 washer, wrap a few rounds of fine copper wire in the groove and up against the washer, and secure it with a little JB Weld. Mount the horn to the stab.



Step 9. Just in front of the stab slot remove a little of the ladder crutch and the former so that the pushrod can be inserted through the rear fuselage.

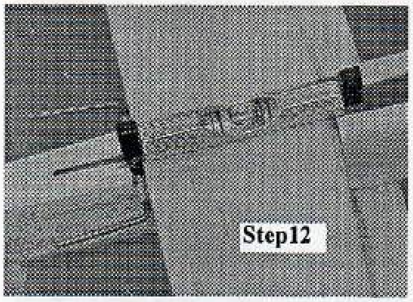


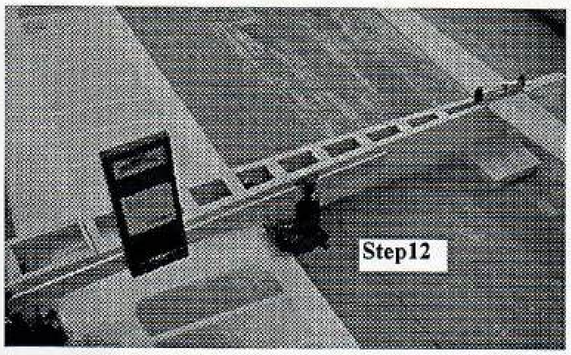
Step10. Drill a small hole into the inboard fuselage side directly opposite the lower hole in the flap horn so that a ball driver can be inserted to secure the pushrod to the flap horn.

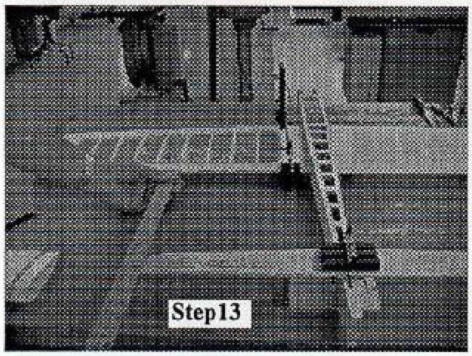


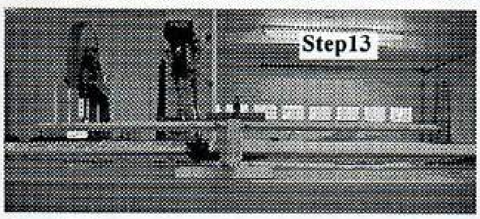
Step11. Connect the pushrod to the flap horn. Hold up the pushrod with needlenose pliers. You could remove part of the ladder crutch and formers if you need more room.

Step12. Glue the horizontal stab to the fuselage. Level the wing using an incidence meter. Mount the stab with just a slight positive incidence using a line level.

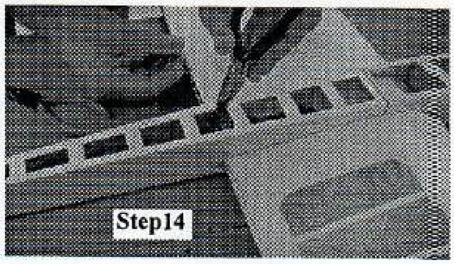




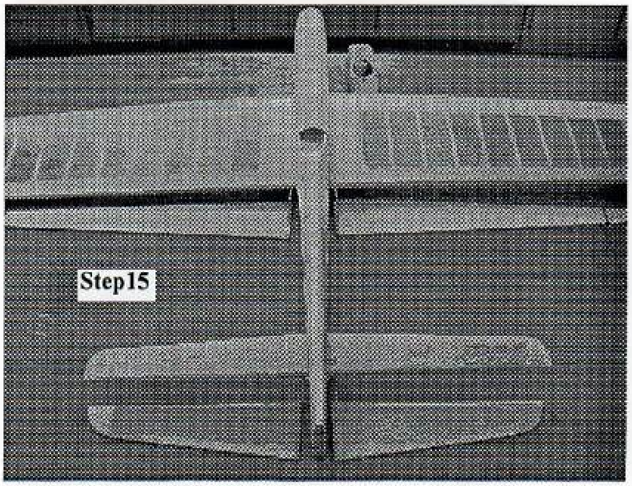




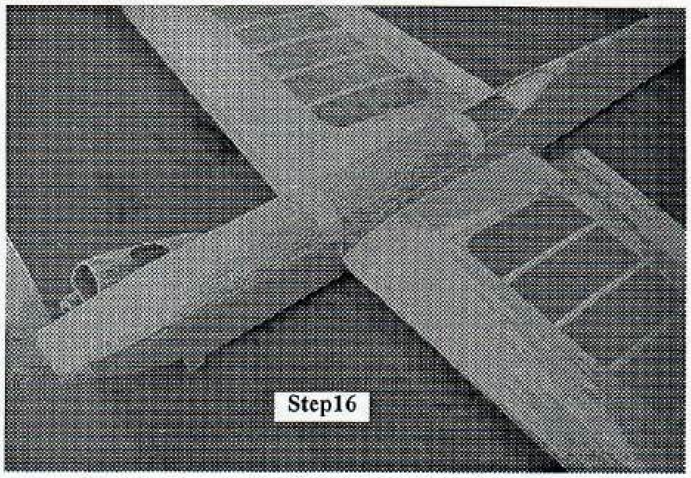
Step13. Insure the stab is exactly parallel to the wing trailing edge and exactly level with the wing.



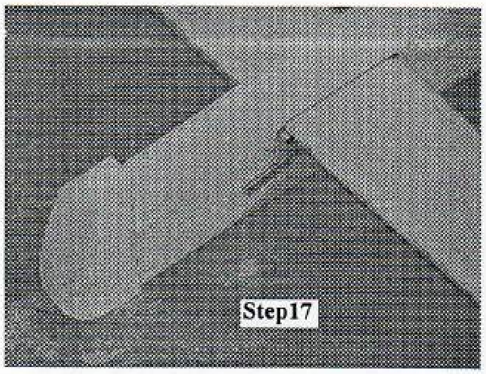
Step14. Adjust the ball link at the flap horn until you get perfect neutral of the flap horn and elevator horn. Use center lines to verify.



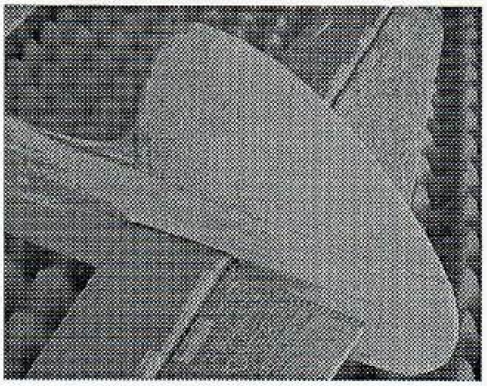
Step15. Cover the wing, stab, elevators, flaps and tabs. Brush on three coats of dope.

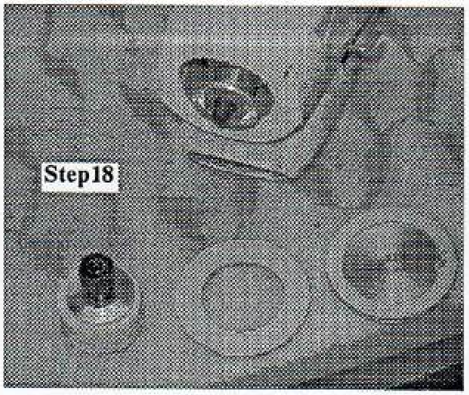


Step16. Glue on the nose top block and the turtle deck.

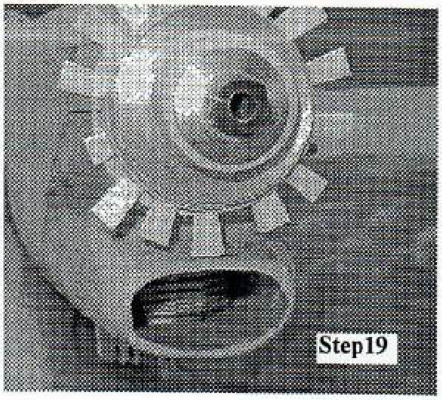


Step17. Glue on and rough shape the rudder. Airfoil the verticle fin on the inboard side. Glue on the verticle fin. Finish shape the rudder to match the verticle fin.



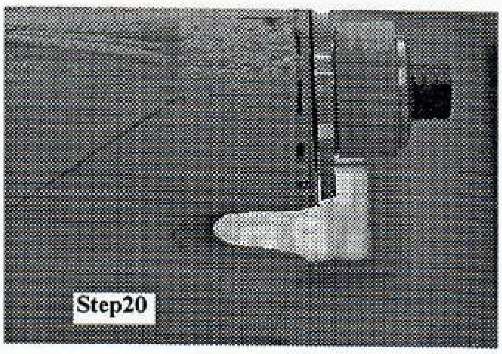


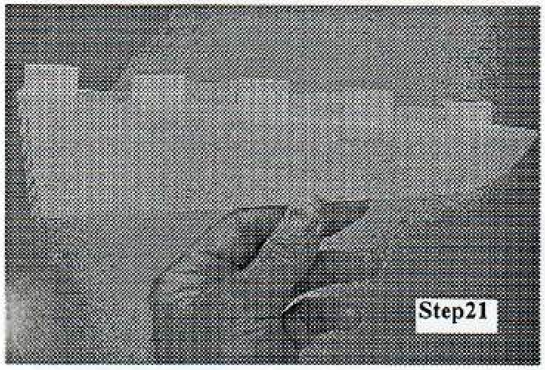
Step18. Mount the engine and shaft extension. Tack glue a 1/32" ply nose ring spacer to the back of the spinner back plate. Place the 1/16" ply nose ring then the spinner back plate on the engine shaft and secure with extension bolt. There should be about a 1/16" gap behind the ply nose ring.



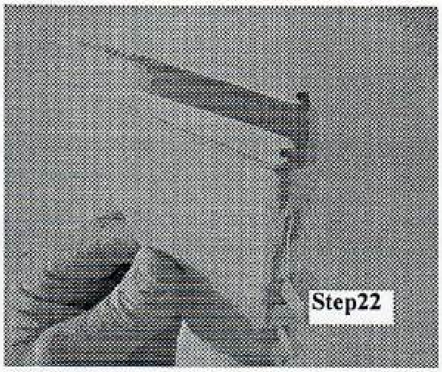
. Glue in small wedges behind the nosering pressing the nosering against the spacer and backplate. Glue the wedges all the way around.

Step20. When dry cut off the excess pieces of the wedges. Remove the back plate and the spacer. Remount the back plate and note the perfect gap between the nose ring and the backplate! Fill in the remaining gaps and sand and shape the nose to the nosering.

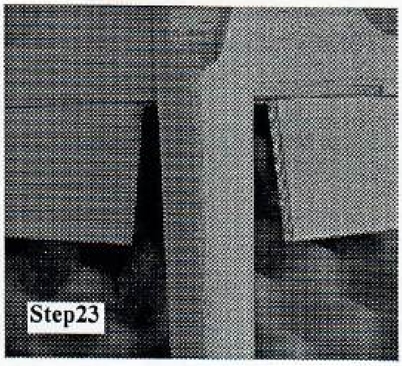




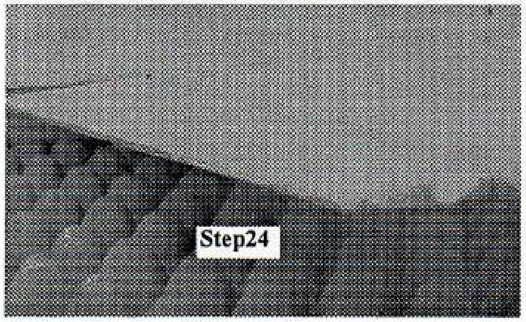
Step21. Using dope brush on the dacron hinges to the elevators, flaps and outboard adjustable tab. One up, one down, shoulder to shoulder.



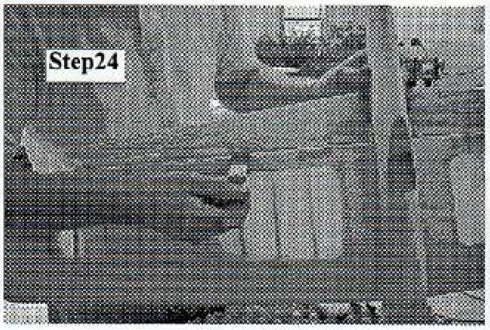
Step22. With a dremel tool carve a notch in the flaps and elevator horn clips for clearance of the horn wire.



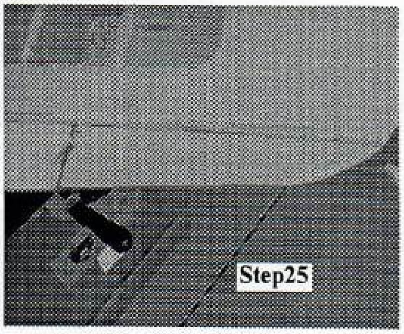
Step23. Slip the flaps and elevators over the horn wire and pin in place. Make sure they are centered on the fuselage.



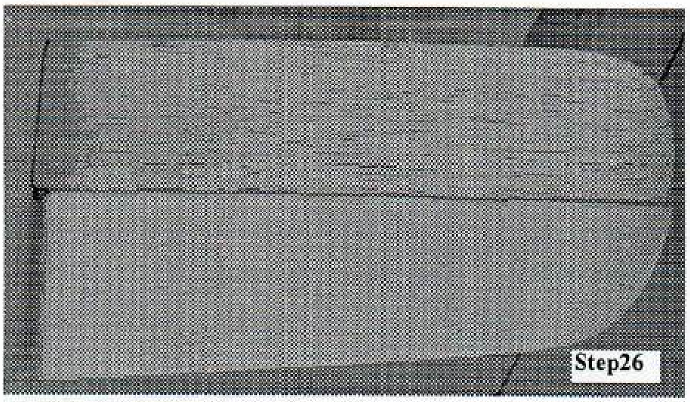
Step24. Pin in place at the tip and make sure the hinges are alternating up and down.



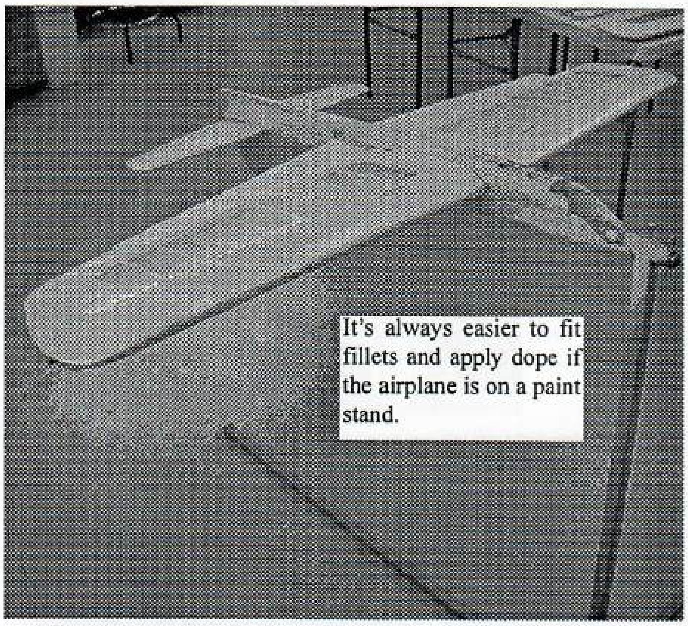
Step24. Looking down at the trailing edge dope down the hinges tot he trailing edge of the wing. Make sure the flap is exactly centered on the trailing edge of the wing.

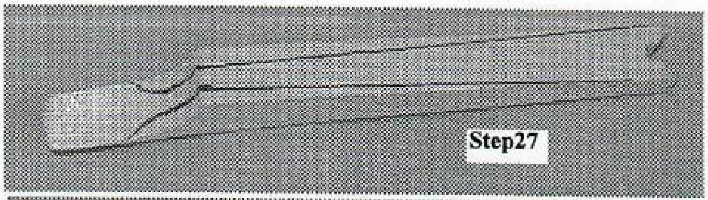


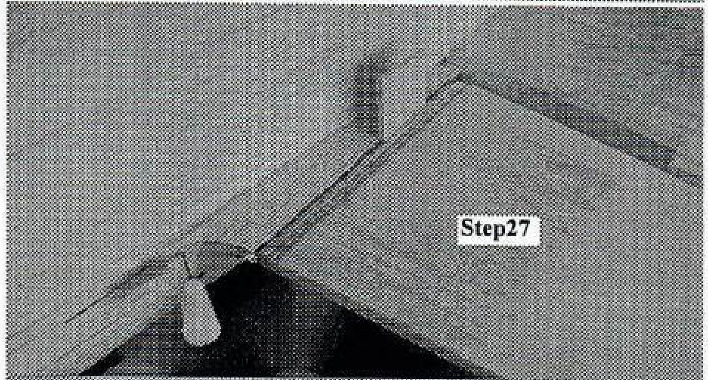
Step25. Mount the outboard adjustable tab. Note: The hinges are heat shrinkable. So, if you get a little wrinkle in some of them just apply a little heat and they will tighten right up.

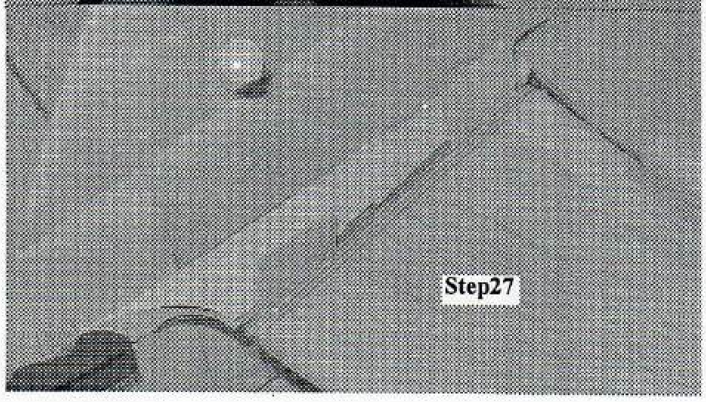


Step26. The final product. Now is the time to even up and shape the tips to match perfectly.

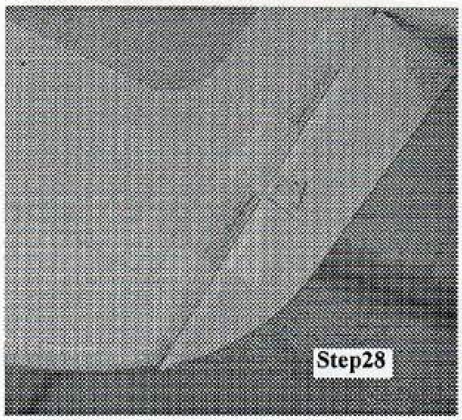




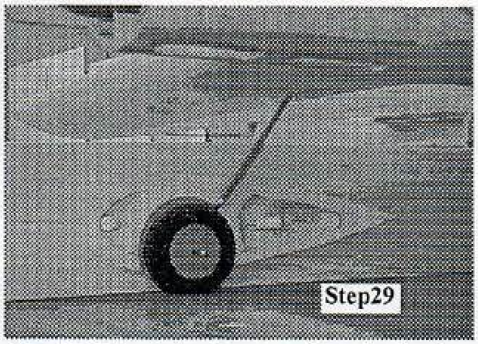




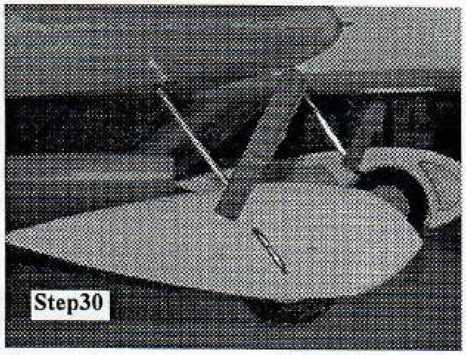
Step27. Shape the fillets. I find it helpful to draw lines on the fillet to where I want to shape it. A 3/8" round file does a nice job of shaping fillets. Fit the fillets to the flap and to the elevators. Note the fillets are held in place by small wedges in the gaps.



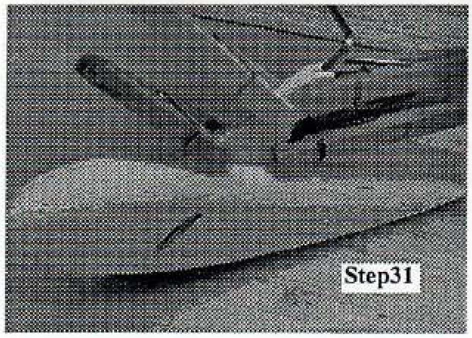
Step28. The adjustable tab is hinged just like the flaps and elevatord.



Step29. Now it is time to adjust the landing gear and fit the wheel pants to the landing gear. With the gear all lined up level the airplane on the work bench. I find it very helpful to pin the wheelpant to a scrap balsa board to accurately mark where the wire crosses the pant.



Step30. Cut grooves for the wire and epoxy in place. The wheel pants should be level with the wing and each other and be pointing straight ahead. Note the groove is cut along the top of the pant as well a through the side.



Step31. After the epoxy has dried glue on the inboard half of the wheelpant. Make sure the wheel is turning freely. Add washers aas spacers oneach side of the wheel. Let the axel stick through the inboard pant half. Cut off the excess later. Mark, groove and glue on the skirts.

Step32. Apply fillets around the fuse/wing joint, the fuse/stab joint and the wheelskirt/wheelpant joint.

Step33. Complete your cockpit detail and mount the canopy.

Step34. Apply your favorite paint finish.

Assembling Your Ball Link Control System.

Your ball link control system has been carefully engineered. It offers many advantages over conventional control systems:

1) It allows free movement with a minimum of play

2) It combines great strength with light weight.

3) It provides easy assembly (no solder joints are required).

4) It's easy to adjust.

Making the Pushrods

1) With the bellcrank, flap horn, and elevator horn all positioned at neutral measure the distance between the holes where each pushrod will mount. For the flap pushrod measure the distance between the hole in the bellcrank and the hole in the flap horn. For the elevator pushrod measure the distance between the hole in the flap horn and the hole in the elevator horn. These are very important measurements. Be sure everything is at neutral and double check your measurements.

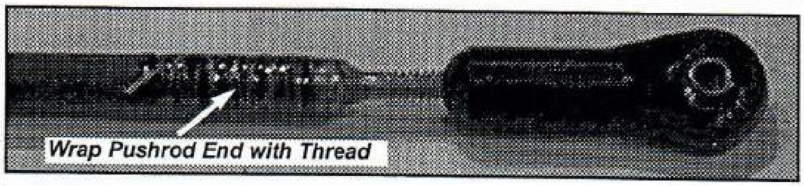
2) The measured distance will be the finished length of the pushrod assembly (carbon-fiber tube, threaded ends, and ball links). Determine the length of each carbon-fiber tube by subtracting 2 5/8 (2.625) inches from the measured distance. (This will provide a pushrod assembly with nominal thread engagement of 1/2-inch in each ball link and 1/8-inch of adjustment in either direction for each ball link). See Photo below.

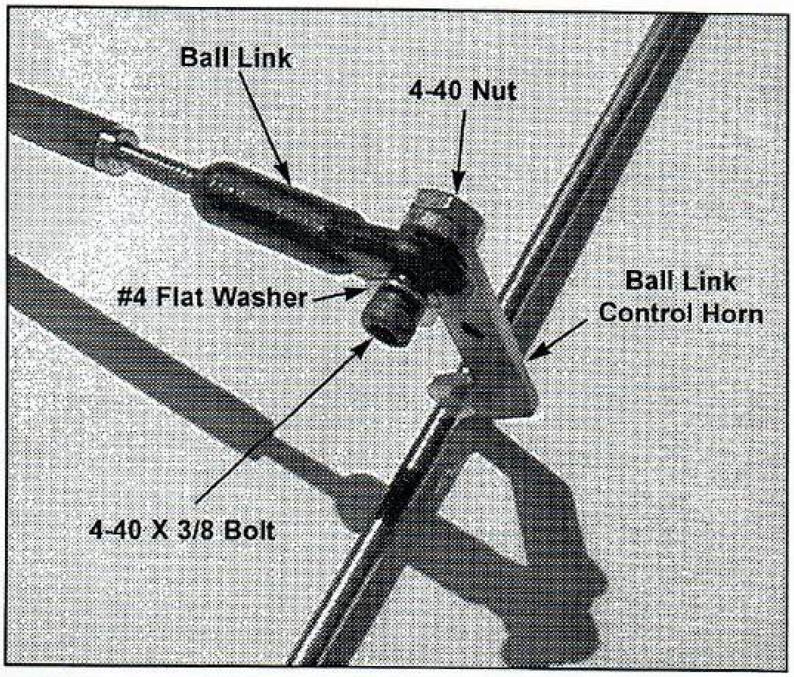
3) Cut the carbon-fiber tubes (a razor saw works well for this). Be careful not to split the ends. Make the saw cut slightly long and then sand the tube to make the ends square and exactly the right length. (Sanding the ends square makes a tight fit for the threaded ends).

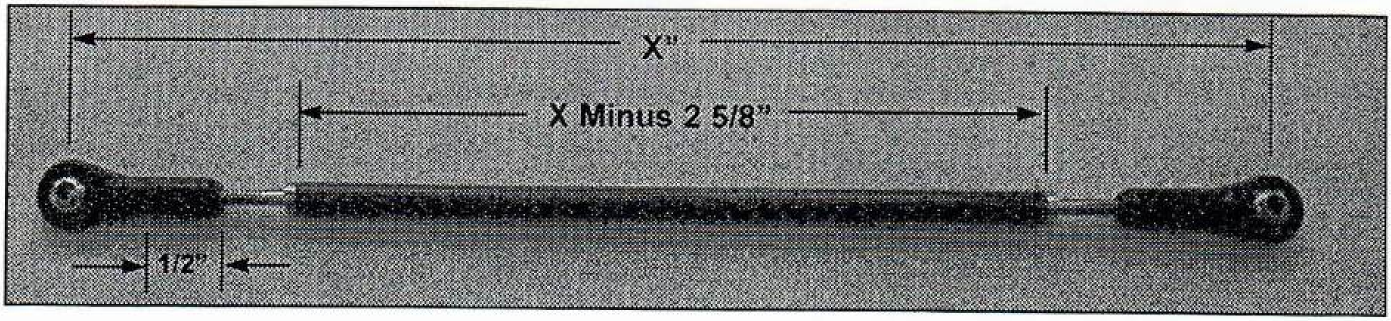
4) Prepare the threaded ends and carbon-fiber tubes for gluing by cleaning them carefully with acetone, lacquer thinner or alcohol.

5) Use J.B.Weld epoxy (available at most hardware stores) to glue the threaded ends into the carbon-fiber tubes. Don't use JB Weld Quick. Wipe off any excess J. B. Weld. With epoxy on them, the threaded ends act like little pistons in a cylinder. When you push them into the carbon fiber rod, air pressure wants to push them back out. Place the pushrods between building weights to keep the ends in place while the J.B Weld dries (24 hours).

6) Clean and lightly sand the ends of the pushrods around the joint with the threaded inserts. Apply a light coat of epoxy around the joint and about 3/4" of the pushrod end. Wrap strong thread around the insert and the pushrod to insure the pushrod won't split and to give an external lock on the insert (see photo).







Assembling the Control System

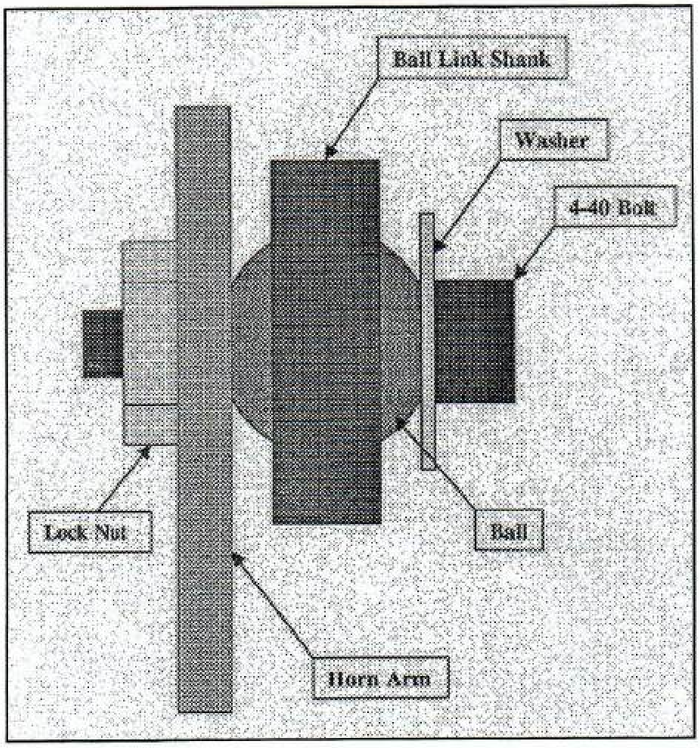
1) Thread the ball links provided (Rocket City #87 or #57) onto the pushrod ends so that 1/2-inch of pushrod threads is inserted into each ball link. (The distance between the ball link centers should now be the distance you measured for the pushrod). Each ball link can be adjusted 1/8-inch (five full turns) in either direction while maintaining a minimum of 3/8-inch thread insertion. Note that no thread locker (either a lock nut or Locktite) is required since the threads in the barrel of the ball link are very tight.

2) Assemble the pushrods to the bellcrank and horn arms in accordance with the drawing below. Check for proper alignment of the system. With the flap horn at neutral both the bellcrank and elevator horn should be at neutral. (Tip — this is easier to see if you slide a straight 12-inch piece of brass tubing over the horn arms). Adjust the ball links until proper alignment is achieved (disassembly is required to turn the ball links). Remember to maintain at least 3/8-inch thread insertion in each ball link.

3) When the pushrods have been adjusted to the proper length you're ready for final assembly:

4) Mounting ball link to horn.

a) First put the washer and ball link onto the 4-40 bolt. Thread the bolt into the threads in the horn arm. Put a drop of Locktite 271 (red) on the threads of the bolt where it comes through the horn arm. Work the bolt back and forth until Locktite is in the threads of the horn. Tighten the 4-40 bolt and wipe off any excess Locktite. Make sure that no Locktite is on the ball part of the ball link. Don't go crazy and over-tighten the bolt — there is only 1/16-inch of threads in the horn arm..



b) Finally, install the lock nut. This can be a plain 4-40 hex nut. If you want extra security use a nylon-insert lock nut. In either case use a drop of Locktite 271 and work it into the threads. Tighten the nut with a wrench and you're done. Again wipe off any excess Locktite. After the Locktite sets up (1/2-hour, 24 hours for full strength) you will have an assembly that will never come loose. It can be disassembled, but only with great difficulty.

5) Mounting ball link to bellcrank.

a) Ball links are normally mounted to the bellcrank arm the same as to the horn arm (ball directly against the arm). However, if your flap pushrod makes a steep angle with the horizontal, you may need a conical standoff between the ball and the bellcrank to provide clearance. Don't use a standoff unless it's needed to prevent the shank of the ball link from hitting the bellcrank arm.

b) If your bellcrank is threaded, a plain 4-40 nut can be used as a lock nut. If it is not threaded, a nylon-insert lock nut is recommended. In either case use Locktite 271 as was done on the horn arms.

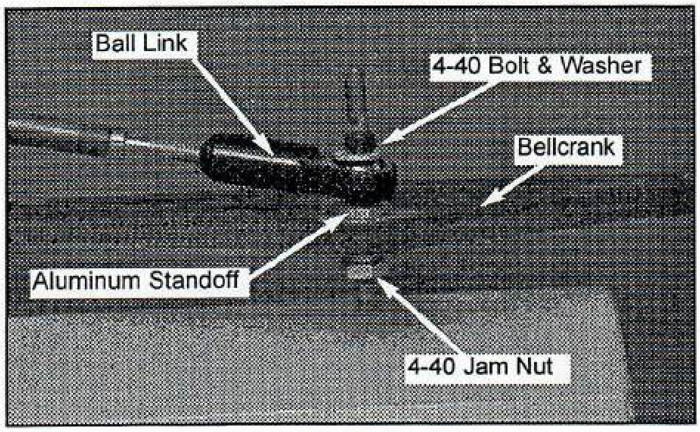
Notes

1) It's important to use J.B.Weld epoxy to assemble the pushrods. Substitutes are not recommended.

2) It's also important to use Loctite 271 threadlocker when assembling your control system. This ensures that your system will never come apart.

3) Don't forget the washer when mounting your ball links. This ensures that the shank cannot jump off the ball under load.

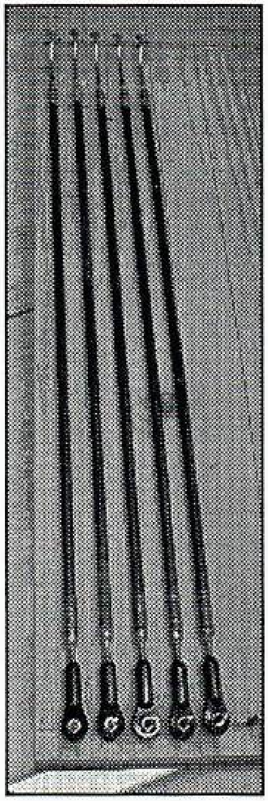
4) If extra space is needed (like in a narrow tail section) the width of the assembly can be reduced by using a button-head cap screw instead of a regular socket-head cap screw.



Assembling Combination Pushrods

Ball links have many advantages. However, they do require a space about 1/2 inch wide. Some airplanes are very narrow in the vicinity of the elevator horn. So, a traditional 3/32 inch wire into a bushed elevator horn is approriate. To keep the adjustability we use a ball link at the flap horn and a 3/32" wire at the elevator horn. Thus we have a combination pushrod. Today, hundreds of these pushrods are operating succesfully in all types of control line stunt models.

As with an all ball link pushrod you cut the 3/16" carbon fiber pushrod about 2 1/2 inches shorter than the horn hole-to-hole measurement. Clean the inside of the pushrod with acetone or dope thinner to remove any release agent that might remain. Clean the 4-40 pusrod insert and clean the 3/32" wire insert with sandpaper.



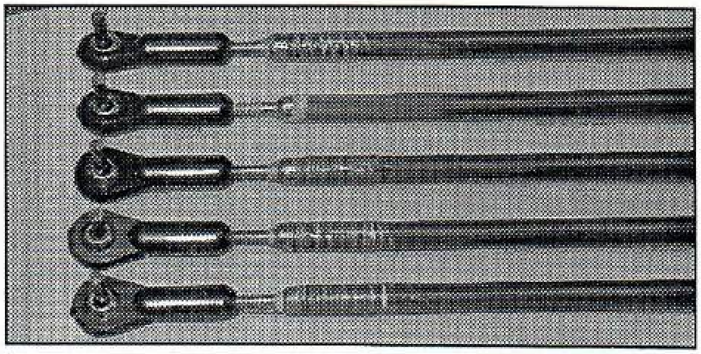
The easiest way to assemble a combination pushrod is to use a piece of board slight ly longer than the pushrod. Drive a nail into one end for the ball link to go over. Drill a 7/64" hole the correct hole-to-hole distance away from the nail for the 3/32" wire insert to go into. this will hold the correct length while the JB Weld cures.

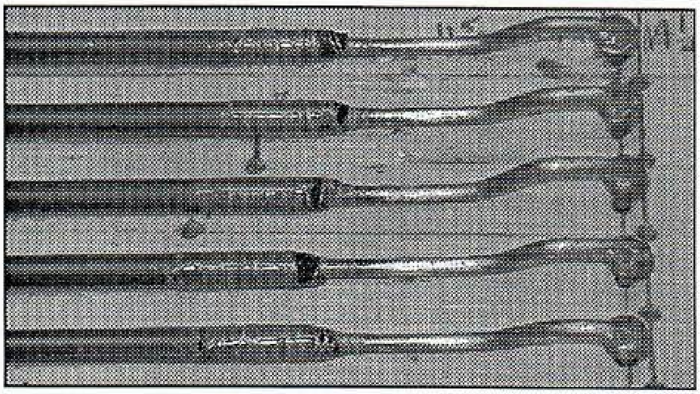
To give the JB Weld something to get a good grip on, with a dremel tool and cut off disk, rough up the part of the 3/32" wire insert that will be inside the pushrod. Coat the wire with JB Weld and insert in into the pushrod rotating it as you do.

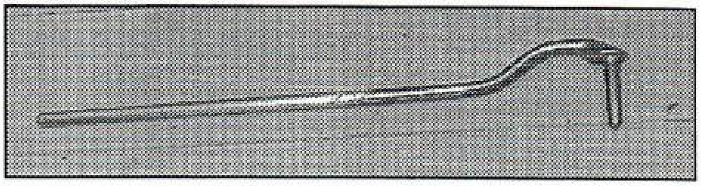
Thread the ball link onto the 4-40 pushrod insert until there is about 1/4" of thread showing. Put JB Weld on the insert and twist it into the pushrod. If it is a little tight just tap in in with a hammer.

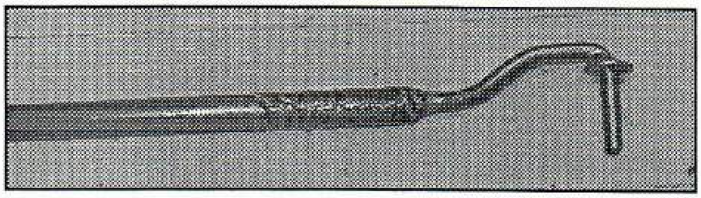
Wipe off any excess JB Weld and place the assembled pushrod onto the jig. Let it cure undisturbed for eight hours.

Next prepare the ends of the pushrod for outside wrapping. With a wire wheel or sand paper rough up about one inch of the ends of the pushrod. I actually bevel the end of the pushrod on the wire end. Mix up some epoxy, spread it on the ends about on inch, and wrap it with a strong thread. Hold the ends of the thread with masking tape. Wrap the thread down onto the 4-40 insert and about 1/4" onto the wire insert. Smoothe the epoxy with your fingers. Let it cure. Cut off the excess thread.









The purpose of the wrapping is to prevent the pushrod from splitting and to give an additional grip, an external lock, on the 4-40 insert and the wire insert.

If you follow these instructions exactly you will never have to worry about this pushrod failing. These procedures has been thoroughly tested.

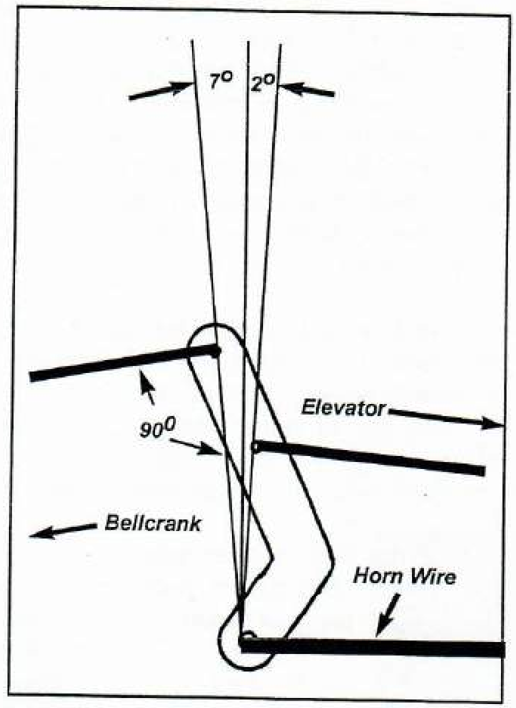
If you have any questions or suggestions:

give me a call at (256) 820-1983 or 6970

or email me at ctmorris@cableone.net

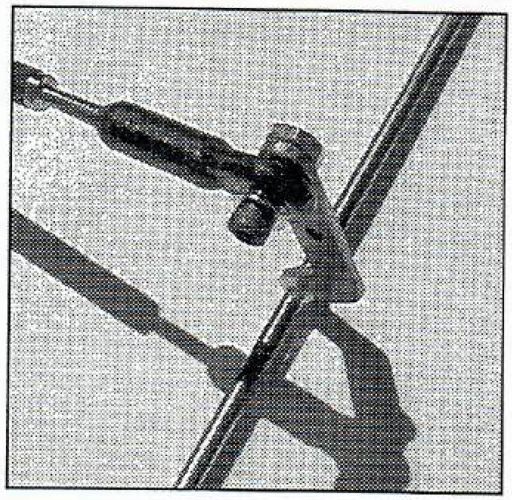
or check out my website at tomsbuildingservice.biz

"True 90" Ball Link Flap Horn



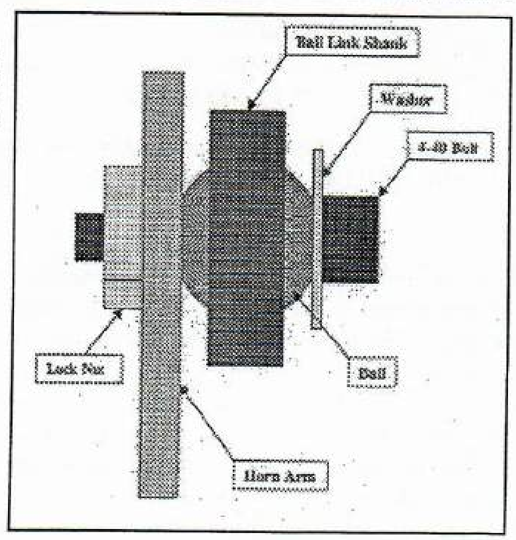
1. Thank you for purchasing the "True 90", Trailing Edge Saving, Ball Link Flap Horn. When properly installed, this horn will deflect the flaps the same number of degrees in each direction and at the same speed. The object is to start out with the flaps in neutral and both pushrods at 90 degrees relative to the pivot point. In other words, each ball link is located in the middle of its arch. The bellcrank pushrod ball link is seven degrees forward and the elevator pushrod ball link is two degrees to the rear of verticle. Of course the geometry is a little different for each airplane, but this configuration was derived at by looking at 20 plans and taking the average. With slight adjustments it will eliminate all or most of the bias in most planes. For the vast majority of planes it's much better than having the bushings directly on top of each other.

2. Another nice feature of this horn is that it is bent away from the trailing edge at 35 degrees thus eliminating the need to cut a deep notch in the trailing edge of the wing to clear the verticle when the flap is full up. Only a small notch is necessary thus saving the strength of the trailing edge.



3. The horn wire is music wire. The horn verticle piece is 4130 aircraft steel, .063" thick. Both ball link holes are tapped 4-40. The horns are brazed using an oxygen acetylene torch bringing the metal to 2000 degrees, cherry red. A 1/16" flux coated bronze brazing rod is flowed into the joint. Twenty-five seconds after the red goes out, the joint is oil quenched. This slightly softens the music wire keeping it strong, but making it tweekable.

4. Bend the horn wire using one of the commercial 3/32" music wire benders such as Harry Higley's or for 1/8" horns the K&S bender works great. Don't bend the horn wire in a vise or with ordinary pliers. Also remember there is a front and a back to this horn. Bend the horn wire towards the elevator!

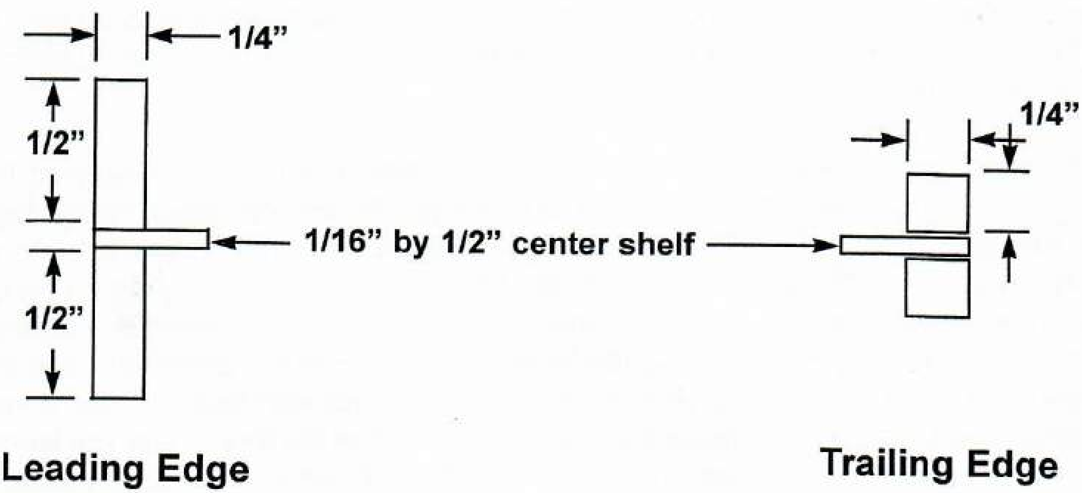


5. Attach the ball link to the control horn with a 4-40 X 3/8" bolt with a #4 flat washer under the head of the bolt. Screw the bolt into the threaded hole in the horn. Put a drop of locktite on the remaining threads and put on a 4-40 lock nut.

6. If you have any questions or suggestions or need help in any way, give me a call (256) 820-6970
or email me at ~~tom_morris@prodigy.net~~
CTMORRIS@CABLEONE.NET

"Lincoln Log" Wing Jig Block Instructions

The openings in the leading edge jig blocks are about 1 1/16" by 1/4". The openings in the trailing edge blocks are about 9/16" by 1/4". In other words, the leading edge should be made with the outside pieces slightly over 1/2" and outside pieces of the trailing edge should be slightly over 1/4". Always make a trial piece from scrap to get a good press fit before you cut the final pieces for the leading and trailing edges.



Spars in the **"Lincoln Log"** wing are 1/8" by 1/4". Cross Bracing is 1/16" by 1/4". See **"Lincoln Log"** wing articles in *Stunt News*: May/Jun '97, pages 54-61; Sep/Oct '98, page 120; Nov/Dec '98, pages 127-128; Jan/Feb '99, pages 126-129; Mar/Apr '99, pages 54-57: Jan/Feb '00. pages 48-49.

Spars in the **"New Millennium"** wing are 1/16" "D" tube sheeted spars. The height is determined by the thickness of the root and tip ribs. Spars can also be built up using 1/16" by 3/8" strips top and bottom and 1/16" by 1/4" cross bracing. See **"New Millennium"** wing articles in *Stunt News*, Jan/Feb '00, pages 50-51.

If you need help call me at 256-820-1983 or 6970 or e-mail me at tom_morris@ prodigy.net.

Instructions for Polyester Tissue

Polyester tissue is a modern replacement for silkspan. It is much stonger than silkspan, but weighs no more than silkspan. It is best used on flat surfaces such as wings, flaps, stabs and elevators. Silkspan is still the best choice for curved surfaces such as wing tips and fuselages.

Polyester tissue is applied similar to silkspan, except it is applied dry and then heat shrunk. Prepare the surface to be covered by applying three coats of dope, lightly sanding between each coat. I prefer using nitrate dope up through the filler/primer coats, but butyrate will work. **Please note that polyester tissue has a definite outside and inside.** The outside is marked with a colored dot. If you put it on with the inside out it will fuzz up when you apply the dope. It will then take several extra coat and sandings to get the fuzz out.

Covering a wing: I recommend you cover the wing with four pieces of tissue overlapping the middle sheeting so that you have two layers of tissue on both sides. Since the tissue has a high tensile strength, this double layer in the middle gives the wing tremendous strength. Don't try to cover the wing tips with poly tissue. Use silkspan for covering the tips. Cover the bottom inboard first, then the bottom outboard, then the top inboard, and finally the top outboard. Lay the polyester tissue over the surface you are covering and apply dope along the trailing edge. Keep the tissue taut and straight as you apply the dope, but don't stretch or distort it. It will lay very flat with few wrinkles if you keep it straight. Next apply dope to the center sheeting area; then out at the tip; then finally the leading edge. I start in the middle of the leading edge and and work outwards. Don't worry if you have a few small wrinkles. They will disappear when you heat shrink it. Cut off the excess tissue. You will overlap the front and back edges with the top pieces, so you can cut off the bottom pieces just short of the leading and trailing edges.

Since you are now going to overlap the center sheeting and the poly tissue you've already applied, you need to lightly heatshrink the tissue in that area so the second layer lays flat. Dope on the bottom outboard just like you did the bottom inboard. Before you put ont he top pieces you need to lightly heat shrink out any wrinkles along the leading and trailing edges where the top sheets will overlap the bottom sheets. Don't shrink the open bays yet. Put on the top sheets wrapping around the leading and trailing edges. You will have to cut the tissue on the leading edge at the center of the wing to get the tissue to lay flat. Now you have the entire wing covered and its time to shrink the tissue.

Shrinking the Tissue: Using a heat gun start shrinking the tissue at the leading edge of the first open bay on the top. Hold the wing up towards a light so you can see the wrinkes disappear. Go slowly with the heat. It is very easy to burn a hole in the tissue if you go too fast. Work from leading edge towards the trailing edge. Next do the same on the bottom of the wing. Then work the second bay from front to rear. Continue to shrink the tissue on both sides of the wing working out towards the tips one bay at a time. This will insure that you don't warp the wing. Finally shrink the other side in the same manner. Apply several coats of dope before you sand. Don't sand into the tissue.

Any questions or suggestions call me, Tom Morris at (256) 820-1983 or
email me at tom_morris@prodigy.net

Dacron Hinges Instructions

I nearly always use nitrate dope up to and including filler coats. But, I have used butyrate once. It worked about as well as the nitrate. Brush on three coats on the entire airplane including the control surfaces (flaps and elevators) sanding lightly between coats.

Attach the dacron hinges to the control surfaces with more dope, one hinge on the top, next hinge on the bottom, shoulder to shoulder, all the way from one end of the control surface to the other. I use 1" X 2" on the flaps and 1" X 1" on the elevators. If you have three good coats on the control surfaces the dacron hinges should readily adhere to the surface. Let dry.

Slide the control surfaces onto the horns and pin in place at the tips and, if necessary to straighten, in the middle between two hinges. Dope down the hinges to the trailing edge of the wing and stab. View from the rear to ensure surfaces are centered. Again, if you have three good coats on the wing and stab, the hinges should just suck right down. Let dry.

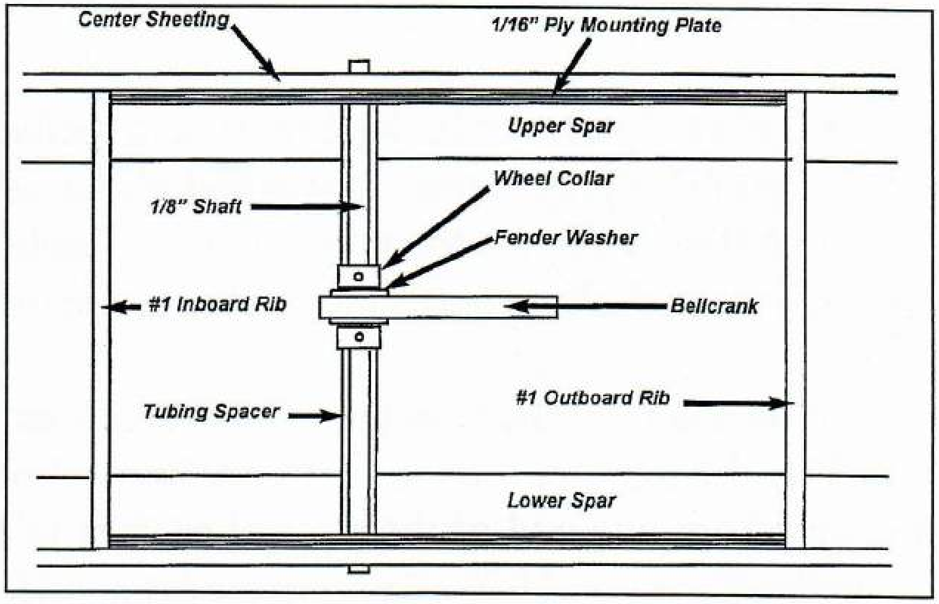
Take out any wrinkles in the hinges with a heat gun.

Recoat the hinge area with a couple of talcum powder or mico balloons and dope filler coats. Sand in between coats with a sanding block and 400 grit paper and the hinges will disappear.

How to Mount a Bellcrank

By Tom Morris

A couple of folks have lost airplanes recently because the bellcrank came lose on the mounting shaft causing the controls to lock up. Here is a simple and effective method to insure your bellcrank stays put.



I like to use fine linen phenolic for making bellcranks. It has a very high strength to weight ratio and, as a bonus, is slightly porus. Once it is oiled it acts like apermanently lubed bearing. We have examine these bellcranks after a thousand flights and they show no wear of the shaft hole or where the leadouts are connected

However, phenolic will melt if subjected to too much heat. For that reason I never solder the bellcrank retainers to the mounting shaft. Soldering also creates a long term corrosion problem.

Several years ago I was introduced to JB Weld and all its wonderful uses. It is really strong,, never corrodes, and has many good uses in our models.

For awhile I mounted the bellcrank to its shaft using eyelets. The eyelets were adhered to the shaft with good ole JB Weld. But the eyelets never gave the bellcrank the strong support I wanted. So I switched to wheel collars. They support the bellcrank very firmly. I still used JB Weld on the shaft just incase a wheel collar set screw were to loosen up.

Well, I got a call from Ed Capitanelli that the controls tried to lock up on his Cavalier. He found that both the wheel collar and JB Weld had failed. Recently Roy Tranthum lost a plane for the same reason. It appears that over time JB Weld exposed to air becomes brittle and simply vibrates off the mounting shaft.

So, I discussed it with some of the top fliers and builders and this is what we recommend to assure that your bellcrank stays put on the mounting shaft.

1. Thoroughly clean the 1/8" music wire bellcrank mounting shaft. Insert the shaft into the bellcrank pivot hole. Make sure it is snug but not tight. It will free up after it is lubricated. Secure the bellcrank firmly in the center of the shaft with the wheels collars on top of the fender washers. **Always use a thread locker on the set screws.** Do not cut a slot for the set screw in the music wire. That would severly weaken the mounting shaft.

2. Drill a 1/8" hole in the 1/16" mounting plates where you want the bellcrank to pivot. Cut two pieces of tubing to make spacers to go between the wheel collars and the ply mounting plates. The tubing can be fuel tubing, aluminum tubing, brass tubing, carbon fiber tubing, anything that will keep the bellcrank firmly centered no matter what. Slide the two spacers onto the mounting shaft. Slide the two pieces of 1/16" ply on the mounting shaft over the spacers. This captures the bellcrank and wheel collars.

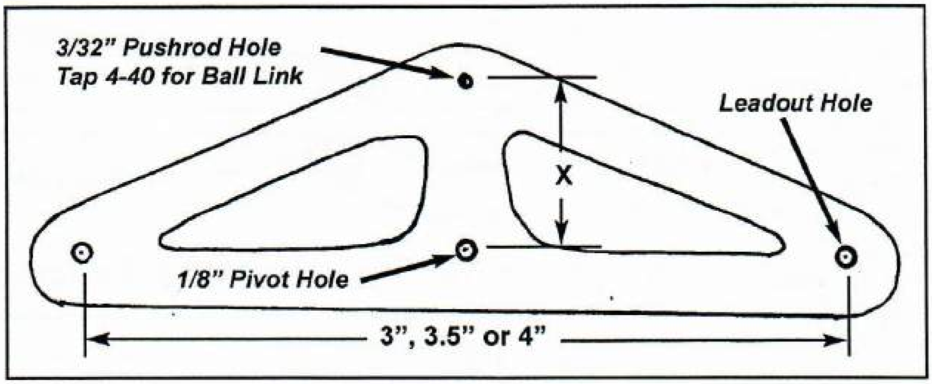
3. Remove 1/16" of the spar between the #1 ribs. The ply mounting plates will be inset into the spar just under the center sheeting.

4. Glue the bellcrank and ply plates assembly between the #1 ribs and on top of the spars.

5. Cut the excess 1/8" bellcrank mounting shaft off flush with the center sheeting. Cap the ends of the mounting shaft with a small piece of ply after the center sheeting is in place. This captures the shaft.

I'm sure there are many ways to successfully mount bellcranks. But, this way is simple, light, very strong and should never fail.

Perma-Lube Bellcrank Assembly

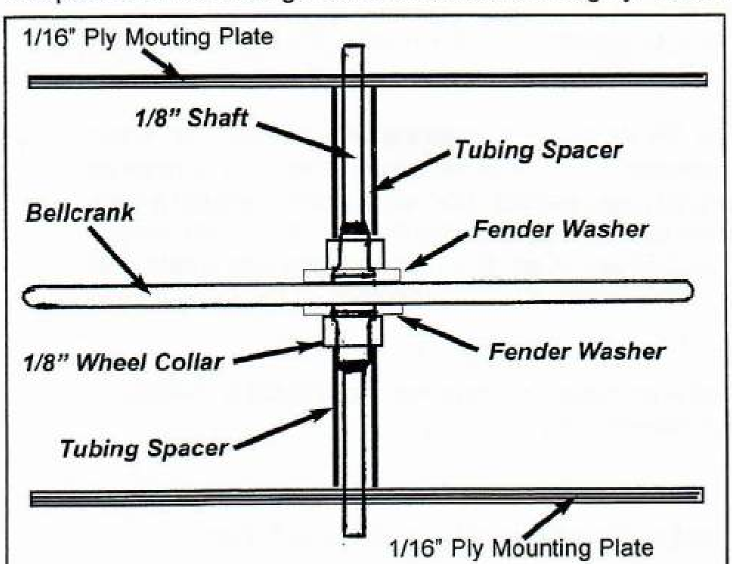


1. Thank you for purchasing the "Perma-Lube Bellcrank Assembly". This assembly has been very thoroughly tested and has proven to be plenty strong and long lasting. When used in conjunction with the other components of my "30/30 Matched Control System" you will get a slow and smooth movement of the control surfaces which matches the movement of your wrist. The leadouts are separated 3", 3.5" or 4". See separate sheet on where to drill your bellcrank.

2. The 4" bellcrank is made from 1/8" or 3/16" fine linen phenolic. The 3.5" and 3" are 1/8" thick. This material has a very high strength to weight ratio and, as a bonus, is slightly porus. Once lubricated it acts like a permanently lubed bearing. It is very friendlly to steel and to brass. We have examined bellcranks after a thousand flights with no evidence of wear. There is no need for additional bushings around the shaft or the leadout bushings.

3. The leadouts are .036", 19 strand, stressed relieved, stainless steel wire rope. They have an exceptionally smooth surface which minimizes the wear on the adjustable leadout eyelets. They are also very strong, 325 pound tensile strength, and very flexible and durable. The leadout connection at the bellcrank is a 3/32" copper swagging collar swagged with the appropriate swagging tool under 2000 pounds pressure. The leadout is doubled back against itself before swagging. It is then encapsulated in 3M 3/16" heat shrink tubing. You'll never have to worry about this connection coming apart. It is permanent! The wire will break before the connection!

4. Included in the assembly is a 1/8" by 3" music wire shaft, two #4 fender washers, two 1/8" id wheel collars and two pieces of fuel tubing. Clean the shaft thoroughly. Slide the shaft through the pivot hole (you may have to file it out a little, the shaft should be a snug but not tight fit at this point, it will loosen after it is lubricated), add the washers and Wheel collars to both sides of the bellcrank. Put on a little thread locker and tighten the set screws in the wheel collars. Put the fuel tubing spacers between the wheel collars and the ply mounting plates. There should be no up and down movement and very little side-to-side wobble.



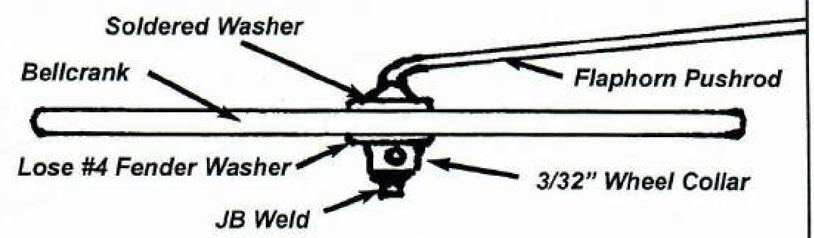
Mounting the Bellcrank to the Shaft

5. Mount the bellcrank assembly to the wing by suspending the bellcrank shaft between two pieces of 1/16" plywood glued to the spar and outer sheeting of the wing either directly under the sheeting or directly over the sheeting.

6. Once permanently installed in the wing, lubricate the pivot point and leadout bushings with a good, modern slicone based super lube and you'll never have to worry about that joint wearing out. The phenolic absorbs the lubricant and acts like a permanently oiled bearing.

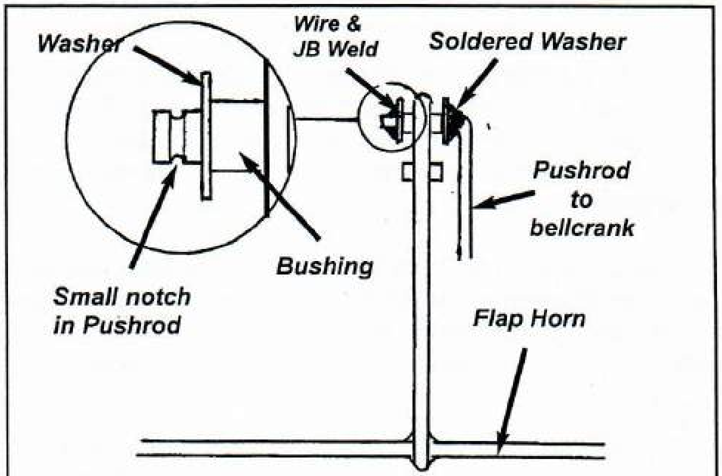
(Continued on other side)

For Bushed Systems



Mounting the Pushrod to the Bellcrank

7. After bending the bellcrank to flaphorn pushrod to the appropriate angle, solder a washer perpendicular to the rod going through the bellcrank. After fitting the pushrod to the flaphorn, thoroughly clean the pushrod, insert it through the hole in the bellcrank, add a #4 fender washer, then a 3/32" wheel collar. Tighten the wheel collar so that there is no verticle play or wobble. Use thead locker on the set screws. Add a little JB Weld below the wheel collar on the shaft. Thoroughly lubricate the pushrod connection top and bottom.



Mounting the Pushrod to the Flaphorn

8. To permanently connect the bellcrank to flaphorn pushrod to the flaphorn, solder a washer perpendicular to the pushrod at the bend. Cut the pushrod to length leaving about 1/8" beyond the flaphorn bushing. With a Dremel cut off disk, cut a shallow notch in the end of the pushrod. Put the pushrod through the bushing, add a #3 flat washer, wrap a few round of fine copper wire into the groove and against the washer. Twist the end tightly together. Add a little JB Weld to the pushrod end, the copper wire and against the washer. Make sure you don't get any behind the washer. Another option is to use a 3/32" wheel collar to secure the pushrod. Always use thread locker on the set screws in wheel collars.

9. When the JB Weld has dried, lubricate the bushings. You should now have a smooth control system with no play in it. And, you'll never have to worry about it coming apart.

10. Use the same method for connecting the flap horn to elevator horn pushrod to the flap horn and to the elevator horn.

11. **Wrapping the Ends.** Two 1/8" ID brass thimbles, 36" of .010 line wrapping wire and two 1/8" X 3/4" pieces of heat shrink tubing have been included to complete the ends of the leadouts where you hook up your control lines. Slip on a piece of heat shrink tubing on each leadout. Run each leadout around a thimble, wrap tightly with the wrapping wire from the thimble to 5/8" away from the thimble. Cut off the excess leadout. Continue wrapping about 1/8" beyond the double wire. Coat this wrapped joint with a light coat of epoxy or JB Weld. Let dry. Slide the heat shrink tubing over the wrapped portion of the leadouts and heat shrink it tightly around the wrap.

12. If you have any questions or suggestions or need help in any way give me a call (256) 820-6970, or Fax me at (256) 820-6977, or E-Mail me at ctmorris@cableone.net

Tom Morris
327 Pueblo Pass
Anniston, AL 36206

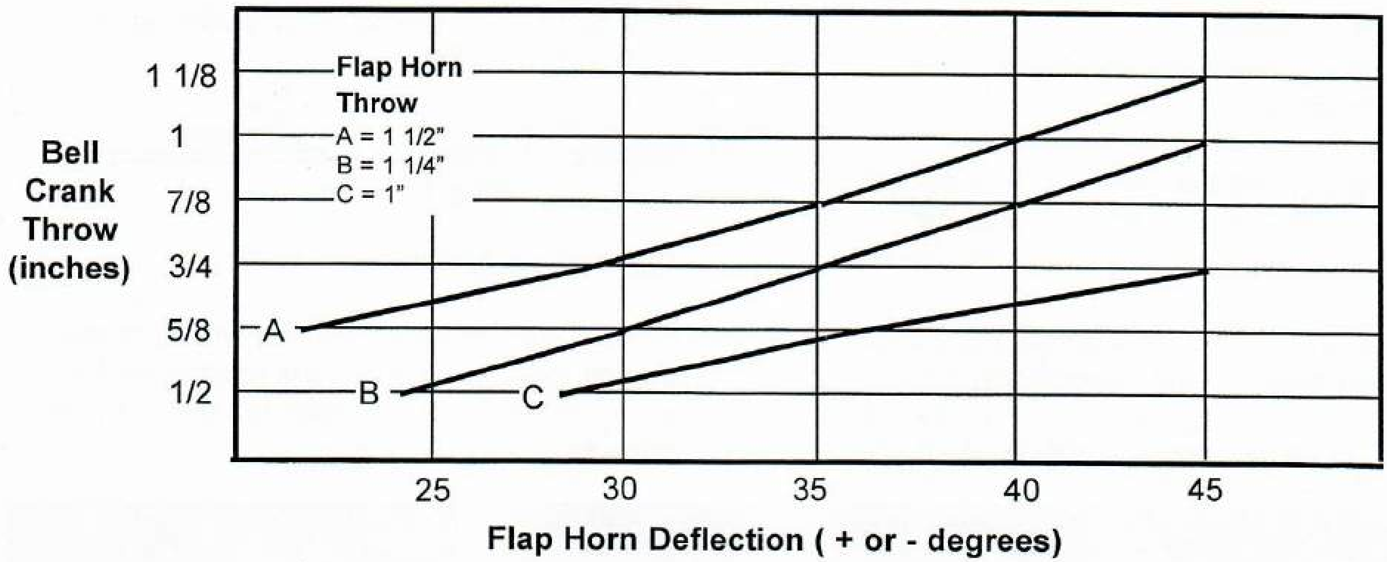
See Separate Instruction Sheet for Ball Link Systems

Where do I drill the hole in my bellcrank?

How much flap deflection do you want? How tall is your flap horn?

I built three Bellcrank to Flap horn models to empirically answer the question "Where do I drill the hole in my bellcrank?" The photo below is the model with the flap horn which has the top hole at 1 1/2". The other two models were for flap horns with the top hole at 1 1/4" and 1". I cut a slot in the bellcranks so I could move the pushrod connection in and out from the pivot. Below is the data.

My personal preference is 30 degrees of flap deflection. That is a good compromise between smooth insensitve flying and having what you need in "an emergency" manuever. To get 30 degrees flap deflection with a flap horn with the top hole (flap horn throw) at 1 1/2" I drill my pushrod connection hole on the bellcrank at 3/4" (bellcrank throw). For a flap horn with a 1 1/4" throw it is 5/8". For a flap horn with a 1" throw it is 9/16". Use this data to determine where to drill the hole in your bellcrank.



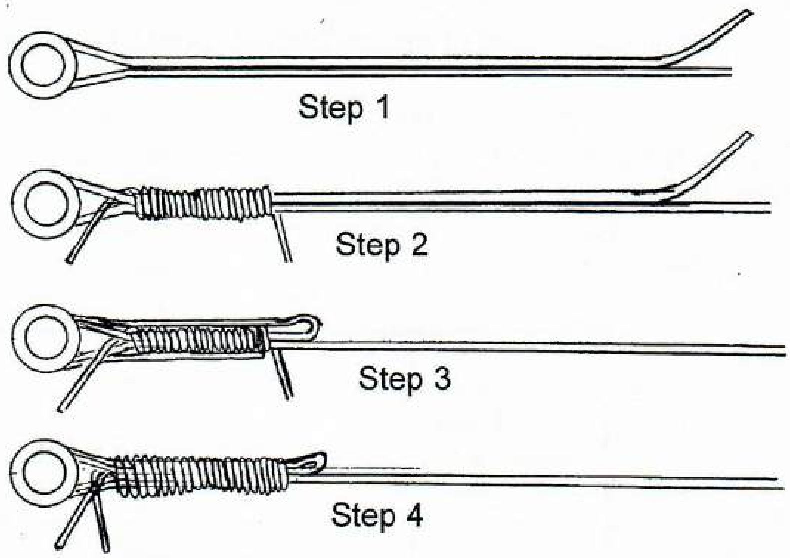
Flap Horn Deflection (+ or - degrees)

Bell Crank Throw (inches)	Flap Horn Throw (inches)		
	1 1/2	1 1/4	1
1 1/8	45.0		
1 1/16	42.5		
1	40.0	45.0	
15/16	37.5	43.0	
7/8	35.0	40.0	
13/16	32.0	37.5	
3/4	29.0	35.0	45.0
11/16	26.0	32.5	41.0
5/8	22.0	30.0	36.0
9/16		27.0	32.0
1/2		24.0	28.0

A slot in the bellcrank let's me change the amount of throw in the bellcrank. This is the model with the flap horn top hole at 1 1/2"

Wrapping Control Lines

It is very important that you correctly wrap your control lines to insure safe flying and to comply with the AMA Rule Book. The rules currently allow a method involving crimping, but experience has proven that wrapping is far superior. Following is a step by step instruction on how to properly wrap your control lines. This method exceeds the rule book requirement in that you have two wires around the thimble instead of one. This method eliminates the weak points next to the thimble.

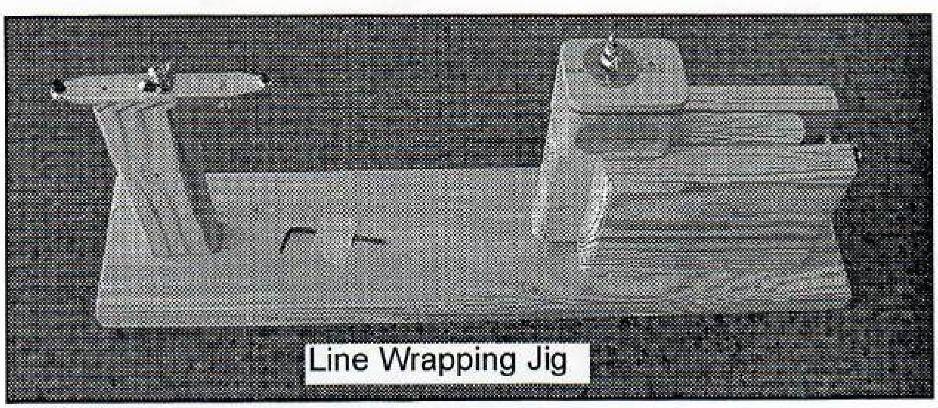


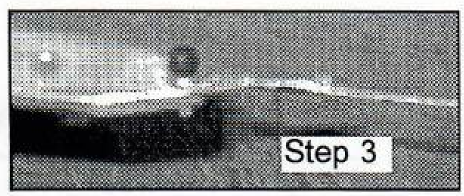
Step 1. Wrap the control line twice around the thimble and lay the short end back against the long line. Leave at least 4" for handling.

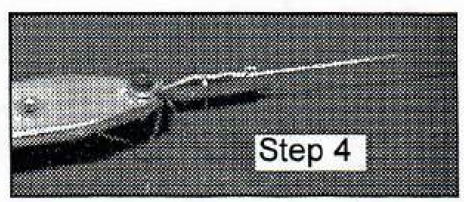
Step 2. Using 18" of fine copper serving wire, tuck one end through the "V" and start wrapping around the two control lines. After a few rounds snug the serving wire towards the thimble, but don't overdo it. Wrap outward a minimum of 5/8".

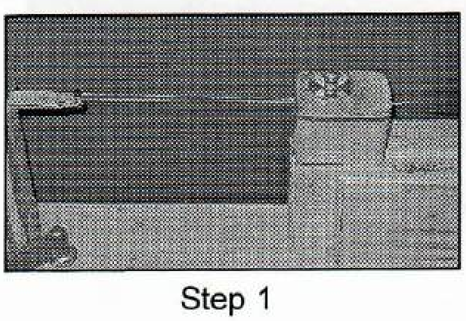
Step 3. Fold the short piece of control line back over the wrap, around the thimble and outward. Cut off the short piece just short of where it was folded back.

Step 4. Wrap back towards the thimble over the two new wires and the previous wrapped wires. Insert the lose end through the "V" and twist the lose ends together. Cut off the excess serving wire. It is not necessary to coat the wrap with a light coat of epoxy or JB Weld, but you may do so. **Do Not Solder.** Soldering will cause corrosin inside the wrap.

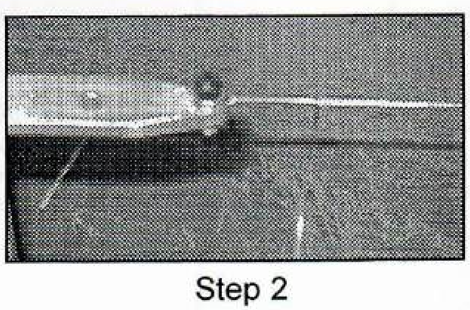




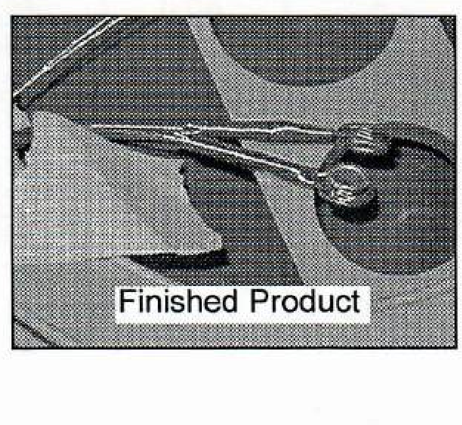




Step 1



Step 2



"Sure Grip" Handle Instructions

Congratulations! You have purchased an affordable, quality handle kit that will allow you to fit the handle to your hand and adjust the handle to your plane. At last, you can afford to have a fully adjustable handle for each of your airplanes. With all the adjustable features of this handle you can trim each plane exactly to your own preferences.

Why a handle kit? Everyone's hand is different. With this kit you can shape and form the handle to fit your hand exactly as you prefer. Why a handle kit? Every plane should have its own set of lines and handle. This handle kit is so affordable you can do just that. Why a fully adjustable handle kit? Each airplane is a little different. With this handle you can trim each of your airplanes exactly to your own preferences.

Parts List

(1) 1/2" X 6" X 3" Plywood Handle Base
(2) 3/8" X 4" X 1 1/4" Spruce Pistol Grips (Pre-shaped and glued to Base)
(1) 1/8" X 1/2" X 6" Aluminum Control Bar
(2) 4-40 Eyebolts
(2) 4-40 Hex Nuts
(2) 4-40 Lock Nuts
(2) 6-32 X 2" Machine Screws
(2) 6-32 Hex Nuts
(2) #6 Flat Washers
(1) # 3 (Medium) Connecting Link
(1) 30" Nylon Safety Thong

Assembly Instructions

Step #1 - Form and shape the handle so that it fits firmly and comfortably in your hand. The 3/4" round Perma-Grit file and the Dremel drum sander do an excellent job of this. If you have large hands, widen the space around the finger grips to fit your hand. Round and smooth all corners. If you have small hands narrow the grip by taking some off the sides of the pistol grips and/or some off the rear of the base and grips. Make it fit your hand.

Step #2 - If it hasn't already been done for you, drill 7/64" holes in the aluminum control bar for the 4-40 eyebolts. Mark the center of the bar and measure out 1 1/2" each way for your first set of holes. Those holes give you 3" line separation. Measure out 2" from the center and drill another pair of holes. Those holes give you 4" line separation. Drill two or three intermediate holes between these two to give you adjustments between the two. That will usually give you all the adjustment you need, but there is plenty of space left on the bar to go wider or narrower if you need to.

(Continued on Back)

Cable Alternative - Try using the eyebolts first. I've used both cable and eyebolts for years and see no difference at all in the feel. I firmly believe the eyebolts are safer than the cables. Cables eventually break, eyebolts don't. The eyebolts make adjustments so much easier. But, if you would prefer to connect you lines to cables rather than eyebolts, drill the two holes shown near the center of the handle. You can then weave the cable through the holes. Be sure and round the edges of all holes so there are no sharp edges against the cable. Your local bike or motorcycle shop has good cable used for brakes and throttles. A 1/16" nylon coated cable is the safest.

Step # 3 - If they haven't already been drilled for you, drill two 5/32" holes in the aluminum control bar for the 6-32 machine screws used to mount the bar to the handle base. These must be lined up with the holes in the handle base.

Step #4 - Mount the control bar to the handle base with the 6-32 machine screws, #6 washers, and the 6-32 hex nuts. Tighten firmly.

Step #5 - Mount the 4-40 eyebolts to the aluminum control bar. Screw one of the 4-40 hex nuts all the way down on the 4-40 eyebolt before inserting it through the hole. The 4" spacing holes is a good choice to start. **Tighten the rear 4-40 lock nut firmly, but don't over tighten.**

Step #6 - Smooth and round the top and bottom of the handle base and aluminum control bar.

Step #7 - Drill a 1/8" hole for the safety thong as shown near the bottom of the handle base.

Step #8 - Apply your favorite finish to your handle. A finish that matches the plane looks real sharp!

Step #9 - While your finish is drying, make up your safety thong. Make a small loop in one end of the nylon cord. Pass the other end through the loop and then through the hole at the bottom of the handle and tie a knot in that end. Now you're ready to hook up you lines to your handle and go fly!

Adjusting/Trimming

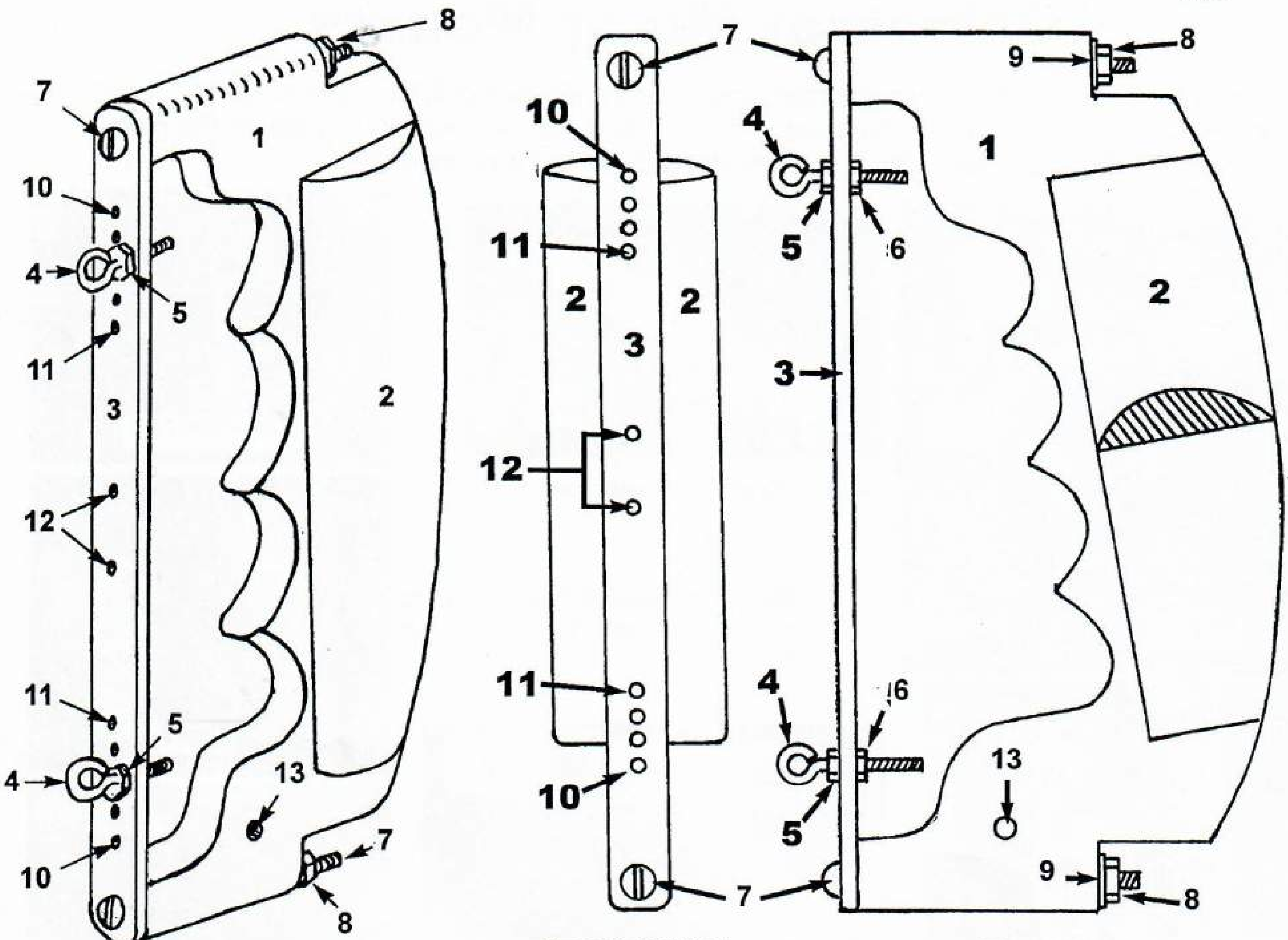
Adjusting to neutral. If you are careful when you make up your lines and leadouts, you will only need a small adjustment to neutral. A #3 (medium) connecting link is included in your kit. Use it on one line and use the large connecting links that came with your lines on the other line to adjust to neutral. You can also adjust by using a scissors type line clip. A second way to adjust neutral is by moving the 4-40 eyebolts in and out. A third way of adjusting neutral is by removing some of the overhang at one end of the handle. Notice that the handle is designed with a 15 degree down bias. That puts neutral exactly in the middle of the arc of your wrist swing.

Adjusting for Sensitivity. Keep moving the 4-40 eyebolts closer to the center thus reducing sensitivity, until your corners and level flight are smooth. If your plane is turning too quickly in one direction, move that connection towards the center until it feels balanced with the other direction. In just a few flights you'll have it feeling just right for you.

Note: After all adjustments have been made, secure all nuts with Loctite 242 (red) thread locker. You can also cut off the excess threads on the 4-40 eyebolts. For maximum security put a little JB Weld on the 4-40 eyebolt threads behind the lock nut.

SAFETY REMINDER: Never fly anywhere near electric lines of any kind!

If you have any problems or suggestions call me at (256) 820-1983 or write me at 327 Pueblo Pass, Anniston, AL 36206, or E-Mail me at tom_morris@prodigy.net



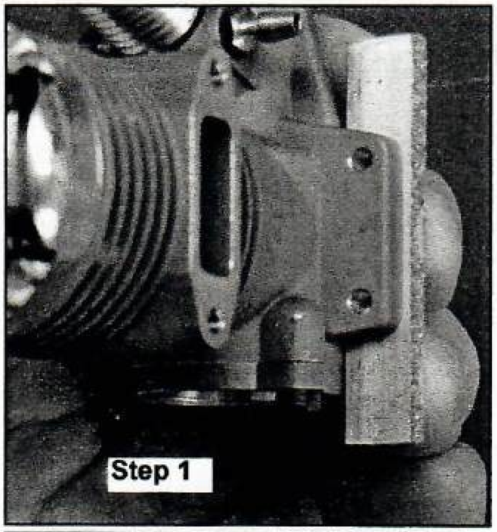
ISOMETRIC VIEW FRONT VIEW SIDE VIEW

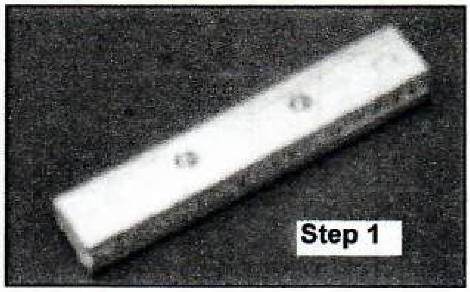
Key

#1 - 1/2" X 6" X 3" Plywood Handle Base
#2 - 3/8" X 4" X 1 1/4" Spruce Pistol Grips (Installed)
#3 - 1/8" X 1/2" X 6" Aluminum Control Bar
#4 - 4-40 Eyebolts
#5 - 4-40 Hex Nuts
#6 - 4-40 Lock Nuts
#7 - 6-32 X 2" Machine Screws
#8 - 6-32 Hex Nuts
#9 - #6 Flat Washers
#10 - 7/64" Holes for 4-40 Eyebolt (4" LIne Spacing)
#11 - 7/64" Holes for 4-40 Eyebolt (3" Line Spacing)
#12 - 7/64" Holes for Cable Option+
#13 - 1/8" Hole for Safety Thong

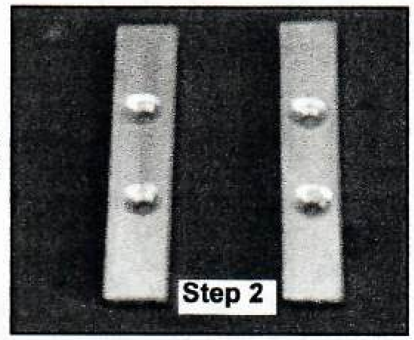
Universal Motor Mounts

This hardware allows you to exchange motors without having to redrill the motor mounts. You mount the motor to the pads with flat head 4-40 bolts and then you mount the pads to the motor mounts. When you exchange motors you use the old pads as a pattern.

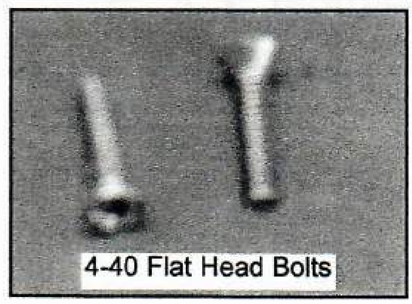


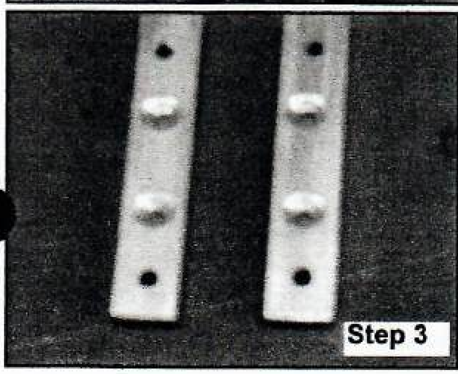


Step 1. Custom drill both pads to the motor.

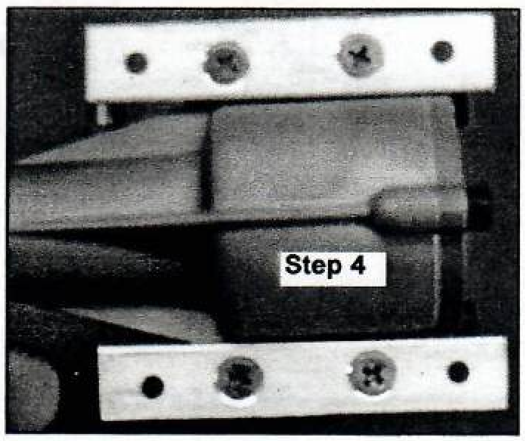


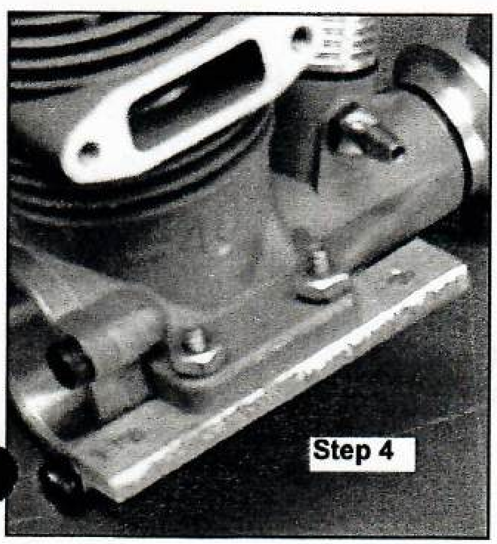
Step 2. Countersink the holes on the bottom of the pads to fit the 4-40 flat head bolts.





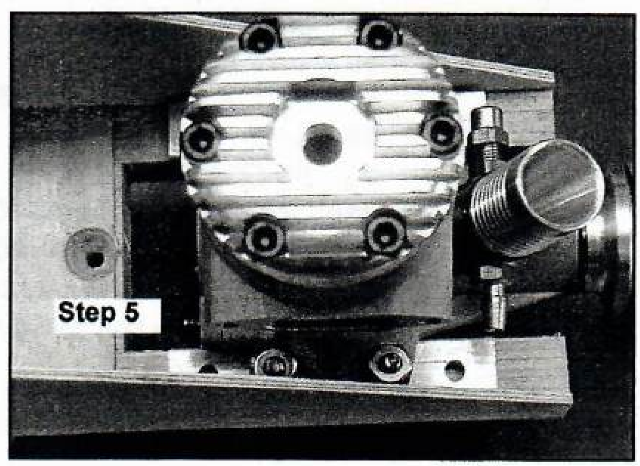
Step 3. Drill the outerholes for the 4-40 hex head bolts that will mount the pads to the motor mounts.





Step 4. Mount the pads to the engine with the 4-40 flathead bolts from below. The engine is secured with a #4 split washer and a 4-40 hex nut.

Step 5. Mount the pads to the motor mounts using 4-40 hex head bolts and 4-40 blind nuts on the bottom of the motor mounts.



Note: Use the old pads as a pattern for drilling the outer holes in the new pads. You will need a new set of pads for each engine.

WING LOADING

Wing Area IN²	Wing Area FT²	VERY LIGHT 9.88 - 10.65 oz/ft^2	LIGHT 10.66 - 11.43 oz/ft^2	MEDIUM 11.44 - 12.21 oz/ft^2	HEAVY 12.22 - 12.99 oz/ft^2	VERY HEAVY 13.00 - 13.77 oz/ft^2
		Airplane Weight In Ounces				
480-490	3.37	33.3 - 35.9	36.0 - 38.5	38.6 - 41.1	41.2 - 43.8	43.9 - 46.4
491-500	3.44	34.0 - 36.6	36.7 - 39.3	39.4 - 42.0	42.1 - 44.7	44.8 - 47.4
501-510	3.51	34.7 - 37.4	37.5 - 40.1	40.2 - 42.9	43.0 - 45.6	45.7 - 48.3
511-520	3.58	35.4 - 38.1	38.2 - 40.9	41.0 - 43.7	43.8 - 46.5	46.6 - 49.3
521-530	3.65	36.0 - 38.9	39.0 - 41.7	41.8 - 44.6	44.7 - 47.4	47.5 - 50.3
531-540	3.72	36.8 - 39.6	39.7 - 42.5	42.6 - 45.4	45.5 - 48.3	48.4 - 51.2
541-550	3.79	37.4 - 40.4	40.5 - 43.3	43.4 - 46.3	46.4 - 49.2	49.3 - 52.2
551-560	3.85	38.0 - 41.0	41.1 - 44.0	44.1 - 47.0	47.1 - 50.0	50.1 - 53.0
561-570	3.92	38.7 - 41.7	41.8 - 44.8	44.9 - 47.8	47.9 - 50.9	51.0 - 53.9
571-580	3.99	39.4 - 42.4	42.5 - 45.6	45.7 - 48.7	48.8 - 51.8	51.9 - 54.9
581-590	4.06	40.1 - 43.2	43.3 - 46.4	46.5 - 49.5	49.6 - 52.7	52.8 - 55.9
591-600	4.13	40.8 - 44.0	44.1 - 47.2	47.3 - 50.4	50.5 - 53.6	53.7 - 56.9
601-610	4.20	41.5 - 44.7	44.8 - 48.0	48.1 - 51.3	51.4 - 54.6	54.7 - 57.8
611-620	4.27	42.2 - 45.5	45.6 - 48.8	48.9 - 52.1	52.2 - 55.5	55.6 - 58.8
621-630	4.34	42.9 -46.2	46.3 - 49.6	49.7 - 53.0	53.1 - 56.4	56.5 - 59.8
631-640	4.41	43.6 - 47.0	47.1 - 50.4	50.5 - 53.8	53.9 - 57.3	57.4 - 60.7
641-650	4.48	44.3 - 47.7	47.8 - 51.2	51.3 - 54.7	54.8 - 58.2	58.3 - 61.7
651-660	4.55	45.0 - 48.5	48.6 - 52.0	52.1 - 55.5	55.6 - 59.1	59.2 - 62.6
661-670	4.62	45.6 - 49.2	49.3 - 52.8	52.9 - 56.4	56.5 - 60.0	60.1 - 63.6
671-680	4.69	46.3 - 49.9	50.0 - 53.6	53.7 - 57.3	57.4 - 60.9	70.0 - 64.6
681-690	4.76	47.0 - 50.7	50.8 - 54.4	54.5 - 58.1	58.2 - 61.8	61.9 - 65.5
691-700	4.83	47.7 - 51.4	51.5 - 55.2	55.3 - 59.0	59.1 - 62.7	62.8 - 66.5
701-710	4.90	48.4 - 52.2	52.3 - 56.0	56.1 - 59.9	60.0 - 63.7	63.8 - 67.5
711-720	4.97	49.1 - 52.9	53.0 - 56.8	56.9 - 60.7	60.8 - 64.6	64.7 - 68.4
721-730	5.03	49.7 - 53.6	53.7 - 57.5	57.6 - 61.4	61.5 - 65.3	65.4 - 69.3
731-740	5.10	50.4 - 54.3	54.4 - 58.3	58.4 - 62.3	62.4 - 66.2	66.3 - 70.2
741-750	5.17	51.1 - 55.1	55.2 - 59.1	59.2 - 63.1	63.2 - 67.2	67.3 - 71.2
751-760	5.24	51.8 - 55.8	55.9 - 59.9	60.0 - 64.0	64.1 - 68.1	68.2 - 72.2
761-770	5.31	52.5 - 56.6	56.7 - 60.7	60.8 - 64.8	64.9 - 69.0	69.1 - 73.1
771-780	5.38	53.2 - 57.3	57.4 - 61.5	61.6 - 65.7	65.8 - 69.9	70.0 - 74.1
781-790	5.45	53.9 - 58.0	58.1 - 62.3	62.4 - 66.5	66.6 - 70.8	70.9 - 75.0
791-800	5.52	54.5 - 58.8	58.9 - 63.1	63.2 - 67.4	67.5 - 71.7	71.8 - 76.0
801-810	5.59	55.2 - 59.5	59.6 - 63.9	64.0 - 68.3	68.4 - 72.6	72.7 - 77.0
811-820	5.66	55.9 - 60.3	60.4 - 64.7	64.8 - 69.1	69.2 - 73.5	73.6 - 77.9

Tap Drill Chart

Bolt/Tap Size	Drill Size	Bolt/Tap Size	Drill Size	Bolt/Tap Size	Drill Size	Bolt/Tap Size	Drill Size
0-80 NF	3/64"	3/16-32 NS	No. 25	9mm-1.00	8.1mm	11/16-11 NS	19/32"
1-64 NC	No. 53	10-24 NC	No. 22	9mm-0.75	8.3mm	18mm-2.50	15.8mm
1-72 NF	No. 53	10-32 NF	No. 21	3/8-24 NF	Ltr. Q	11/16-16 NS	5/8"
2-56 NF	No. 50	5mm-0.90	4.2mm	10mm-1.50	8.7mm	3/4-10 NC	21/32"
2-64 NF	No. 50	5mm-0.80	4.4mm	10mm-1.25	8.9mm	18mm-1.5	16.8mm
3-48 NC	No. 47	12-24 NC	No. 16	10mm-1.00	9.1mm	3/4-16 NF	11/16"
3-56 NF	No. 45	12-28 NF	No. 14	7/16-14 NC	Ltr. U	20mm-2.5	17.8mm
4-36 NS	No. 44	12-32 NEF	No. 13	11mm-1.50	9.7mm	7/8-9 NC	49/64"
4-40 NC	No. 43	14-20 NS	No. 10	7/16-20 NF	25/64"	7/8-14 NF	13/16"
4-48 NF	No. 42	1/4-20 NC	No. 7	12mm-1.75	10.5mm	22mm-1.5	20.9mm
3mm-0.60	2.5mm	14-24 NS	No. 7	12mm-1.50	10.7mm	7/8-18 NS	53/64"
1/8-40 NS	No. 38	6mm-1.00	5.2mm	1/2-13 NC	27/64"	24mm-3.00	21.4mm
5-40 NC	No. 38	1/4-24 NS	No. 4	12mm-1.25	10.9mm	1-8 NC	7/8"
5-44 NF	No. 37	1/4-28 NF	No. 3	1/2-20 NF	29/64"	24mm-2.00	22.3mm
6-32 NC	No. 35	1/4-32 NEF	7/32"	1/2-24 NS	29/64"	1-12 NF	59/64"
6-36 NS	No. 34	1/4-40 NS	No.1	14mm-2.00	12.2mm	1-14 NS	15/16"
6-40 NF	No. 33	7mm-1.00	6.1mm	9/16-12 NC	31/64"	1 1/8-7 NC	63/64"
6-48 NS	No. 31	5/16-18 NC	Ltr. F	14mm-1.50	12.7mm	1 1/8-12 NF	1 3/64"
4mm-0.70	3.4mm	8mm-1.25	6.9mm	14mm-1.25	12.8mm	1 1/4-7 NC	1 7/64"
4mm-0.75	3.4mm	5/16-24 NF	Ltr. I	9/16-18 NF	33/64"	1 1/4-12 NF	1 11/64"
8-32 NC	No. 29	8mm-1.00	7.1mm	5/8-11 NC	17/32"	1 3/8-6 NC	1 7/32"
8-36 NF	No. 29	5/16-32 NEF	9/32"	16mm-2.00	14.2mm	1 3/8-12 NF	119/64"
8-40 NS	No. 28	9mm-1.25	7.9mm	5/8-18 NF	37/64"	1 1/2-6 NC	1 11/32"
3/16-24 NS	No. 26	3/8-16 NC	5/16"	16mm-1.50	14.7mm	1 1/2-12 NF	1 27/64"

Balsa Density Chart

For 3" X 36 Sheets

Weight in Grams

Thickness ↓	Density in Pounds per Cubic Foot											
	Very Light		Light		Medium Light		Medium		Medium Hard		Hard	
Density lbs/ft³ →	4	5	6	7	8	9	10	11	12	13	14	15
1/32"	3	4	5	6	7	8	9	10	11	12	13	14
1/20"	6	7	9	10	11	13	14	16	17	18	20	21
1/16"	7	9	11	12	14	16	18	20	21	23	25	27
3/32"	11	13	16	19	21	24	27	29	32	35	37	40
1/8"	14	18	21	25	28	32	35	39	43	46	50	53
3/16"	21	27	32	37	43	48	53	59	64	69	74	80
1/4"	28	35	43	50	57	64	71	78	85	92	99	106
5/16"	35	44	53	62	71	80	89	98	106	115	124	133
3/8"	43	53	64	74	85	96	106	117	128	138	149	160
1/2"	58	71	85	99	114	128	142	156	170	184	199	213
3/4"	85	106	128	149	170	192	213	234	255	277	298	319
1"	114	142	170	199	227	255	284	312	341	369	397	426

Drill Sizes in Metric and Decimal Inches

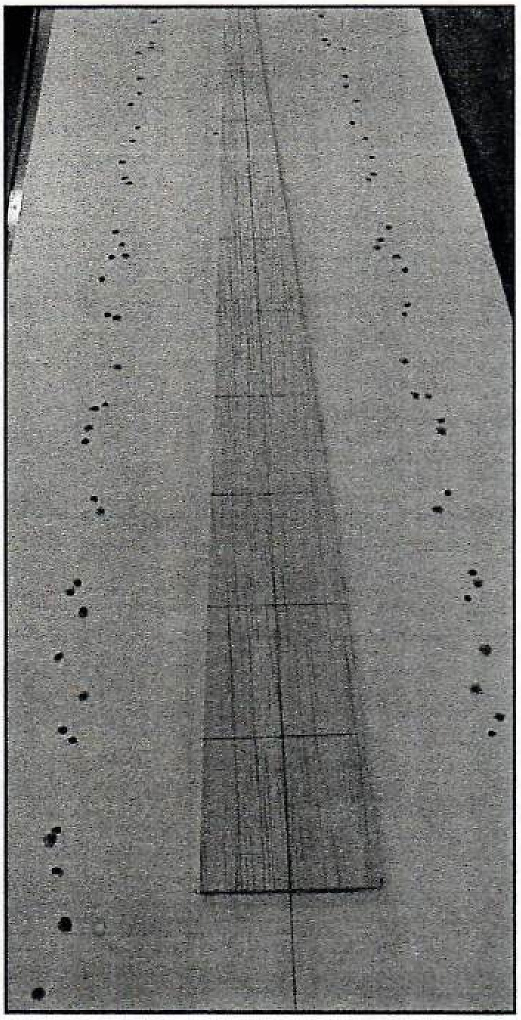
Drill Size	mm	Decimal Inches
-	0.10	.0039
-	0.20	.0079
-	0.25	.0098
-	0.30	.0118
80	0.34	.0135
79	0.37	.0145
1/64	0.40	.0156
78	0.41	.0160
77	0.46	.0180
-	0.50	.0197
76	0.51	.0200
75	0.53	.0210
74	0.57	.0225
-	0.60	.0236
73	0.61	.0240
72	0.64	.0250
71	0.66	.0260
-	0.70	.0276
70	0.71	.0280
69	0.74	.0292
-	0.75	.0295
68	0.79	.0310
1/32	0.79	.0313
-	0.80	.0315
67	0.81	.0320
66	0.84	.0330
65	0.89	.0350
-	0.90	.0354
64	0.91	.0360
63	0.94	.0370
62	0.97	.0380
61	0.99	.0390
-	1.00	.0394
60	1.02	.0400
59	1.04	.0410
58	1.07	.0420
57	1.09	.0430
56	1.18	.0465
3/64	1.19	.0469
55	1.32	.0520
54	1.40	.0550
53	1.51	.0595
1/16	1.59	.0625
52	1.61	.0635
51	1.70	.0670
50	1.78	.0700
49	1.85	.0730
48	1.93	.0760
5/64	1.98	.0781
47	1.99	.0785
-	2.00	.0787
46	2.06	.0810

Drill Size	mm	Decimal Inches
45	2.08	.0820
44	2.18	.0860
43	2.26	.0890
42	2.37	.0935
3/32	2.38	.0938
41	2.44	.0960
40	2.50	.0980
39	2.53	.0995
38	2.58	.1015
37	2.64	.1040
36	2.71	.1065
7/64	2.78	.1094
35	2.79	.1100
34	2.82	.1110
33	2.87	.1130
32	2.95	.1160
-	3.00	.1181
31	3.05	.1200
1/8	3.18	.1250
30	3.26	.1285
29	3.45	.1360
28	3.57	.1405
9/64	3.57	.1406
27	3.66	.1440
26	3.73	.1470
25	3.80	.1495
24	3.86	.1520
23	3.91	.1540
5/32	3.97	.1562
22	3.99	.1570
-	4.00	.1575
21	4.04	.1590
20	4.09	.1610
19	4.22	.1660
18	4.31	.1695
11/64	4.37	.1719
17	4.39	.1730
16	4.50	.1770
15	4.57	.1800
14	4.62	.1820
13	4.70	.1850
3/16	4.76	.1875
12	4.80	.1890
11	4.85	.1910
10	4.91	.1935
9	4.98	.1960
-	5.00	.1968
8	5.05	.1990
7	5.11	.2010
13/64	5.16	.2031
6	5.18	.2040

Drill Size	mm	Decimal Inches
5	5.22	.2055
4	5.31	.2090
3	5.41	.2130
7/32	5.56	.2188
2	5.61	.2210
1	5.79	.2280
A	5.94	.2340
15/64	5.95	.2344
-	6.00	.2362
B	6.05	.2380
C	6.15	.2420
D	6.25	.2460
1/4	6.35	.2500
E	6.35	.2500
F	6.53	.2570
G	6.63	.2610
17/64	6.75	.2656
H	6.76	.2660
I	6.91	.2720
-	7.00	.2756
J	7.04	.2770
K	7.14	.2810
9/32	7.14	.2812
L	7.37	.2900
M	7.49	.2950
19/64	7.54	.2969
N	7.67	.3020
5/16	7.94	.3125
-	8.00	.3150
O	8.03	.3160
P	8.20	.3230
21/64	8.33	.3281
Q	8.43	.3320
R	8.61	.3390
11/32	8.73	.3438
S	8.84	.3480
-	9.00	.3543
T	9.09	.3580
23/64	9.13	.3594
U	9.35	.3680
3/8	9.53	.3750
V	9.56	.3770
W	9.80	.3860
25/64	9.92	.3906
-	10.00	.3937
X	10.08	.3970
Y	10.26	.4040
13/32	10.32	.4062
Z	10.49	.4130
27/64	10.72	.4219
-	11.00	.4331

Drill Size	mm	Decimal Inches
7/16	11.50	.4375
29/64	11.51	.4531
15/32	11.91	.4688
-	12.00	.4724
31/64	12.30	.4844
1/2	12.70	.5000
-	13.00	.5118
33/64	13.10	.5156
17/32	13.49	.5312
35/64	13.89	.5469
-	14.00	.5512
9/16	14.29	.5625
37/64	14.68	.5781
-	15.00	.5906
19/32	15.08	.5938
39/64	15.48	.6094
5/8	15.88	.6250
-	16.00	.6299
41/64	16.27	.6406
21/32	16.67	.6562
-	17.00	.6693
43/64	17.07	.6719
11/16	17.46	.6875
45/64	17.86	.7031
-	18.00	.7087
23/32	18.26	.7188
47/64	18.65	.7344
-	19.00	.7480
3/4	19.05	.7500
49/64	19.45	.7656
25/32	19.84	.7812
-	20.00	.7874
51/64	20.24	.7969
13/16	20.64	.8125
-	21.00	.8268
53/64	21.03	.8281
27/32	21.43	.8438
55/64	21.84	.8594
-	22.00	.8661
7/8	22.23	.8750
57/64	22.62	.8906
-	23.00	.9055
29/32	23.02	.9062
59/64	23.42	.9219
15/16	23.81	.9375
-	24.00	.9449
61/64	24.21	.9531
31/32	24.61	.9688
-	25.00	.9842
63/64	25.00	.9844
1.00	24.40	1.0000

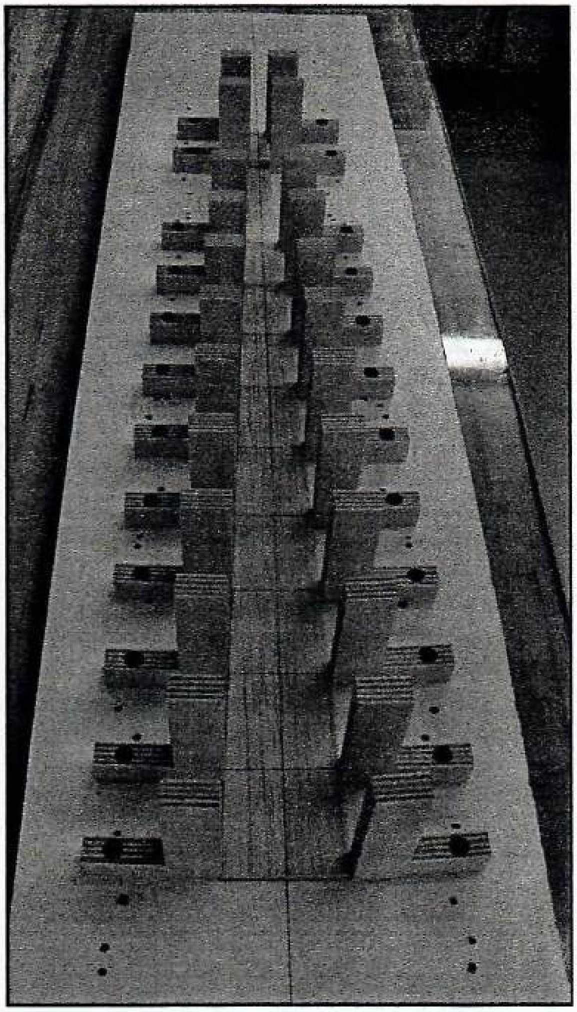
Fuselage Building Jig



Step 1. Make a pattern of the top view of the fuselage using See Temp. Draw the pattern on a hard piece of 1/8" balsa with center lines and with the former locations marked.

Step 2. Find a flat board wide enough and long enough for your fuselage. Draw a center line down the board.

Step 3. Using two sided tape or tack glue the 1/8" balsa pattern to the board matching center lines.



Step 4. Mount the "L" brackets to the board against the pattern and on each side of the former locations.

Step 5. Place your fuselage sides in the jig and fill in the formers and/or geodectic stringers.